



US 20250282242A1

(19) **United States**

(12) **Patent Application Publication**
TONG et al.

(10) **Pub. No.: US 2025/0282242 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **CHARGING SYSTEM**

Publication Classification

(71) Applicant: **Nanjing Chervon Industry Co., Ltd.**,
Nanjing (CN)

(72) Inventors: **Jiajun TONG**, Nanjing (CN);
Dongqing HAN, Nanjing (CN); **Zheng**
GENG, Nanjing (CN); **Yaxin FU**,
Nanjing (CN); **Jingru ZHANG**,
Nanjing (CN); **Yuexiang ZHANG**,
Nanjing (CN); **Yafei CAO**, Nanjing
(CN); **Qiheng Yang**, Nanjing (CN);
Guanzhou Li, Nanjing (CN)

(51) **Int. Cl.**
B60L 53/53 (2019.01)
B60L 53/30 (2019.01)
H02J 7/00 (2006.01)
H02J 7/34 (2006.01)

(52) **U.S. Cl.**
CPC **B60L 53/53** (2019.02); **B60L 53/305**
(2019.02); **H02J 7/00047** (2020.01); **H02J**
7/0013 (2013.01); **H02J 7/342** (2020.01);
B60L 2200/40 (2013.01); **B60L 2210/30**
(2013.01); **H02J 2207/20** (2020.01)

(21) Appl. No.: **19/220,348**

(22) Filed: **May 28, 2025**

Related U.S. Application Data

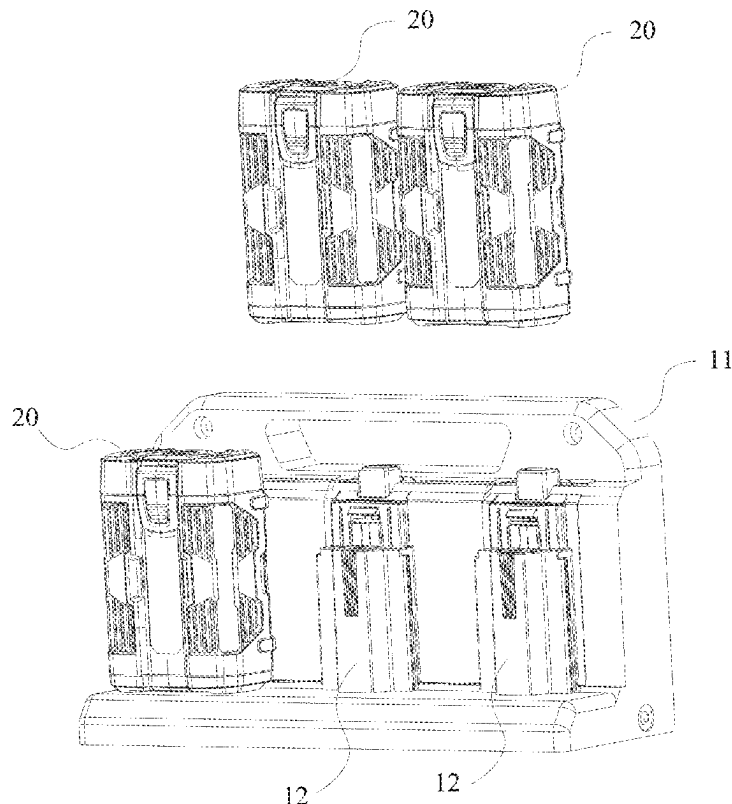
(63) Continuation-in-part of application No. 19/022,865,
filed on Jan. 15, 2025.

(30) **Foreign Application Priority Data**

Jan. 26, 2024	(CN)	202410117497.9
Jan. 26, 2024	(CN)	202410117667.3
Jan. 26, 2024	(CN)	202410117985.X
Jan. 26, 2024	(CN)	202420210815.1
May 30, 2024	(CN)	202410692020.3
May 30, 2024	(CN)	202410692358.9

ABSTRACT

A charging system includes an electrical energy conversion device and multiple electrical energy receiving devices connected in cascade. The charging system charges battery packs electrically connected to the multiple electrical energy receiving devices. The electrical energy conversion device includes a controller; a power interface for connecting an external power supply; a connecting port for connecting one of the multiple electrical energy receiving devices; and a conversion circuit electrically connected to the power interface and the connecting port separately. Each of the multiple electrical energy receiving devices includes an electrical energy input port for connecting the electrical energy conversion device or another of the multiple electrical energy receiving devices; and an electrical energy output port for connecting another of the multiple electrical energy receiving devices. The controller is configured to adjust an output of the connecting port based on the number of the multiple electrical energy receiving devices.



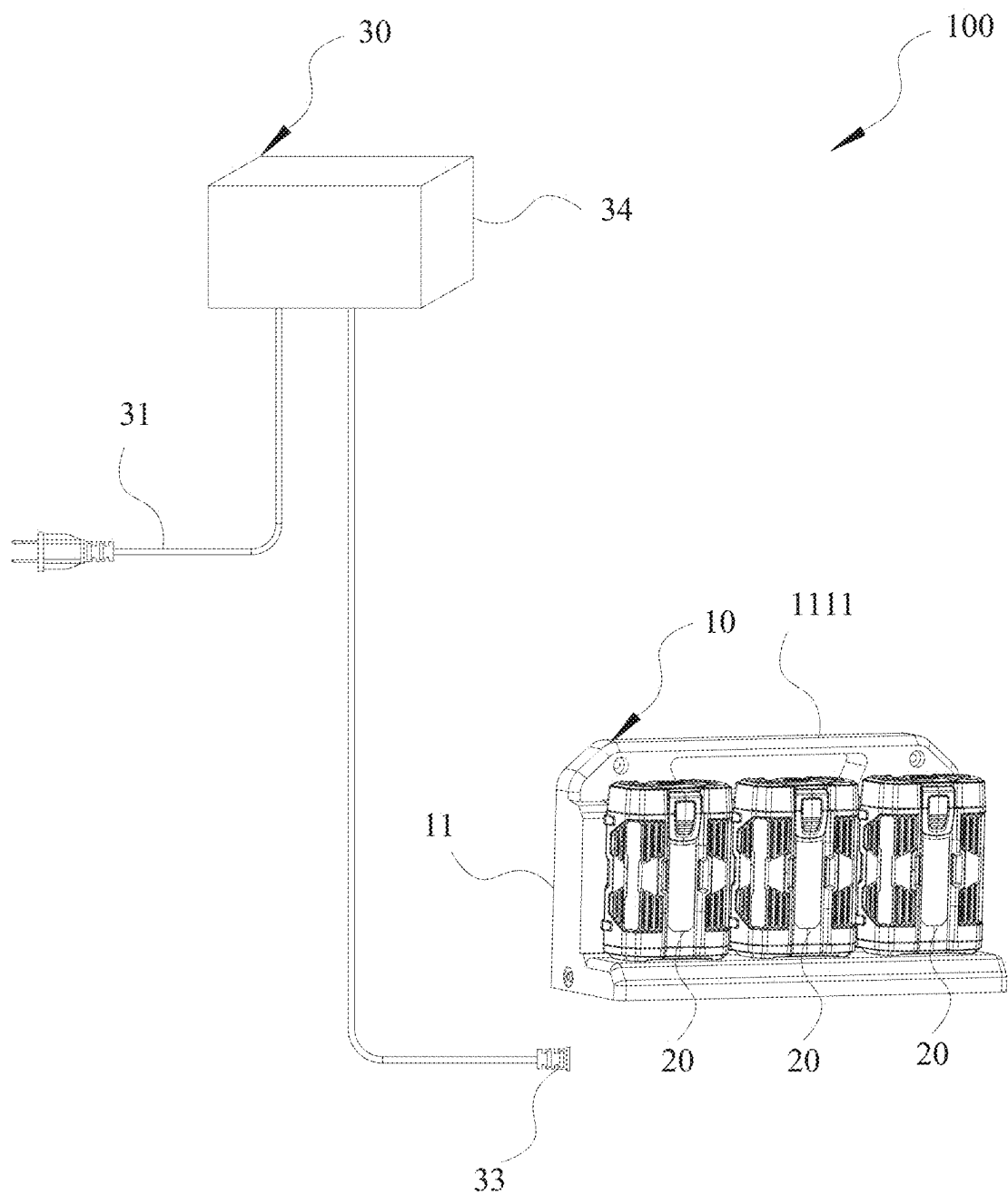


FIG. 1

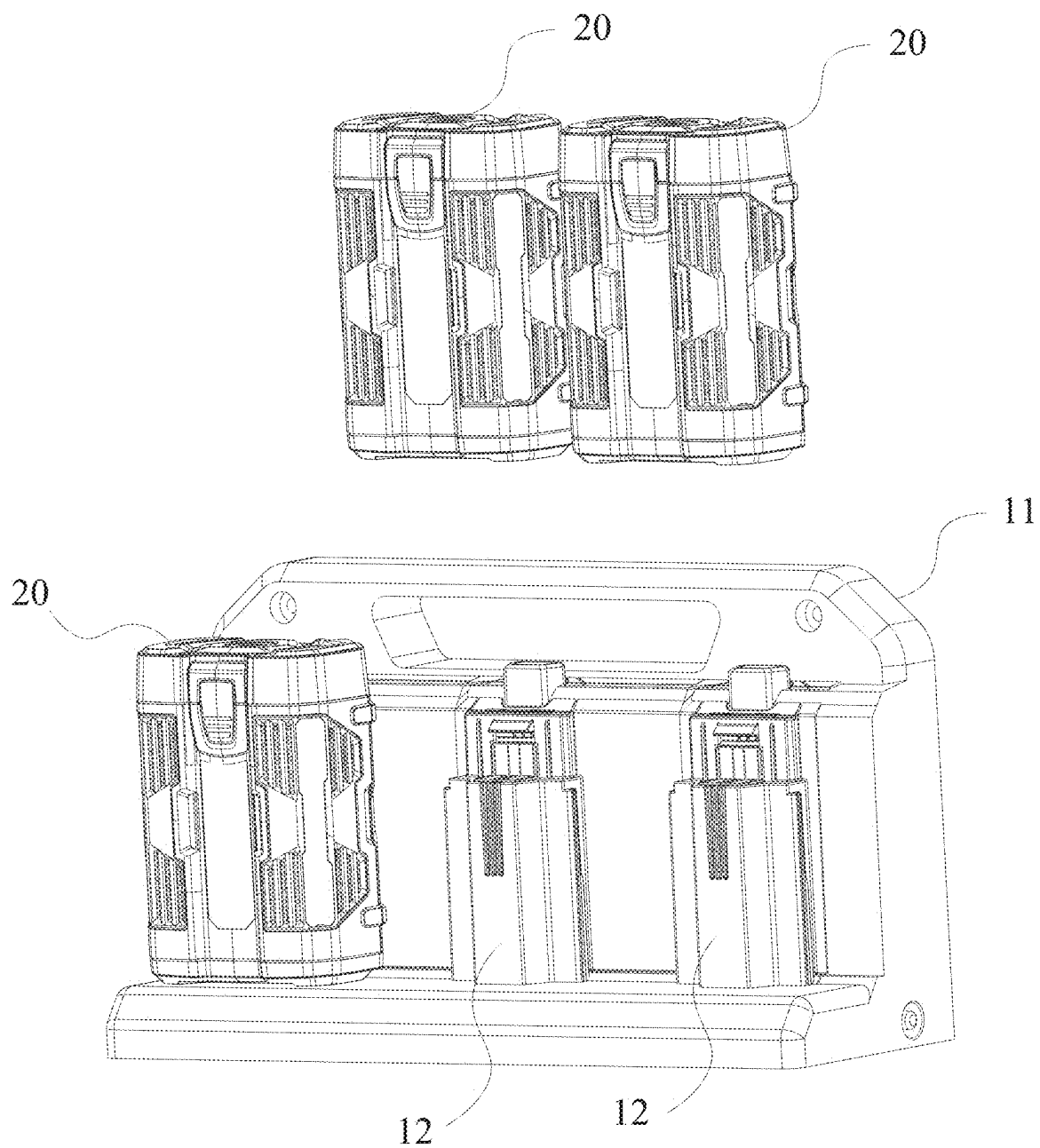
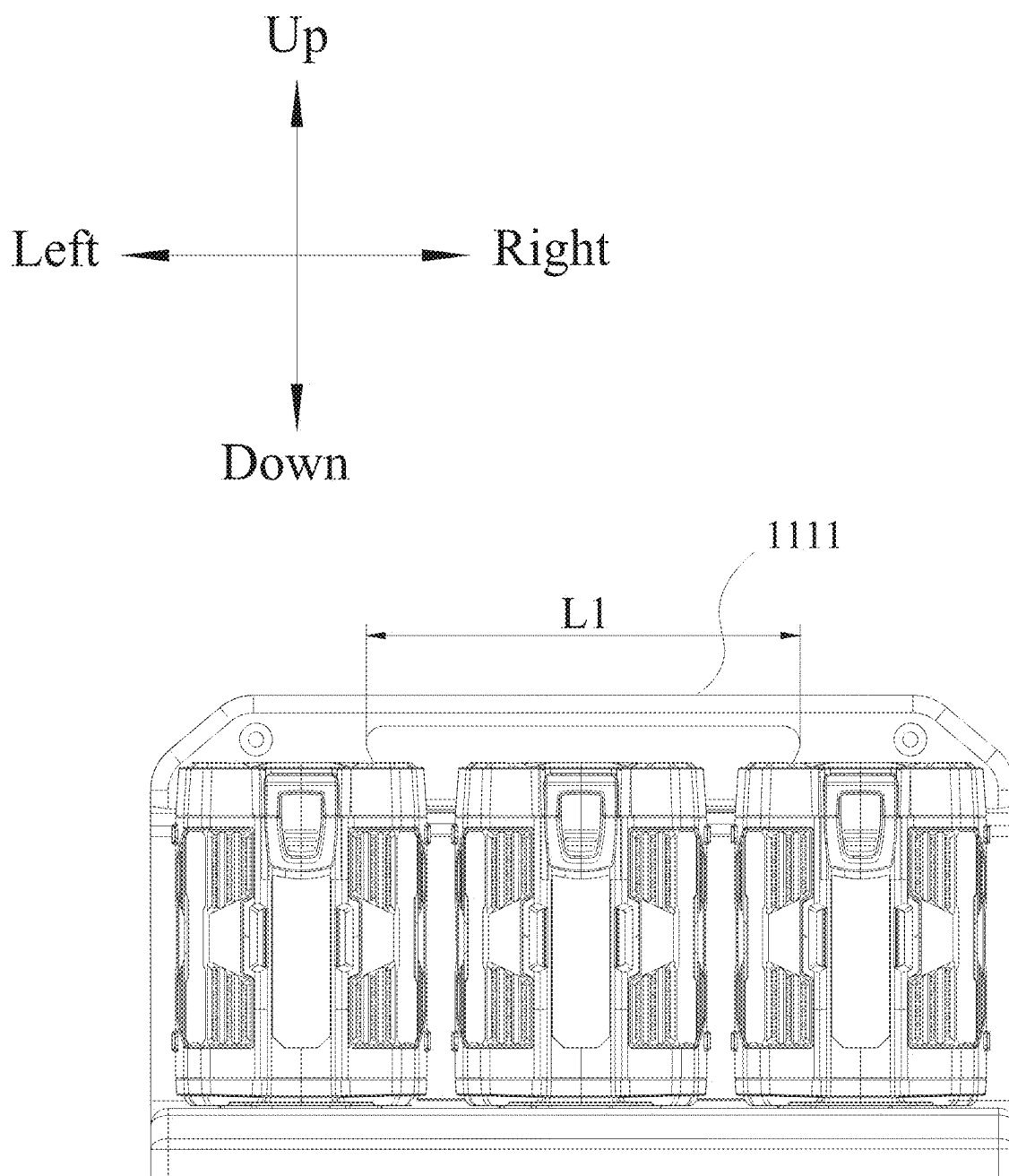


FIG. 2



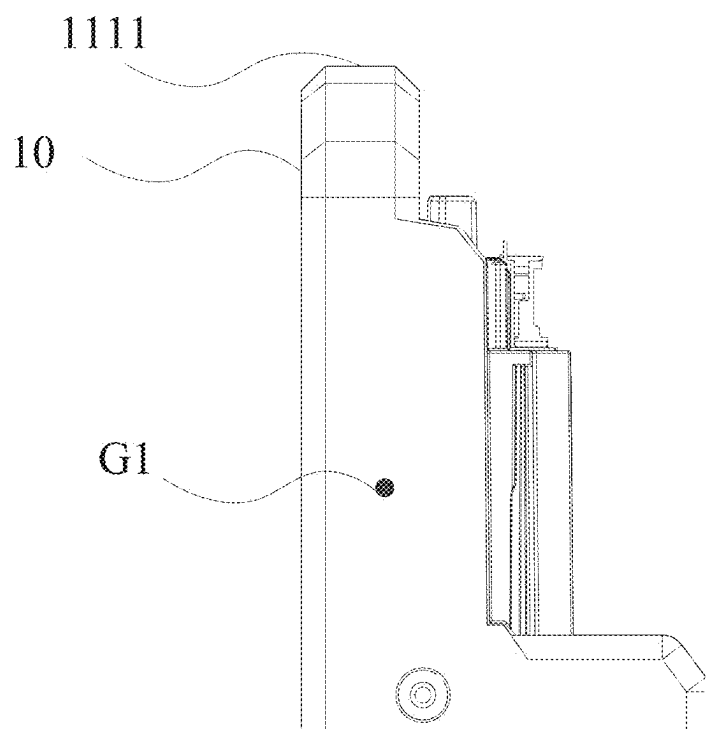


FIG. 4

Rear ← → Front

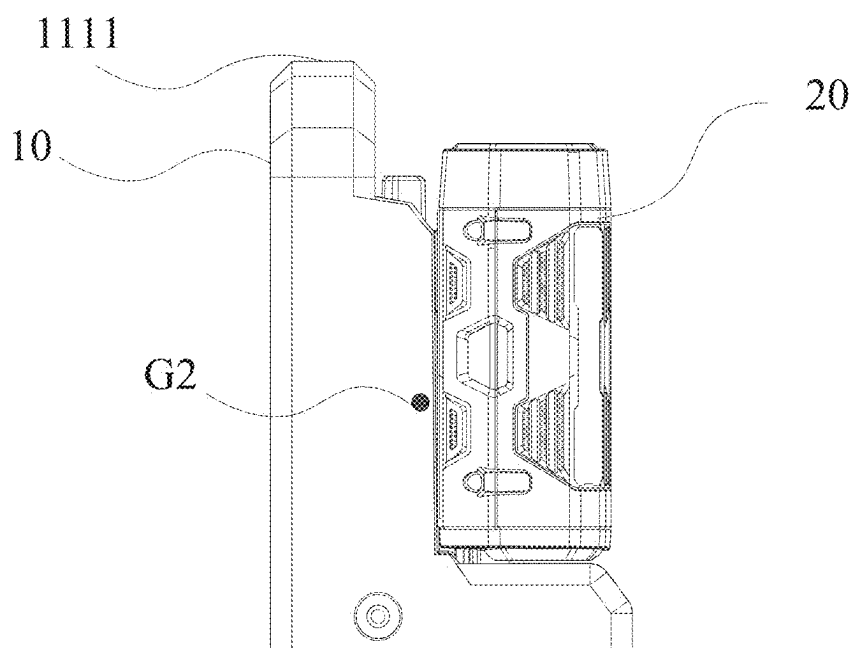


FIG. 5

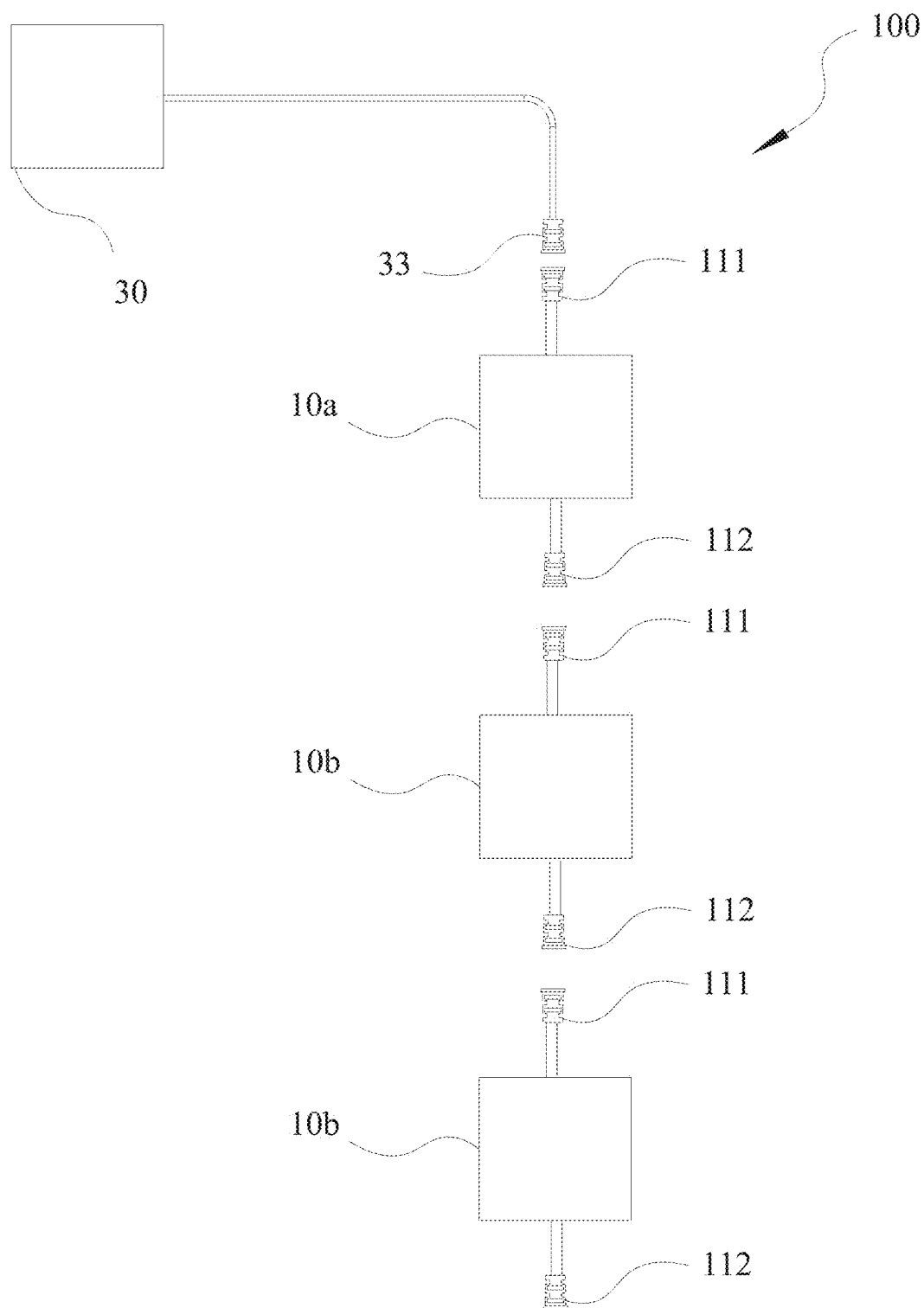


FIG. 6

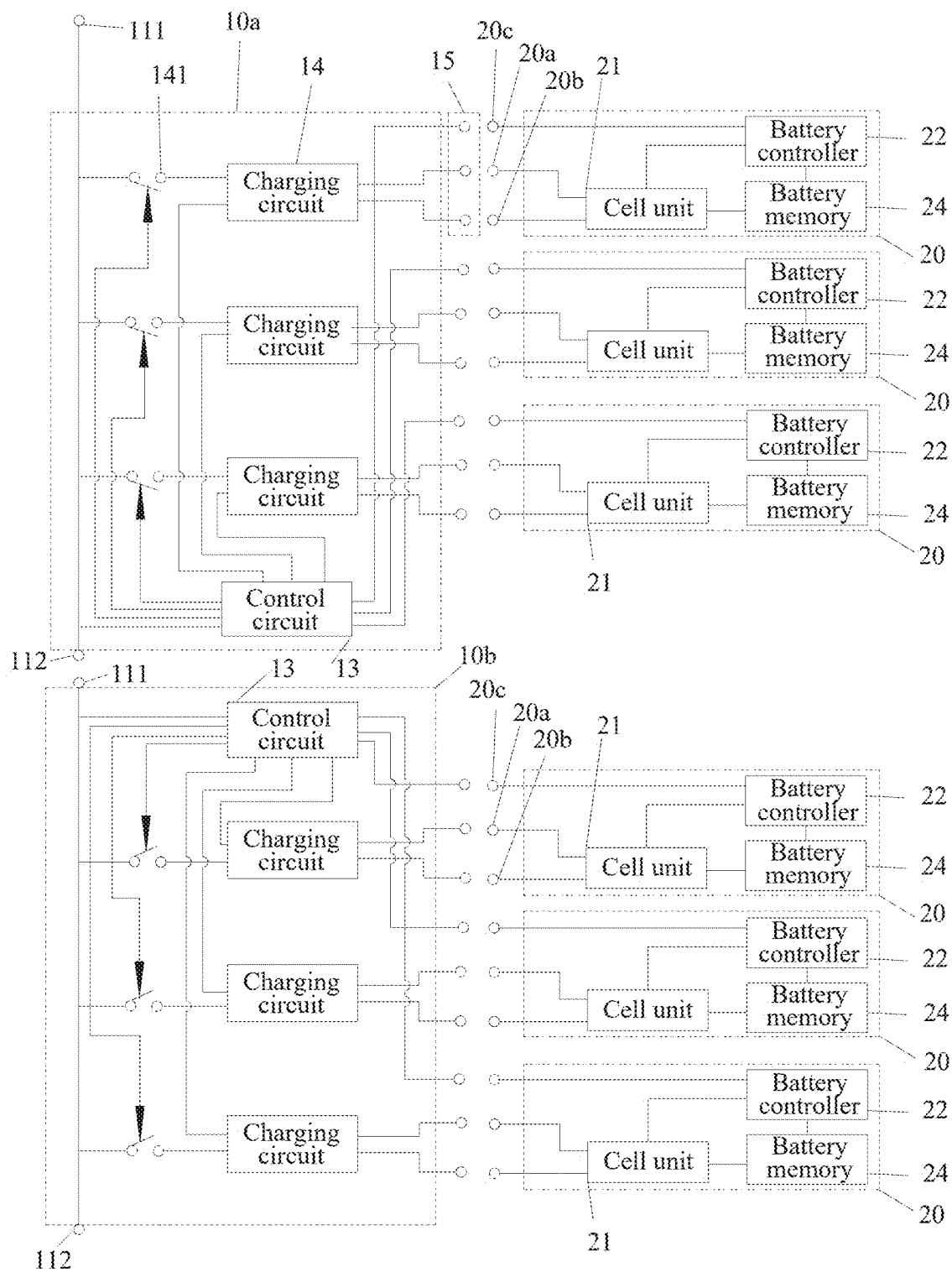


FIG. 7

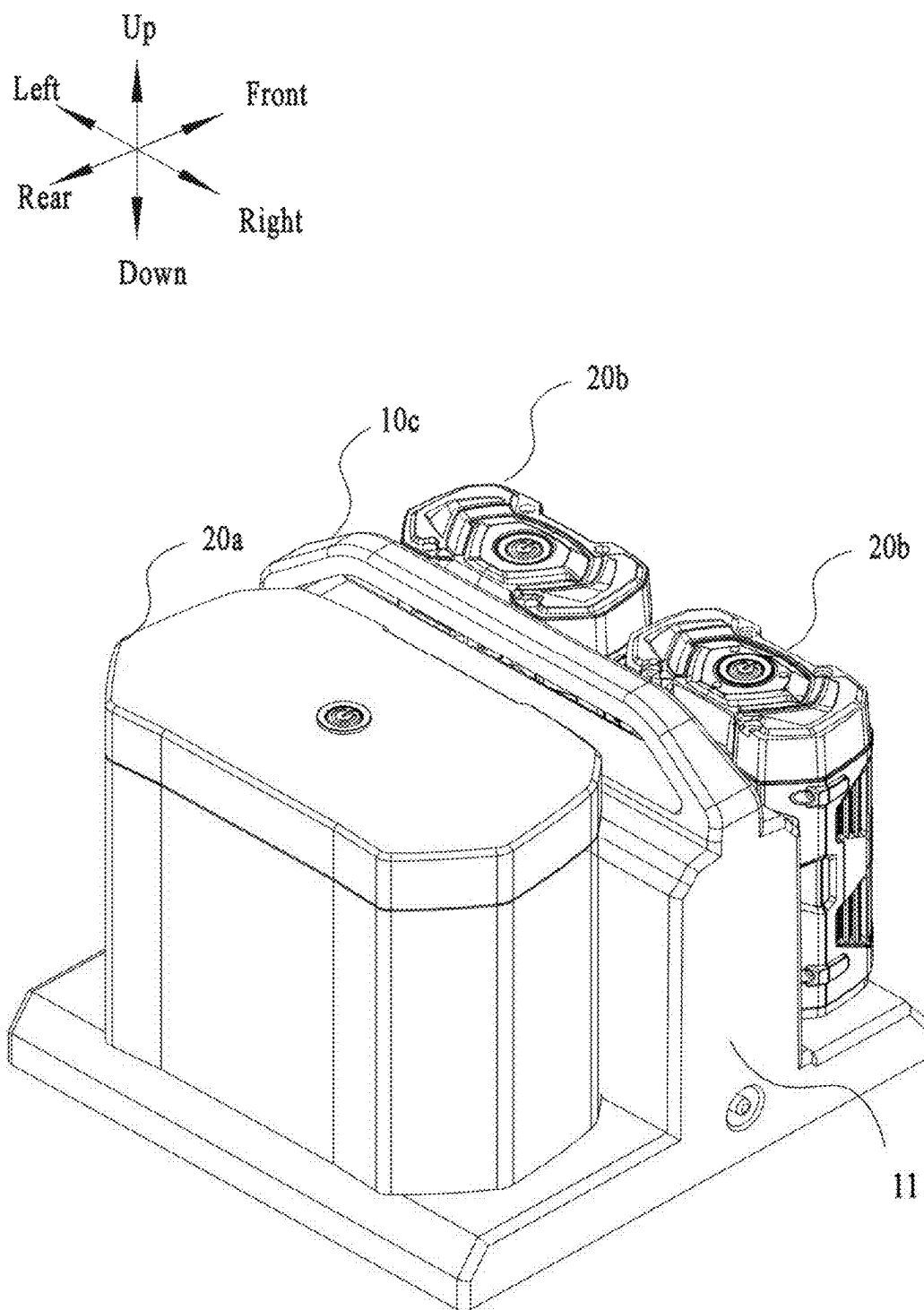


FIG. 8

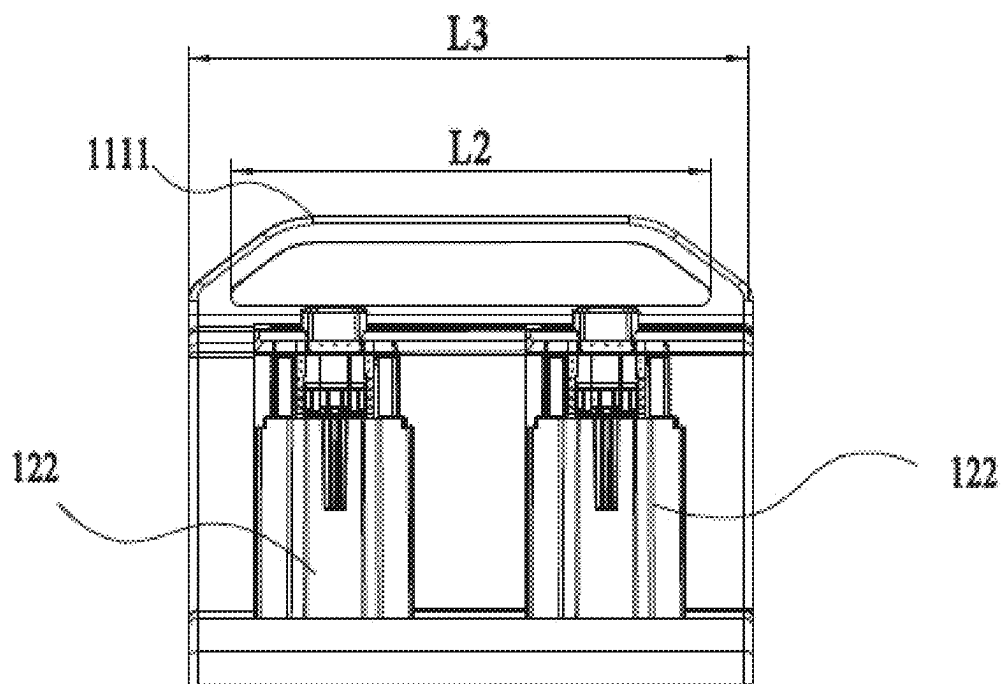


FIG. 9A

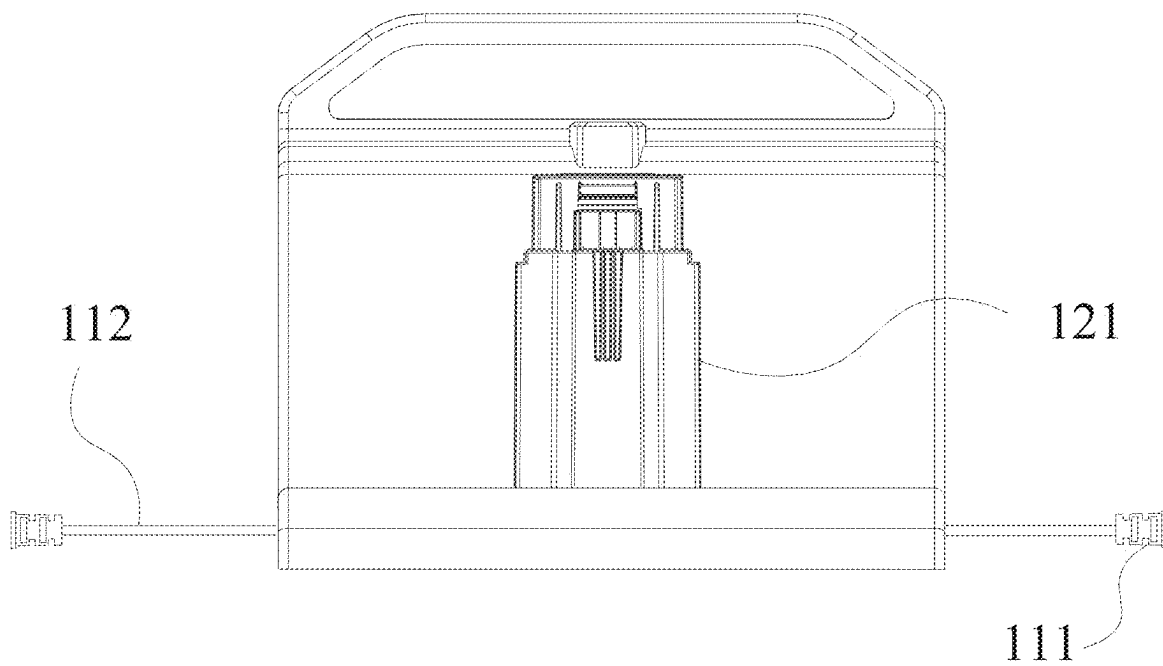


FIG. 9B

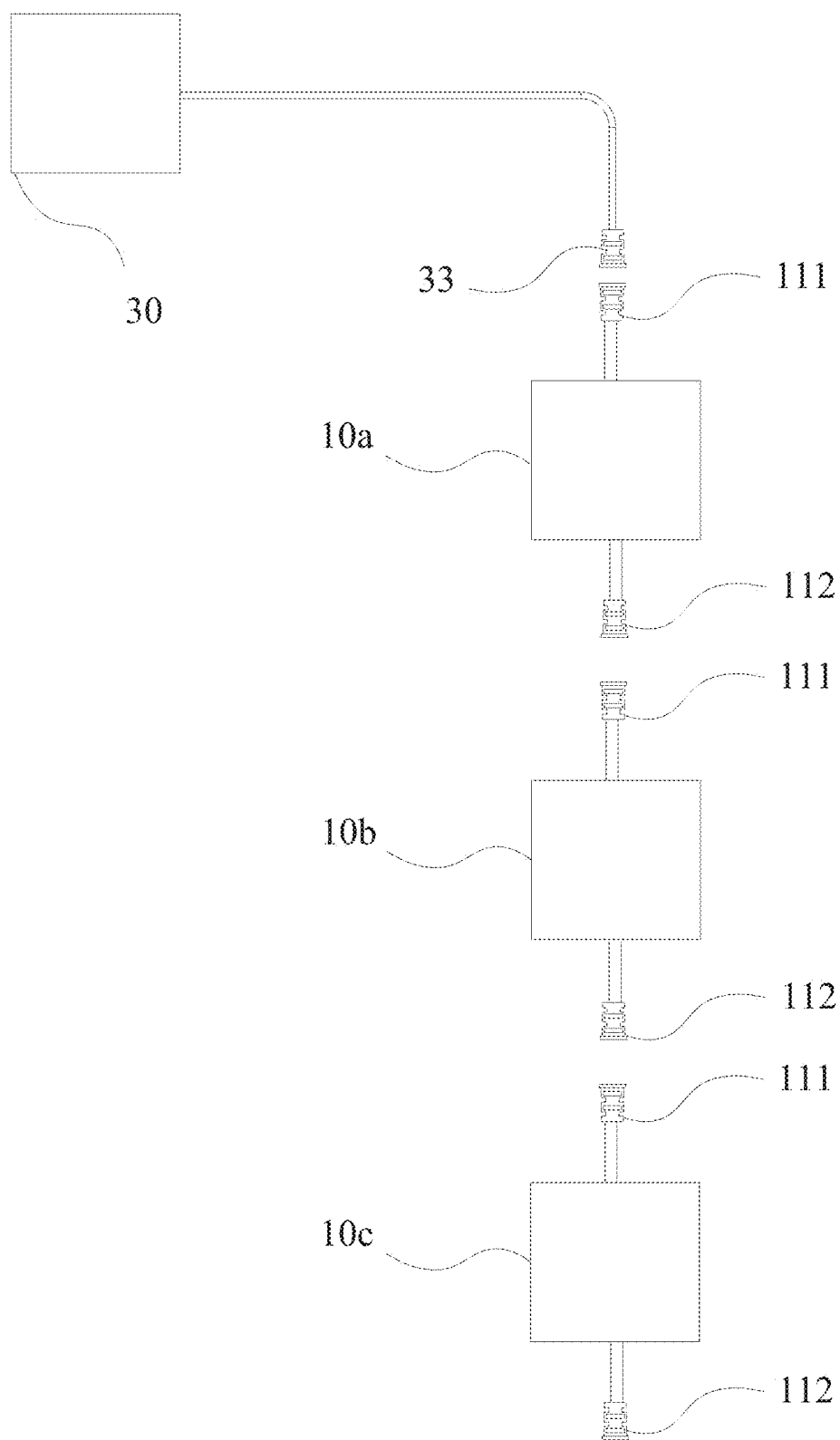


FIG. 10

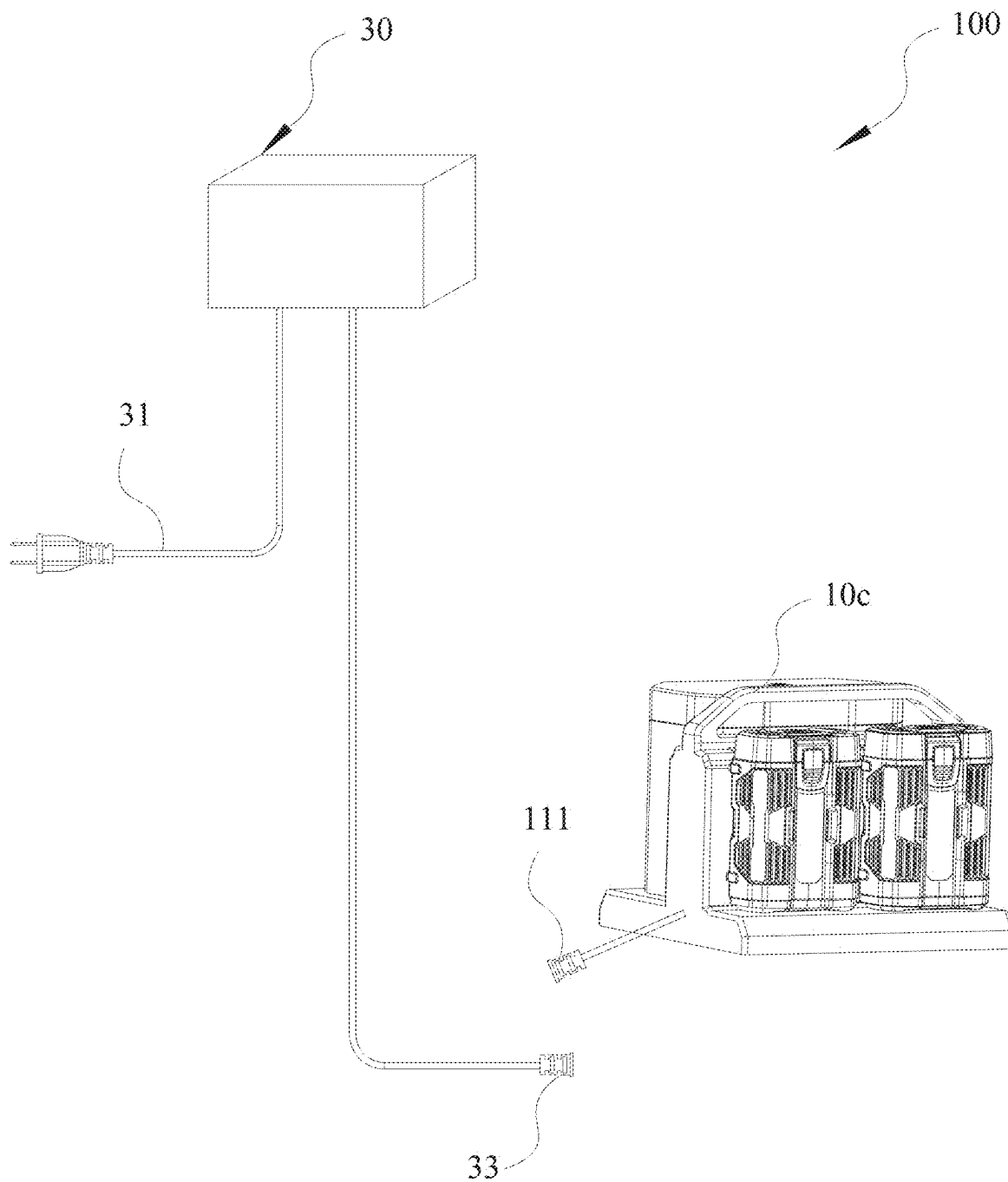


FIG. 11

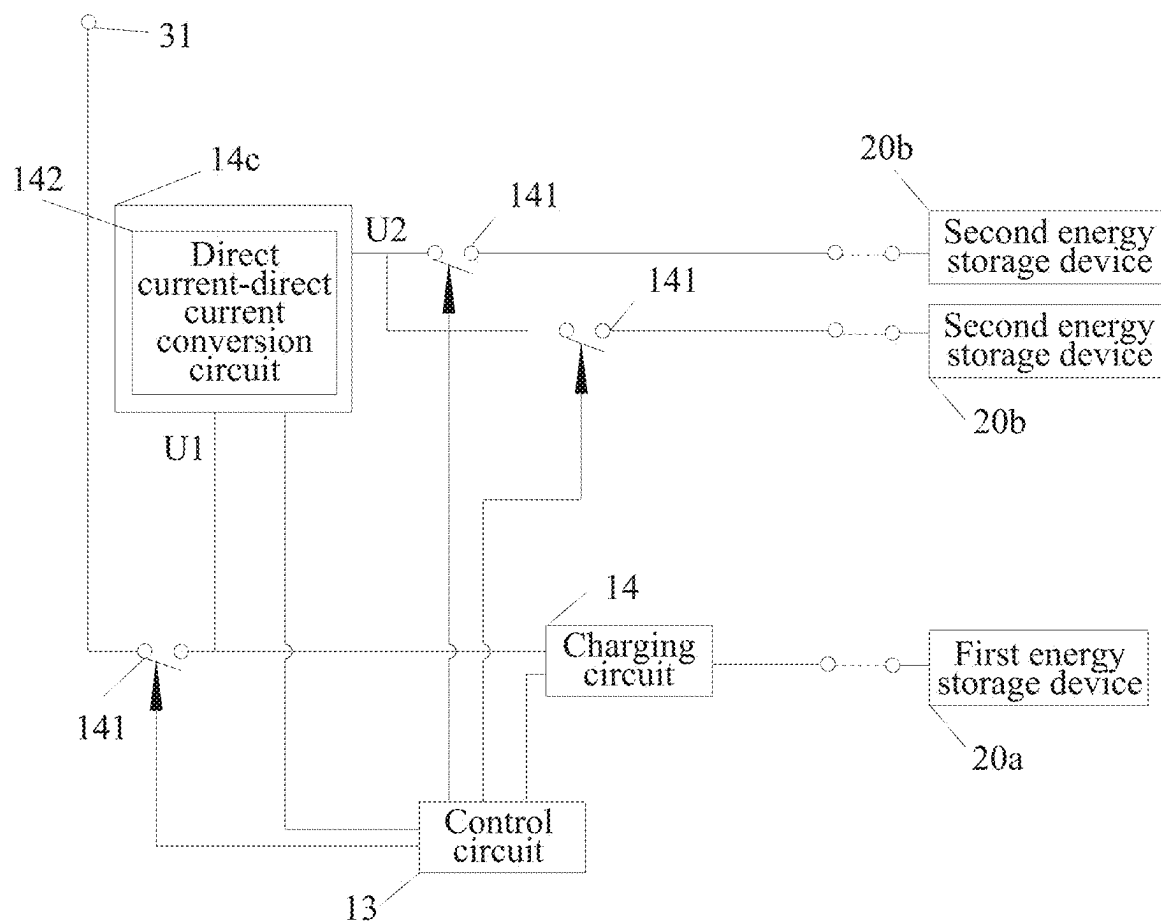


FIG. 12

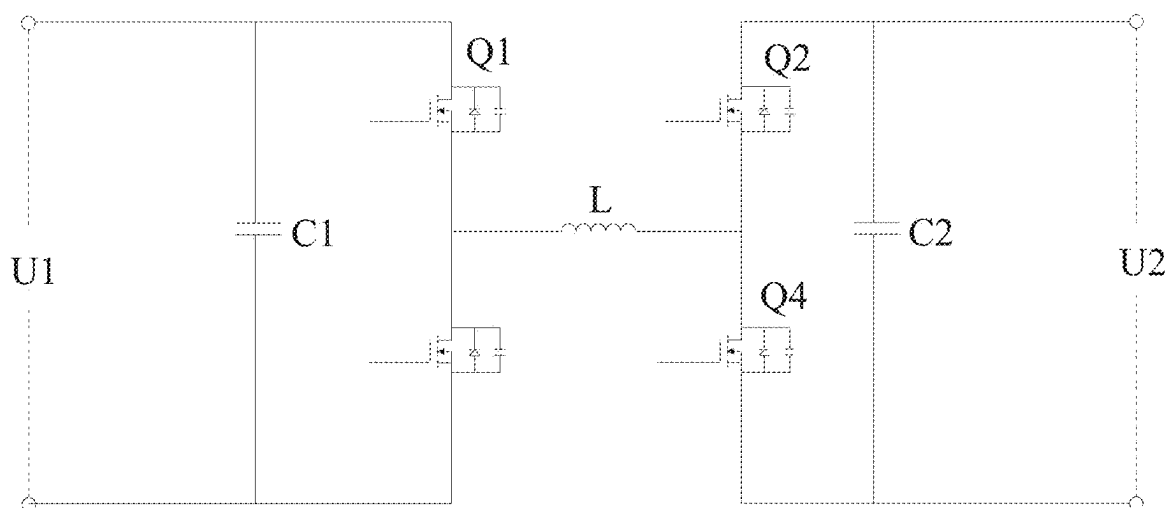


FIG. 13

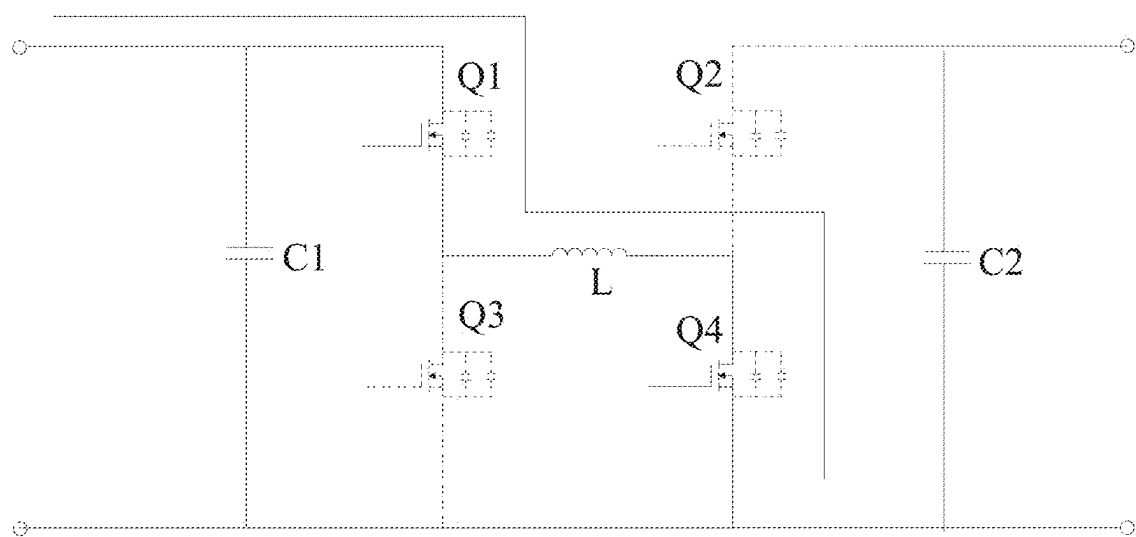


FIG. 14A

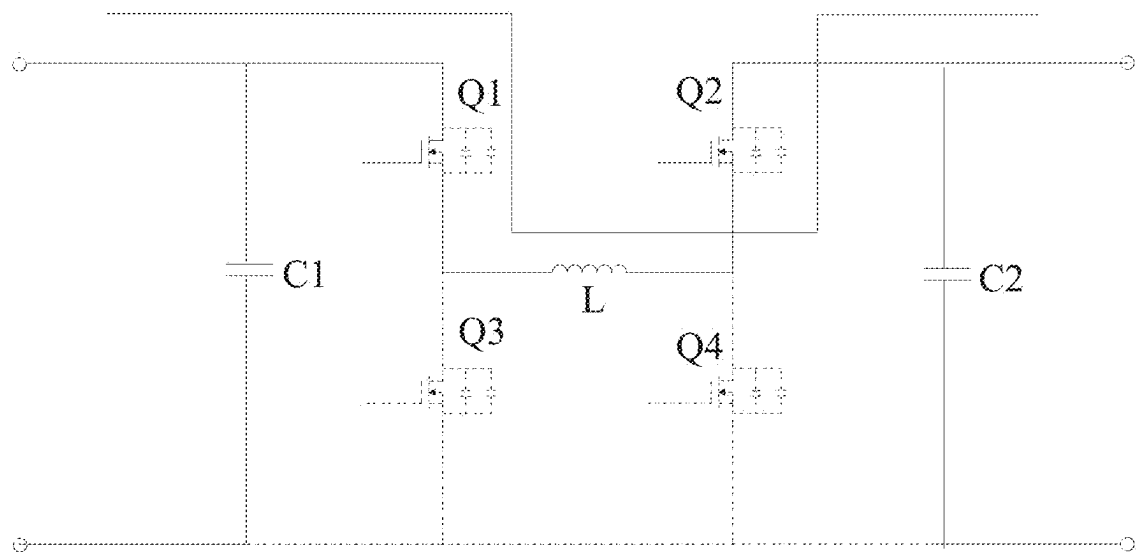


FIG. 14B

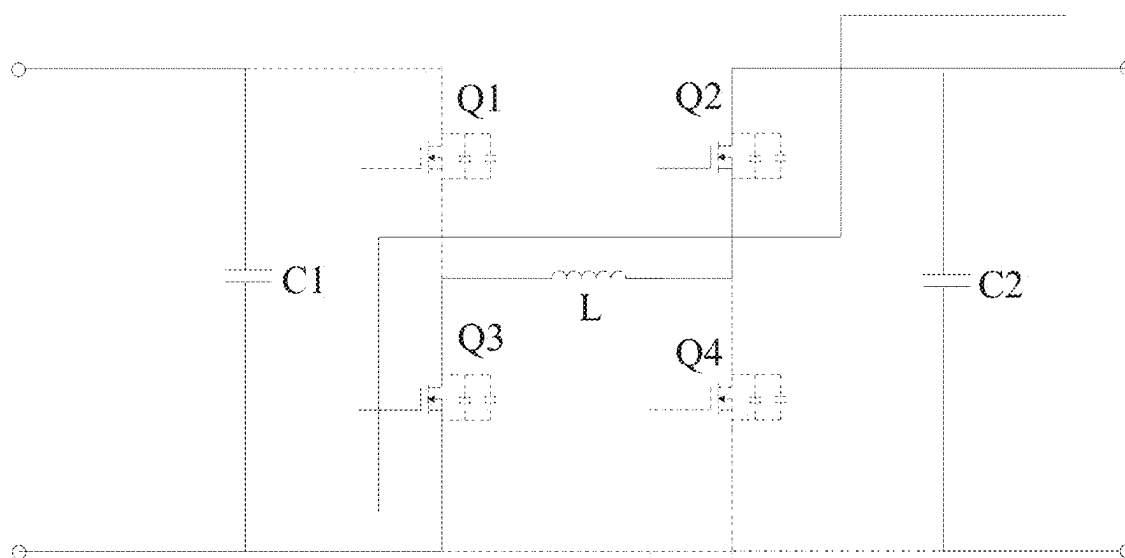


FIG. 14C

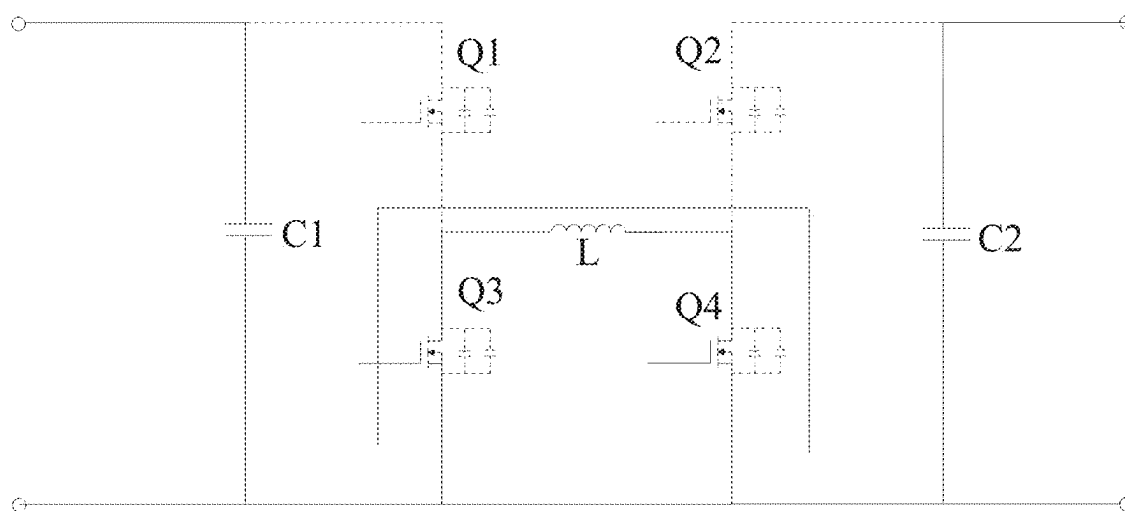


FIG. 14D

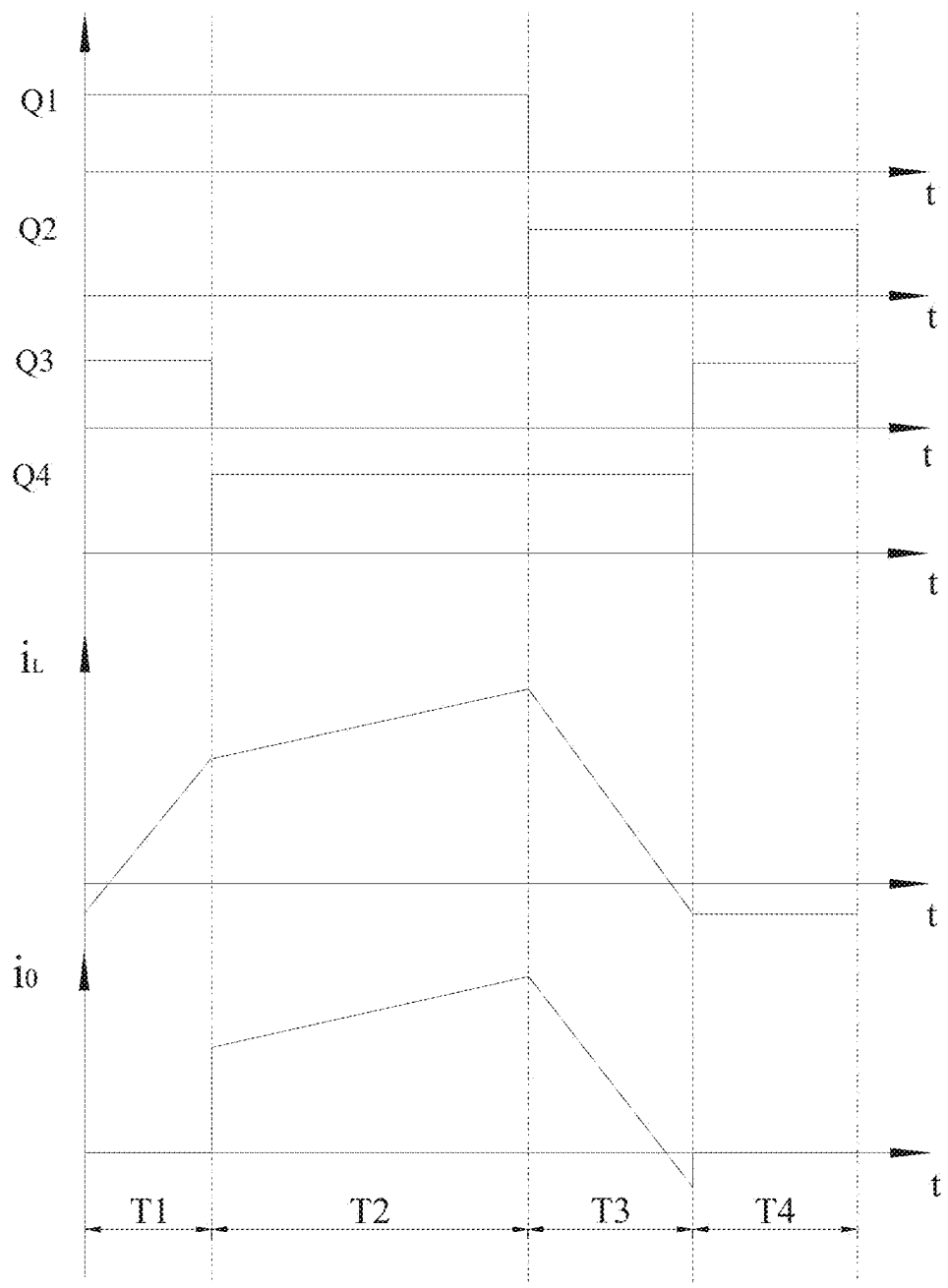


FIG. 15

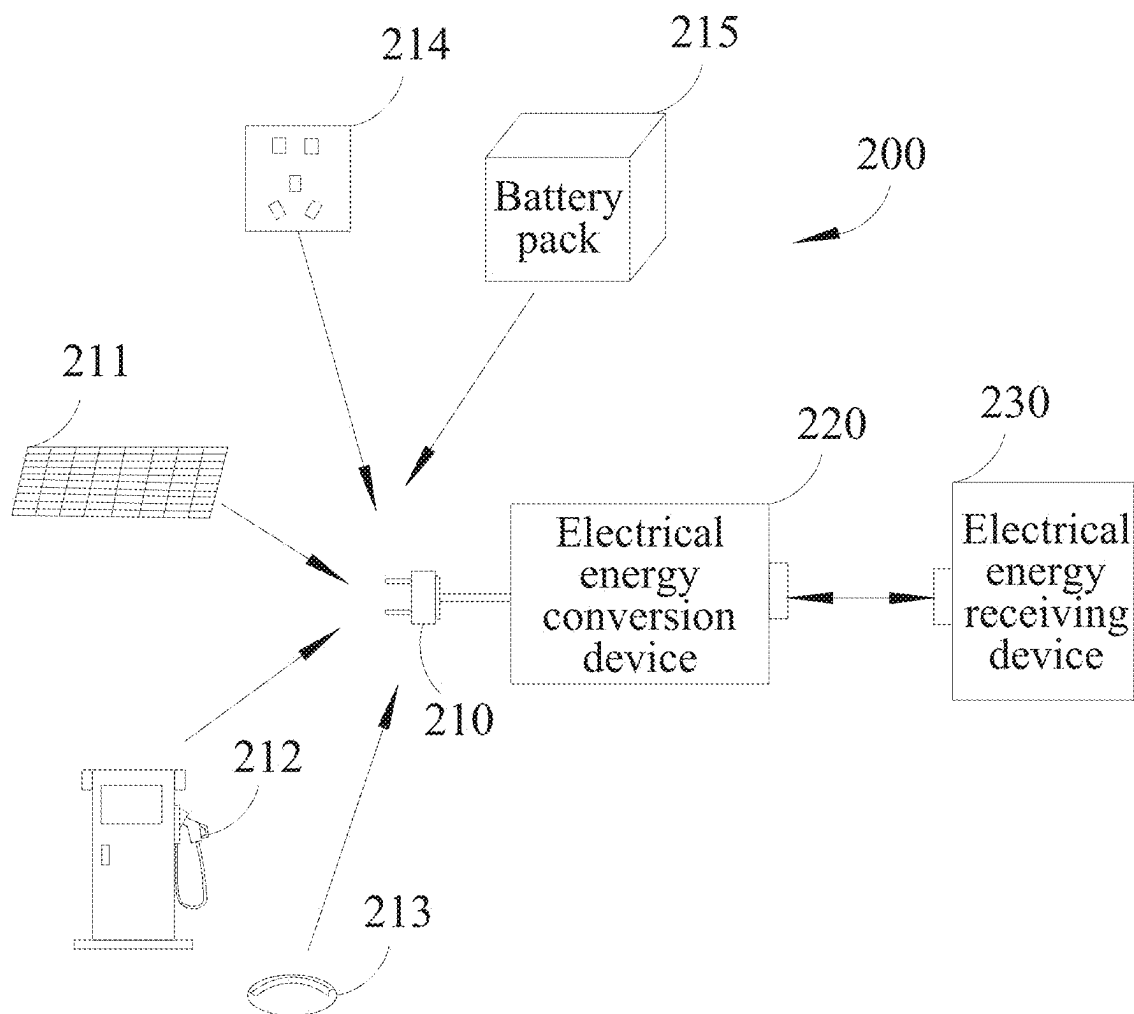


FIG. 16

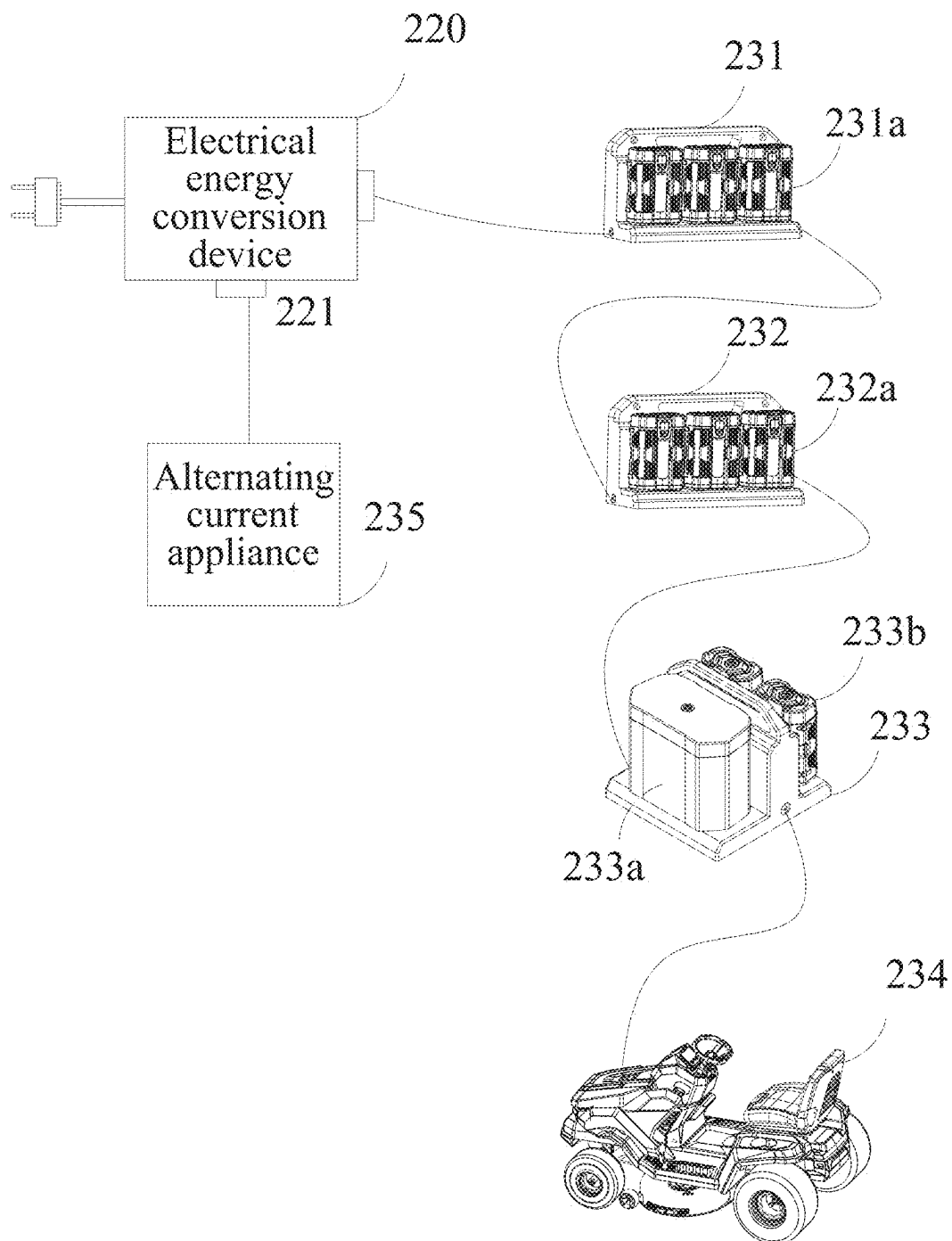


FIG. 17

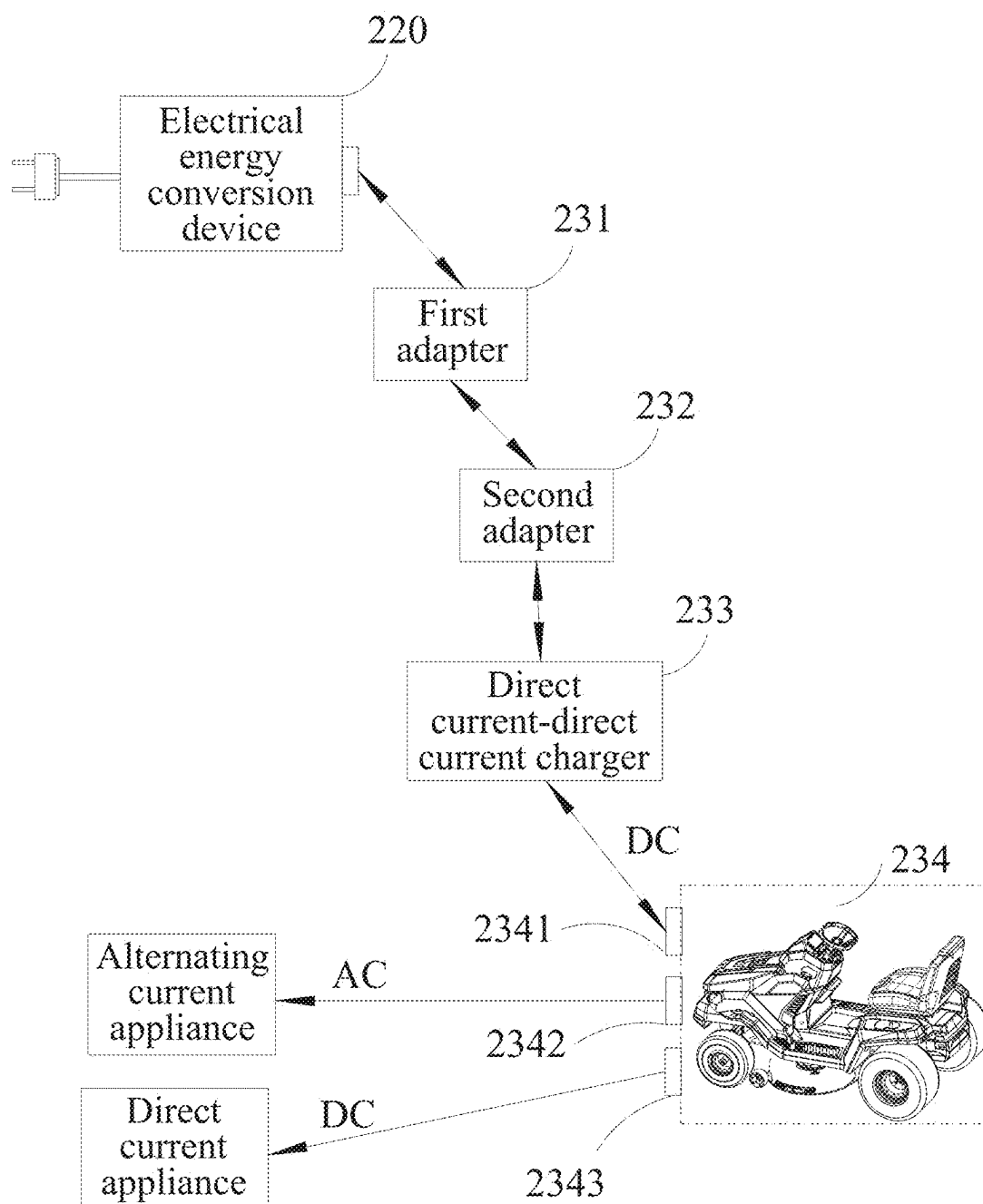


FIG. 18

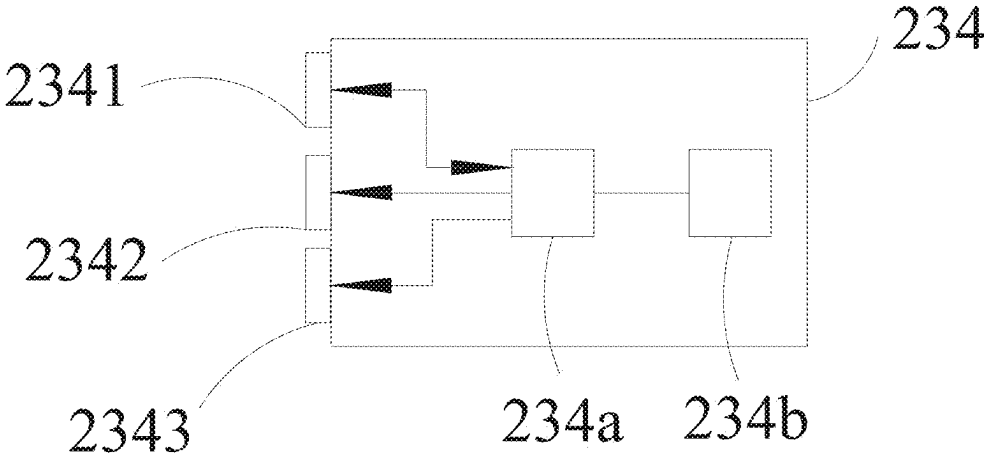


FIG. 19A

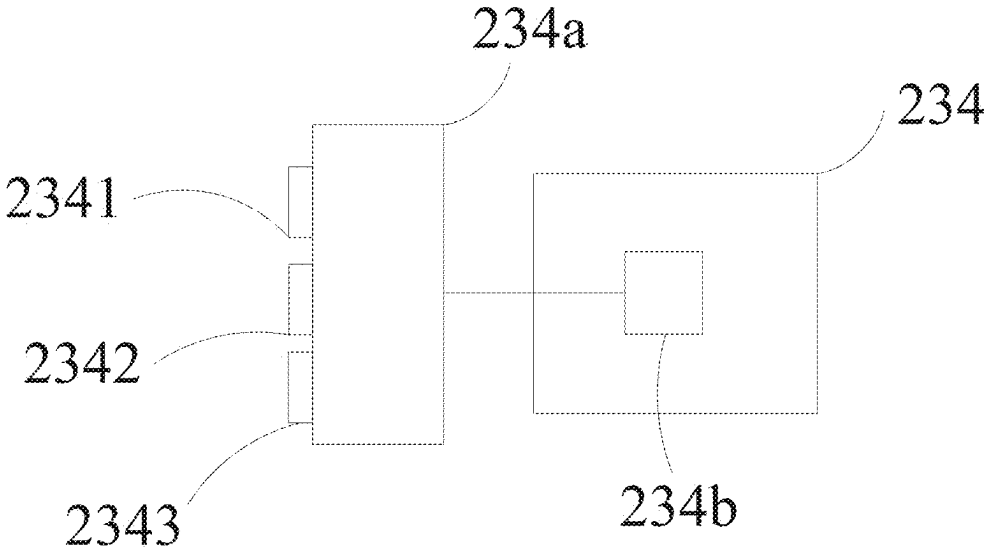


FIG. 19B

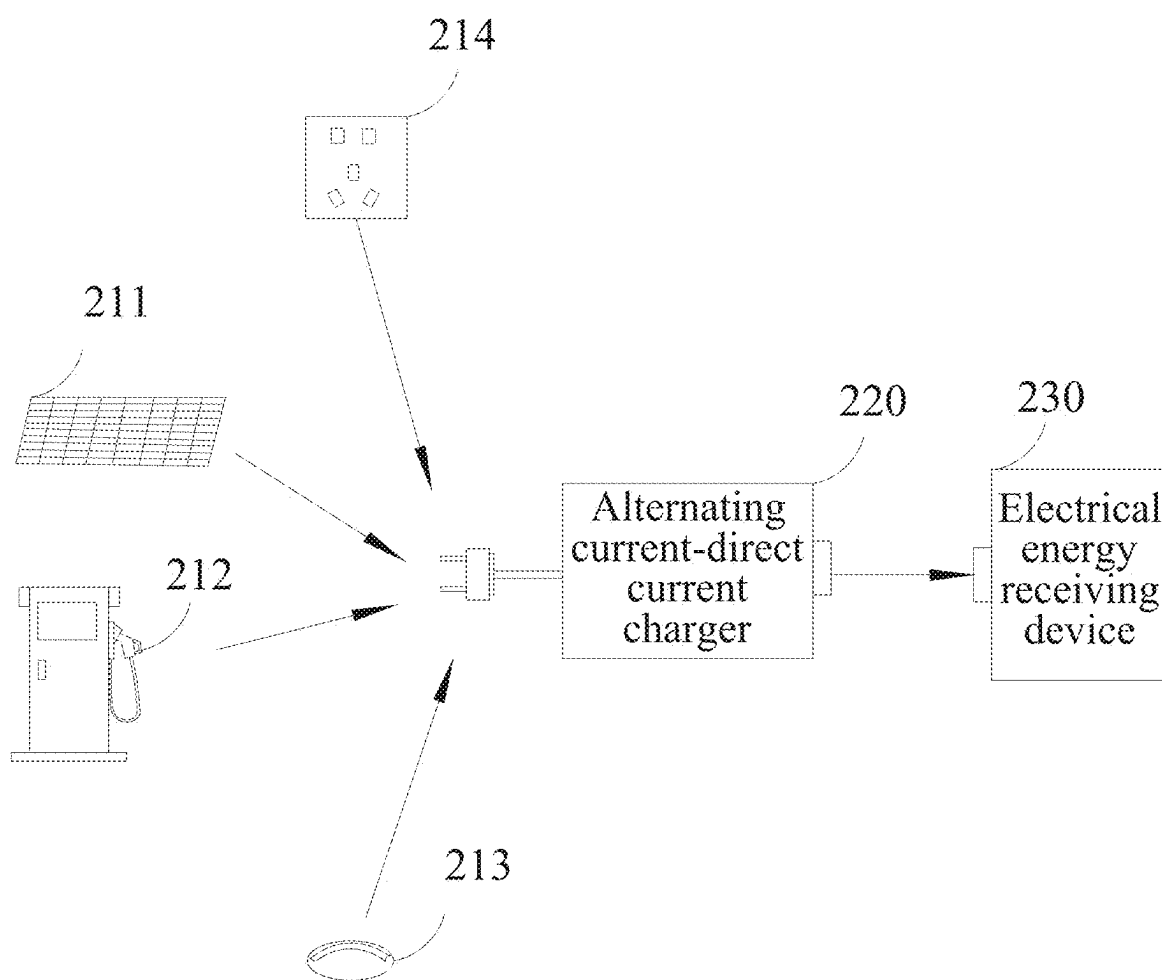


FIG. 20

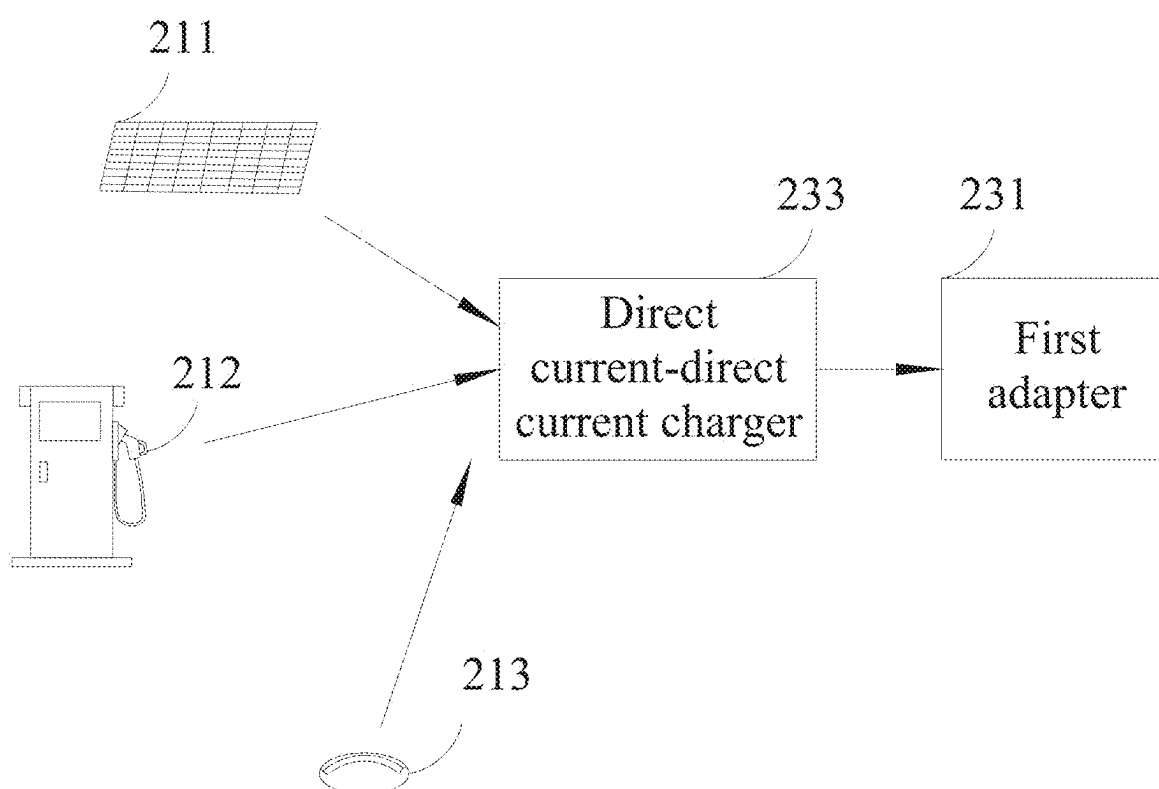


FIG. 21

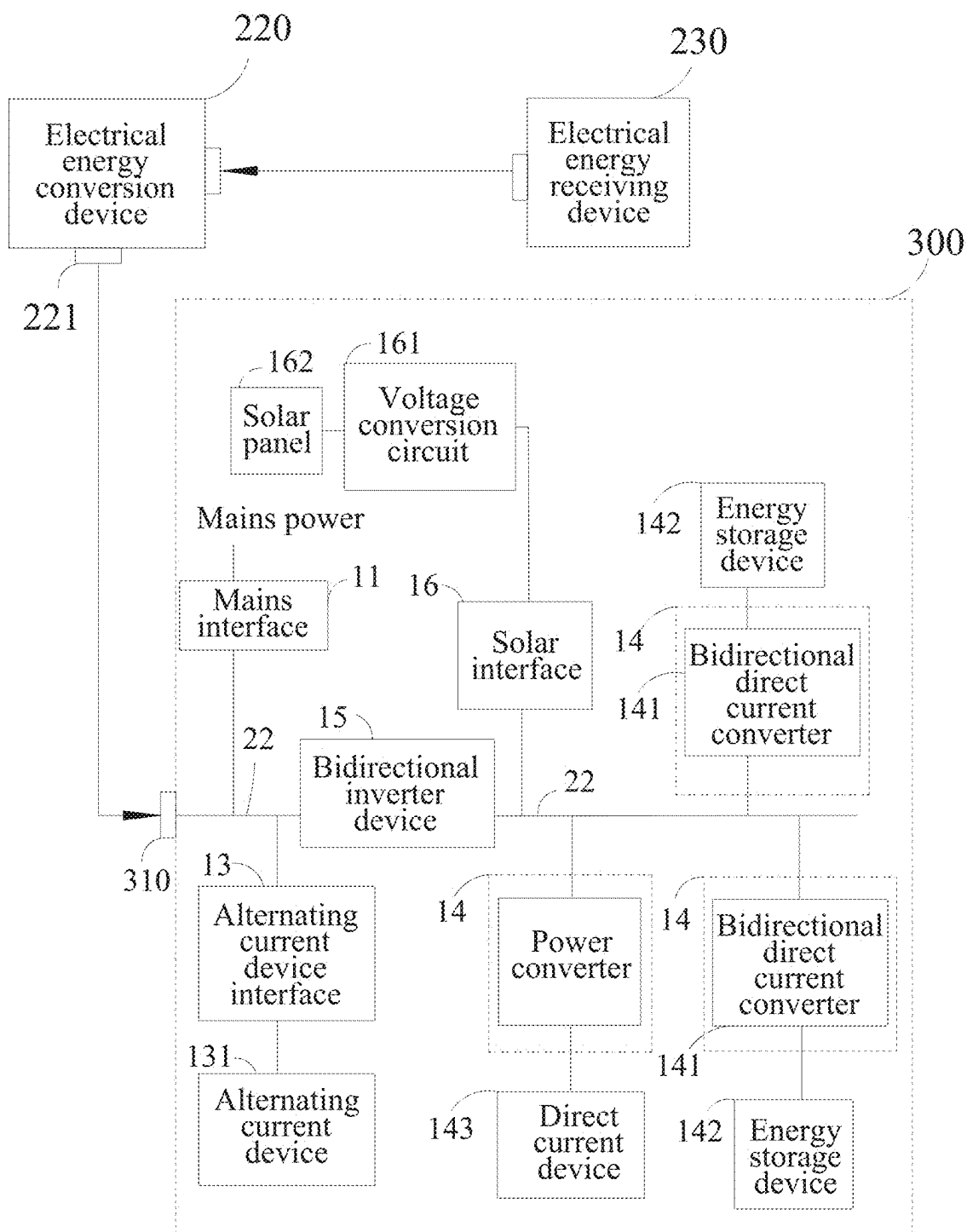


FIG. 22

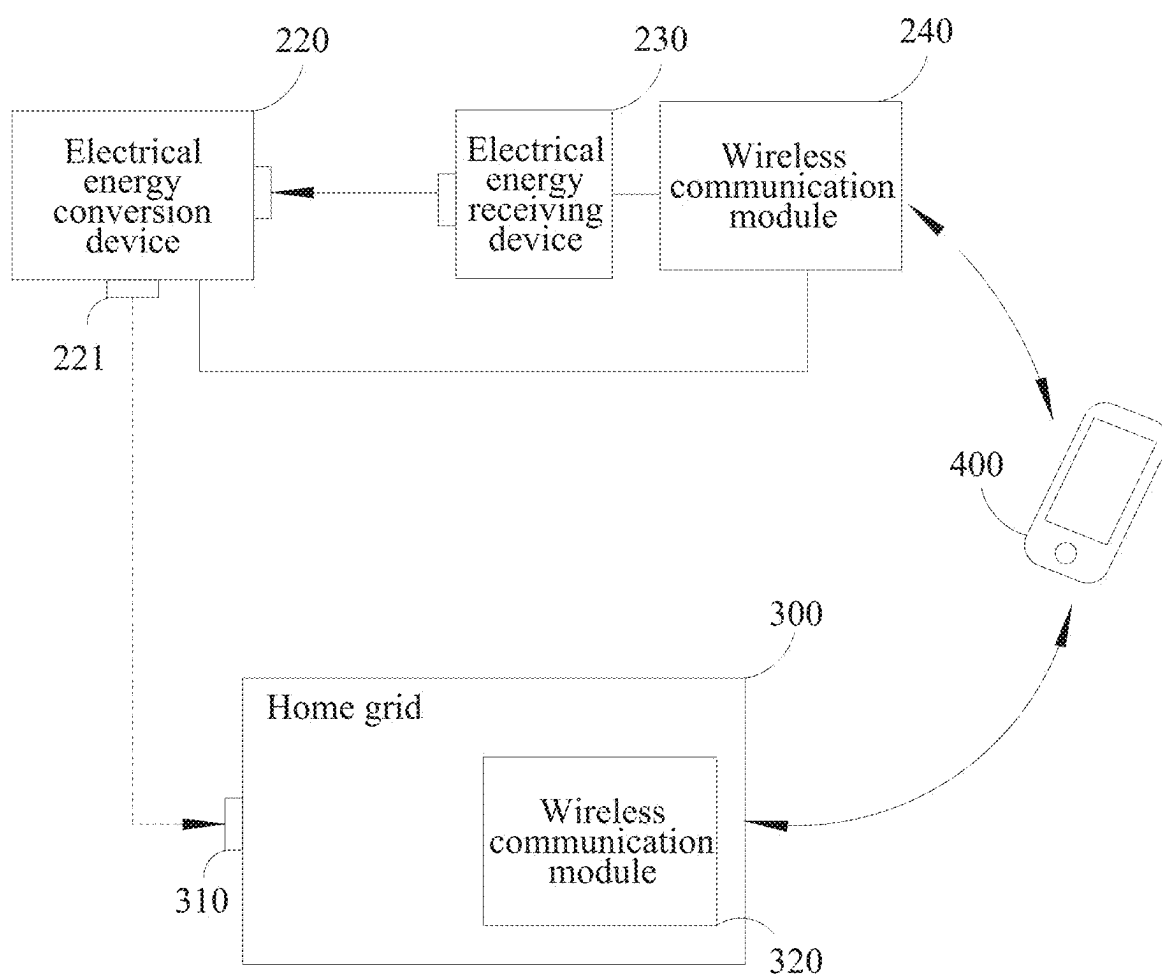


FIG. 23

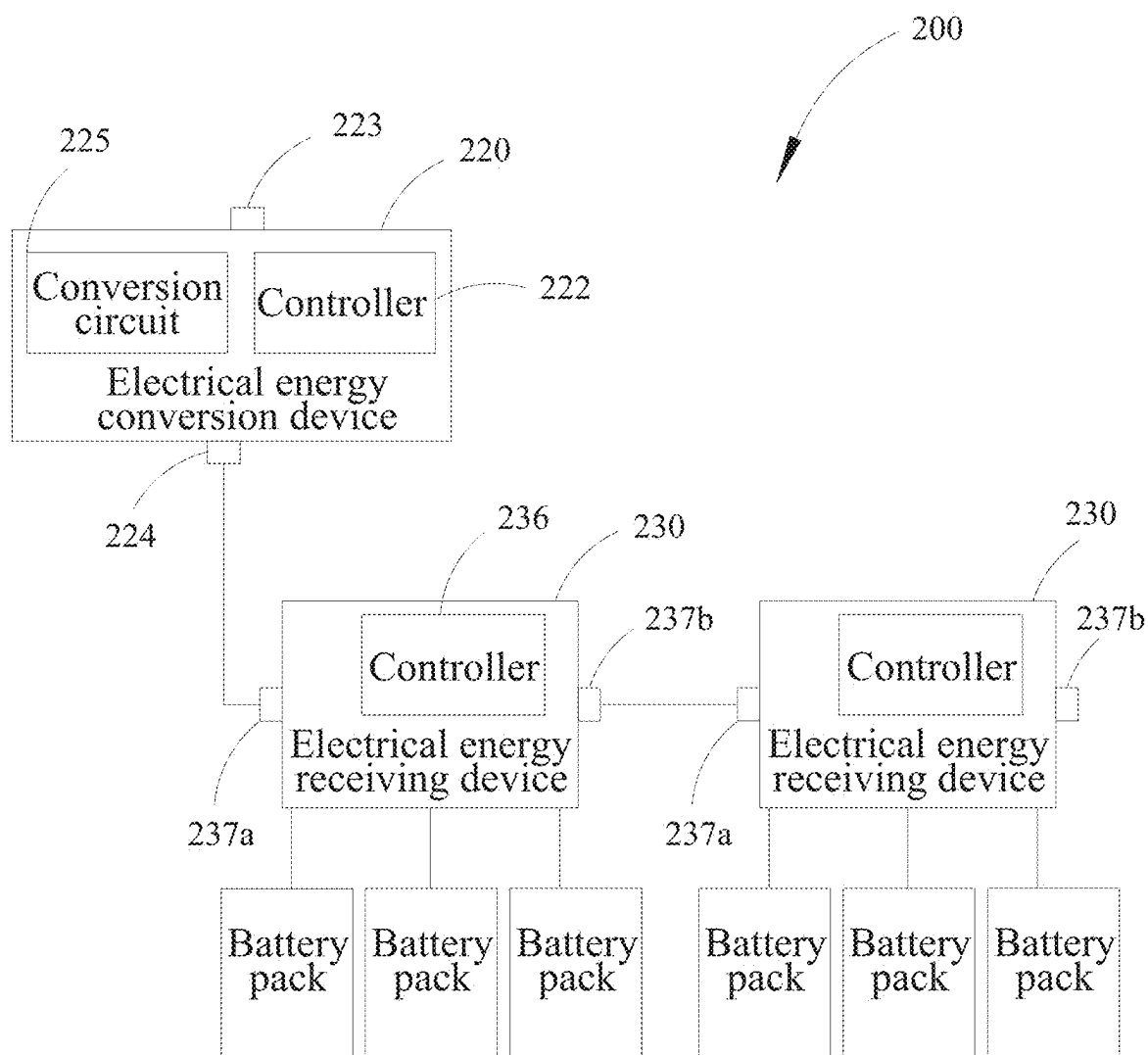


FIG. 24

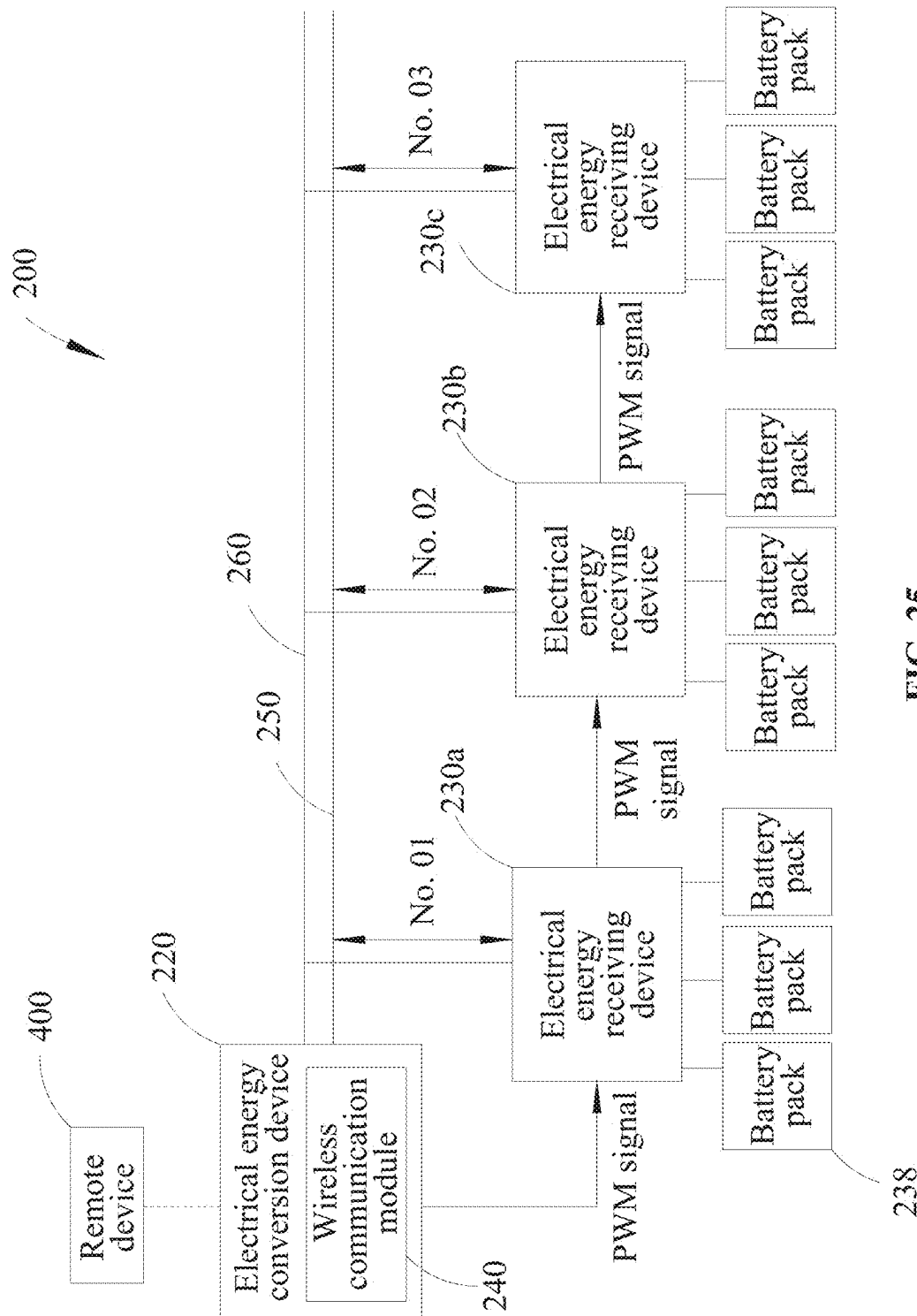


FIG. 25

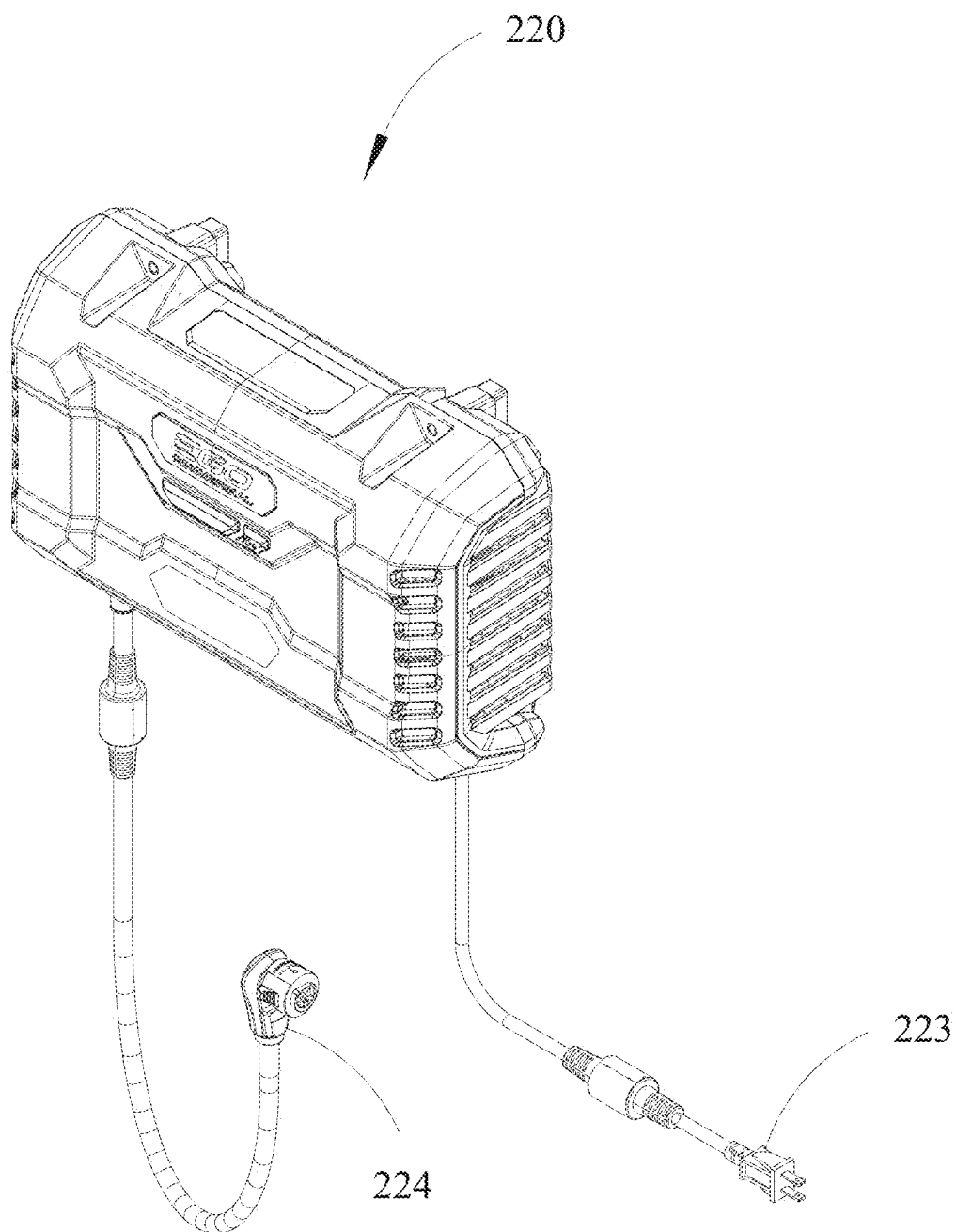
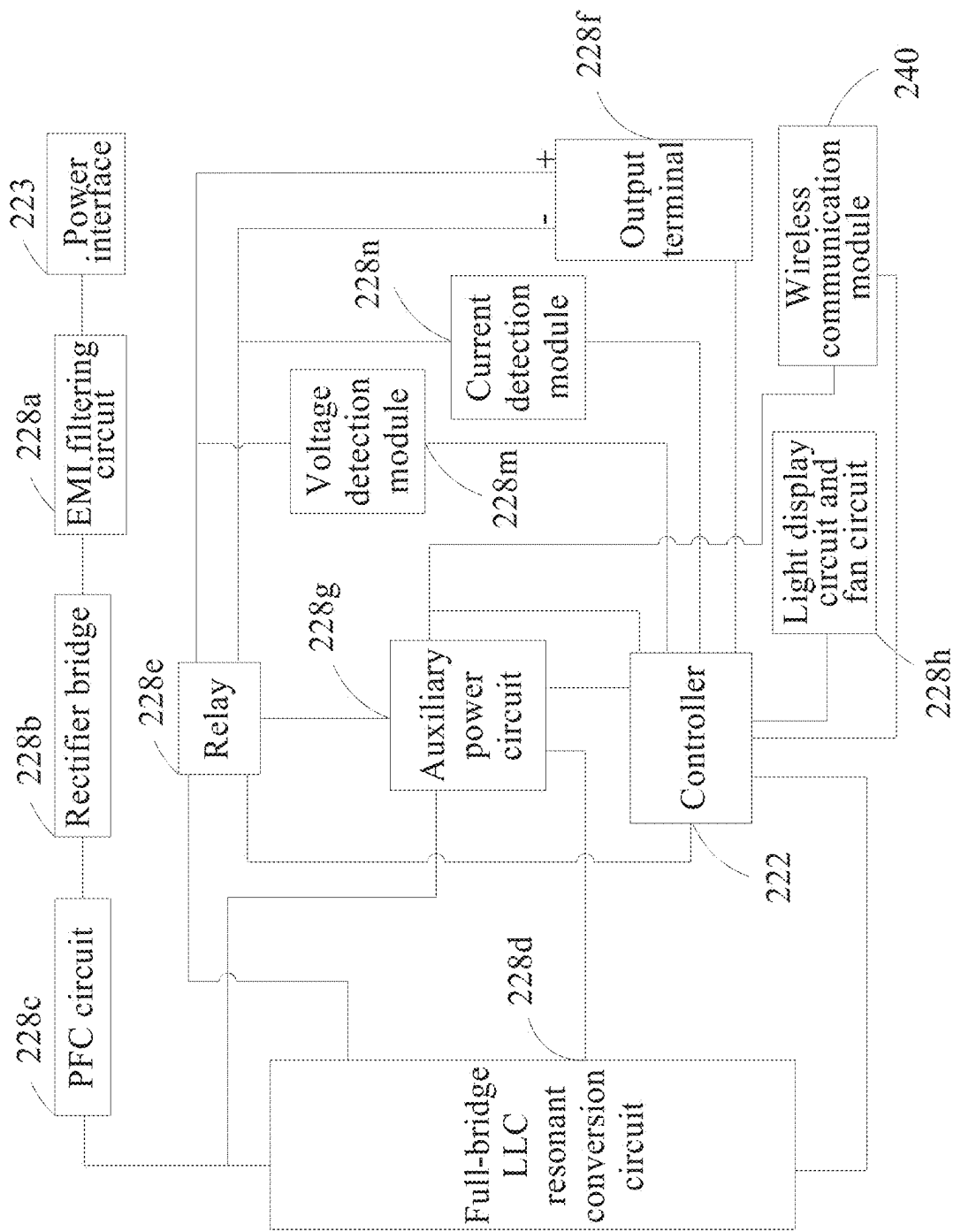


FIG. 26



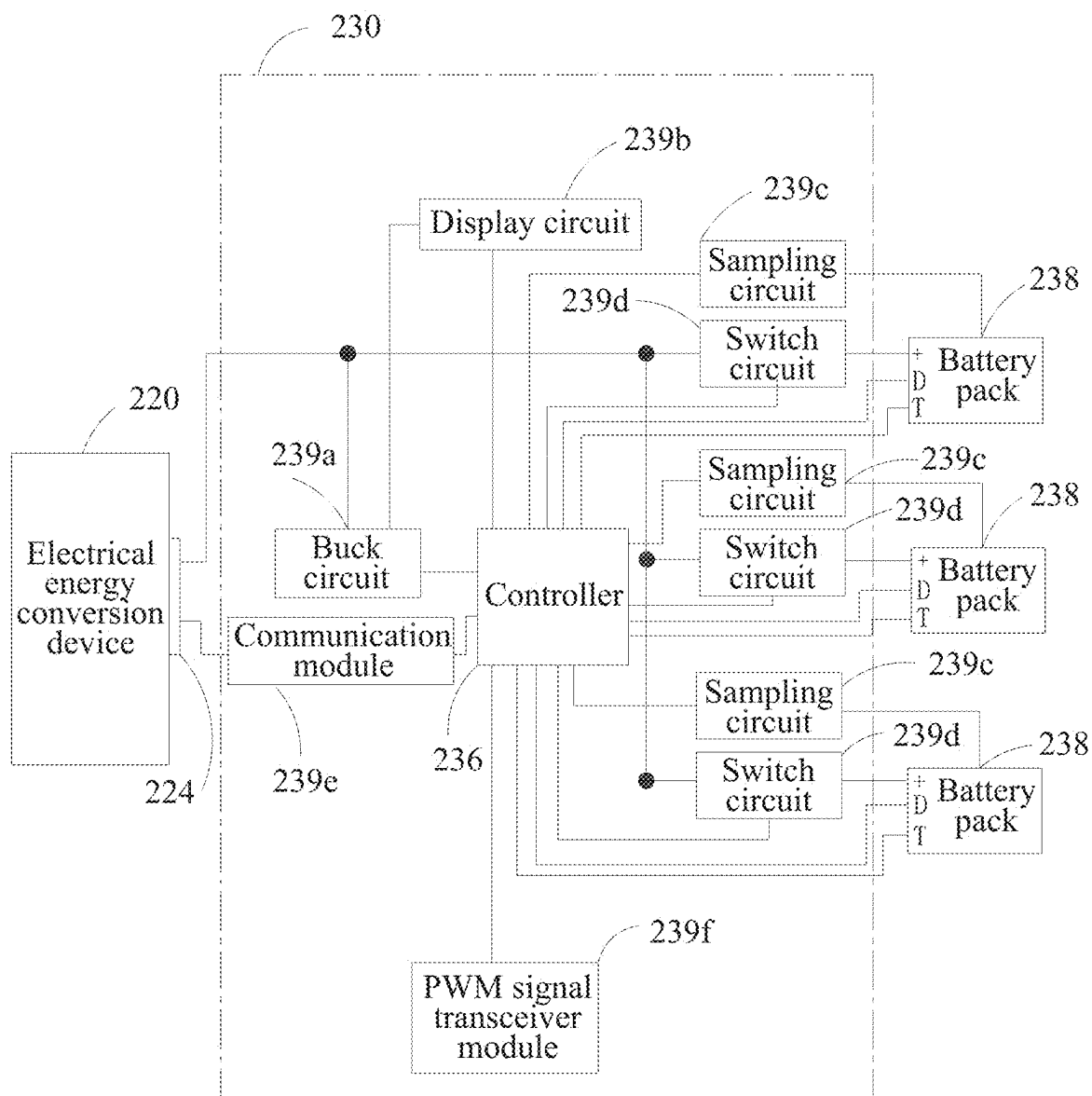


FIG. 28

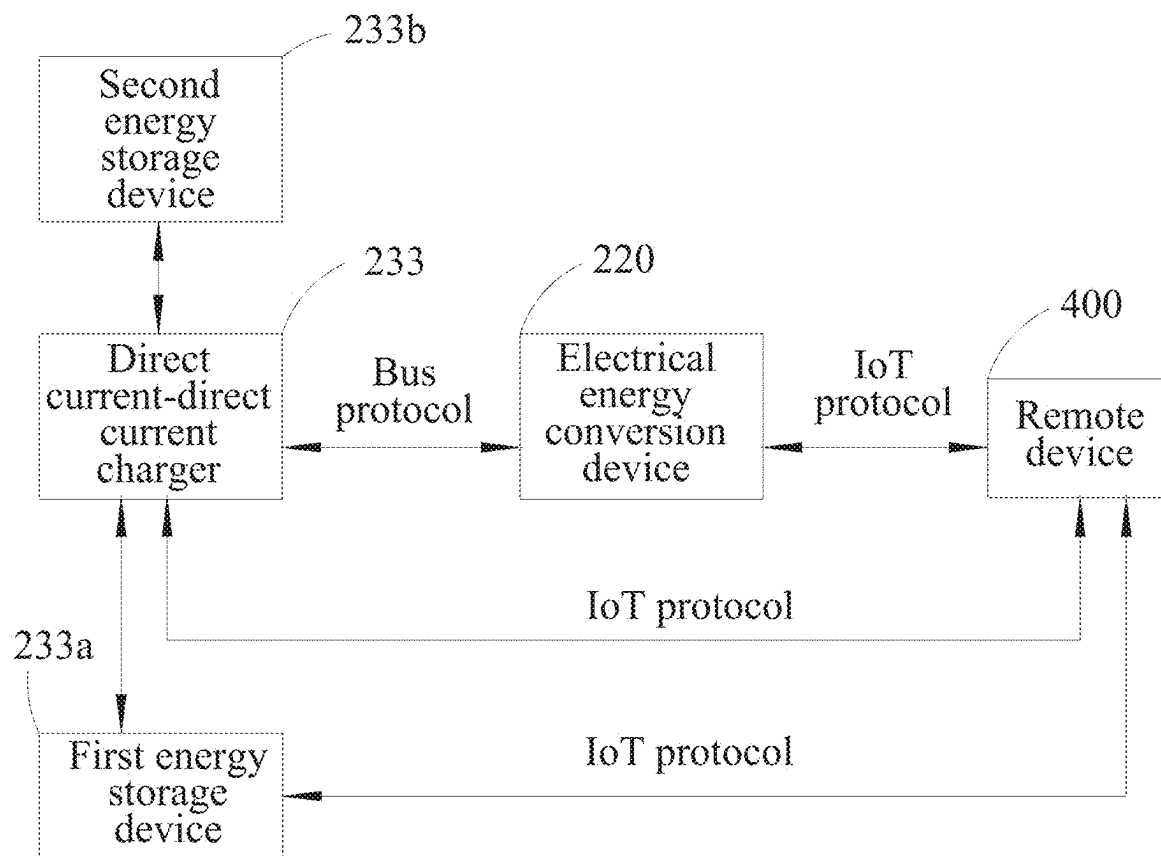


FIG. 29

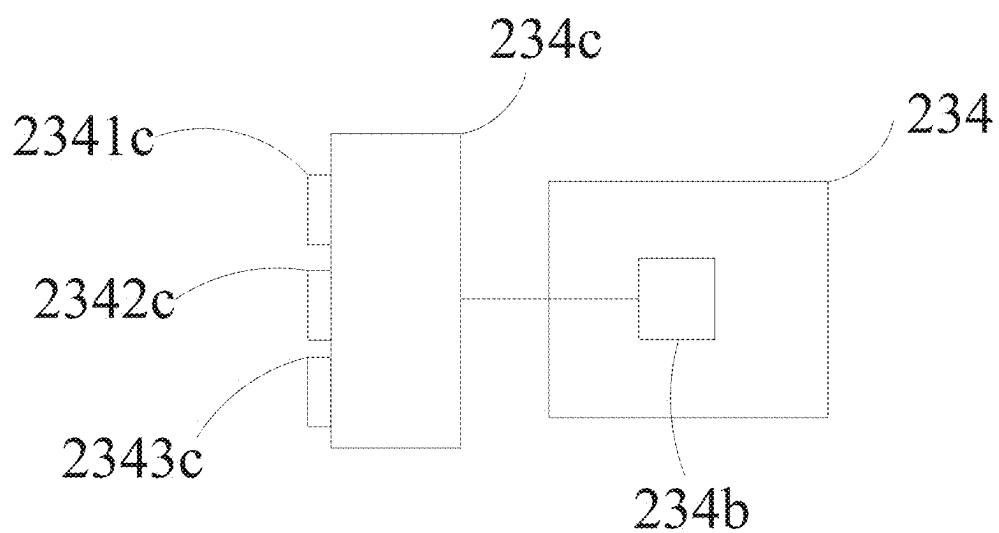


FIG. 30

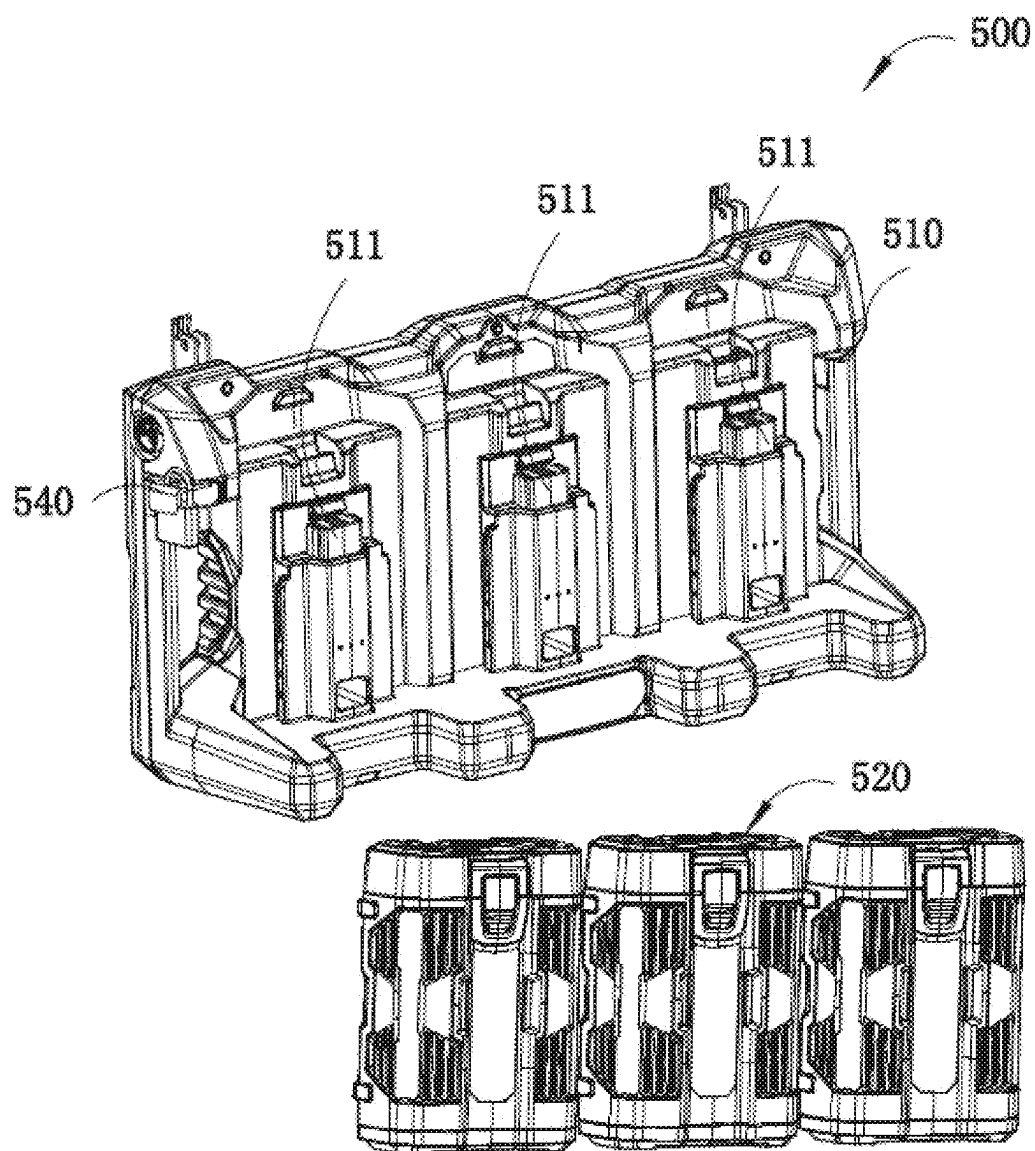


FIG. 31

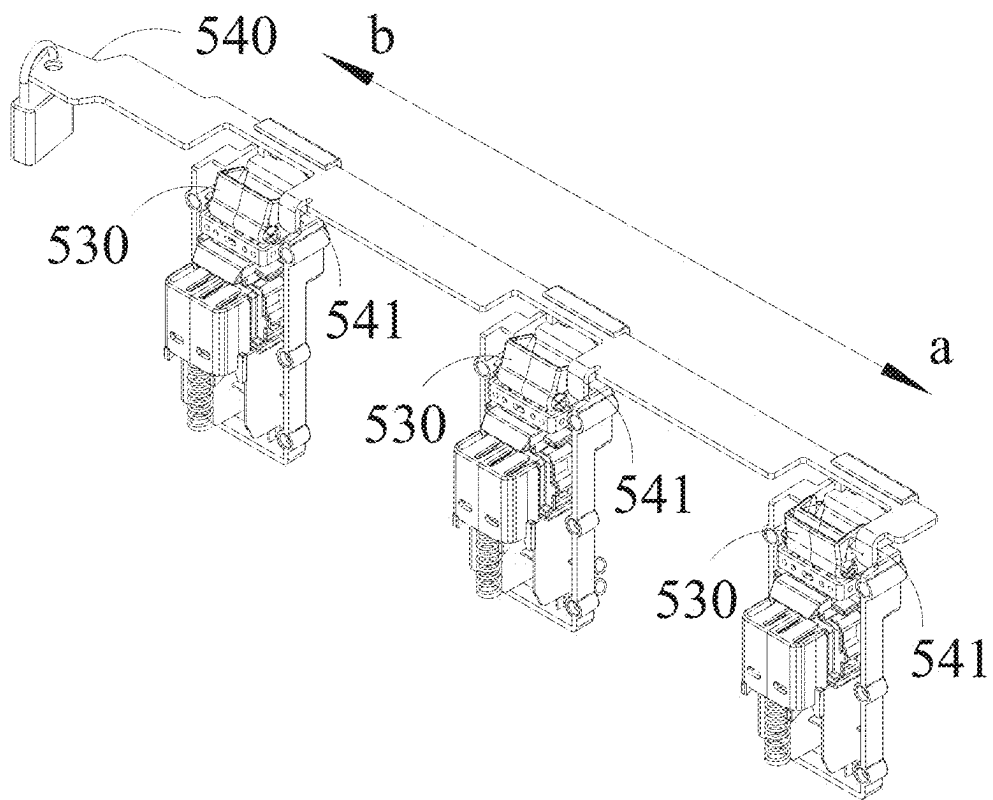


FIG. 32

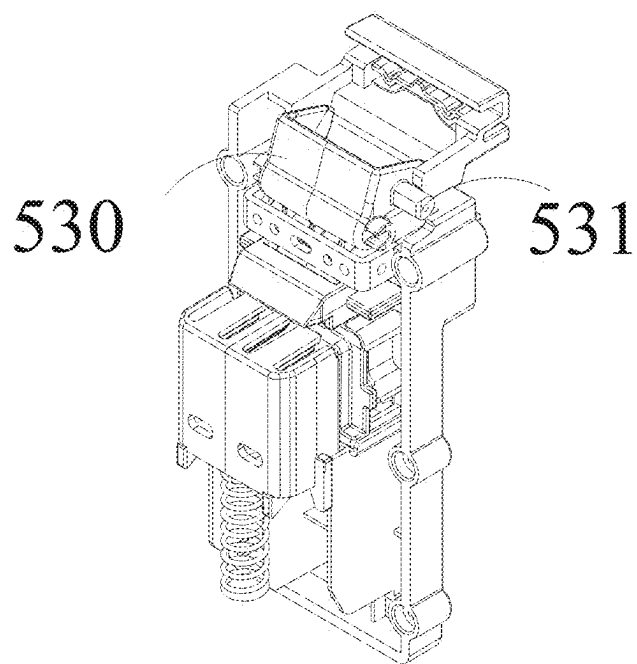


FIG. 33

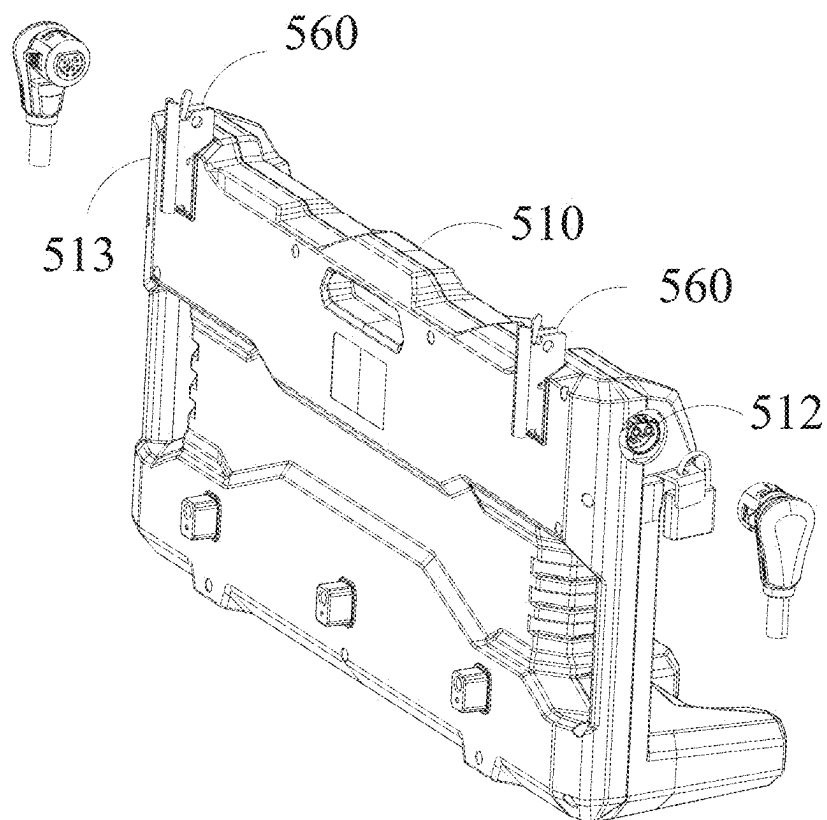


FIG. 34

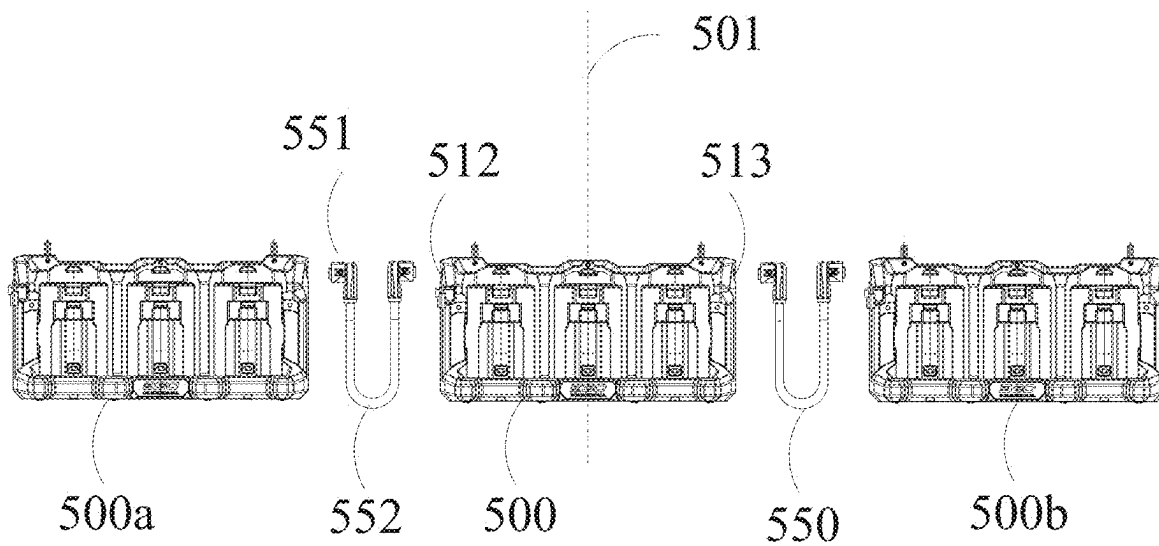


FIG. 35

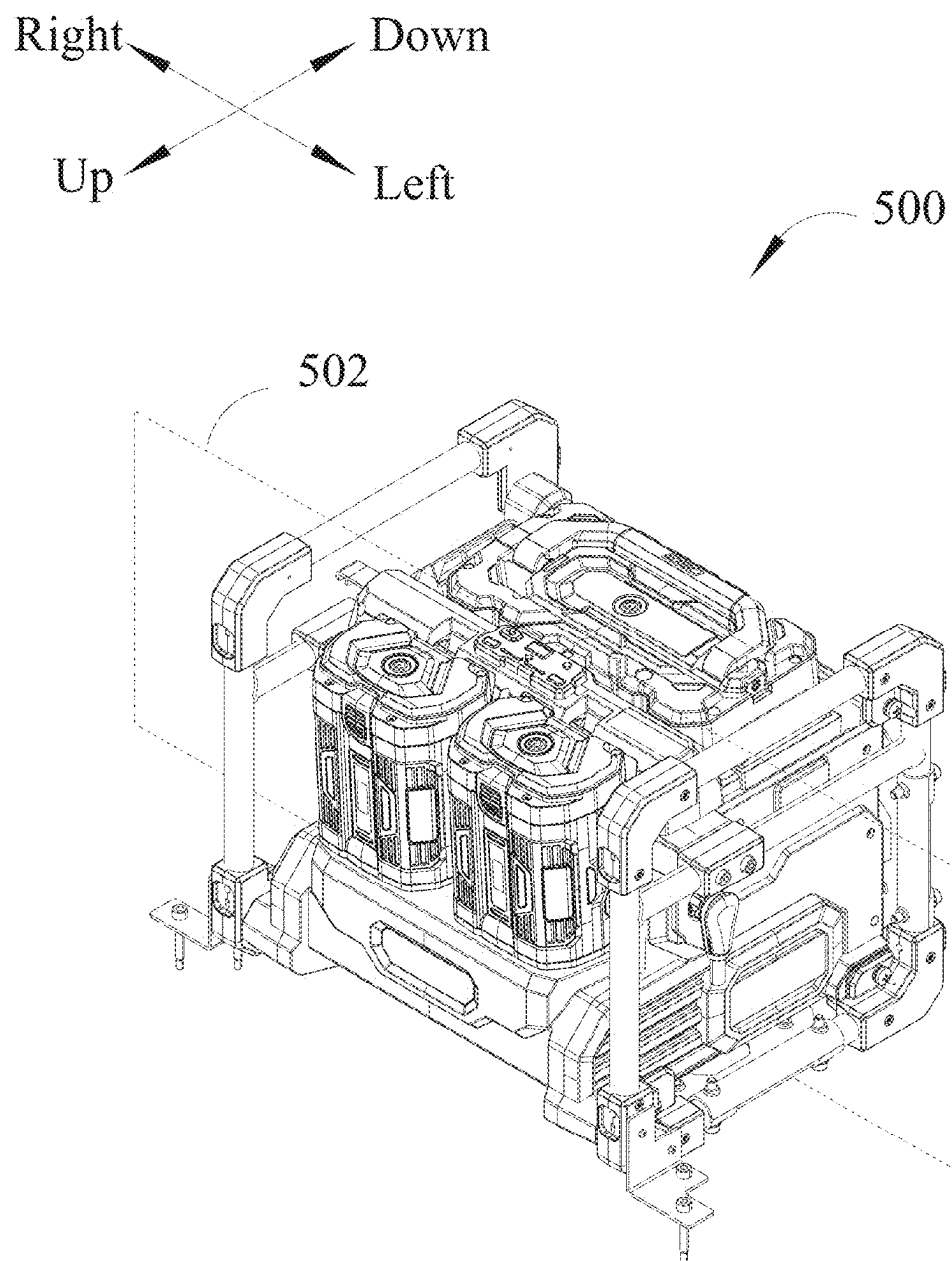


FIG. 36

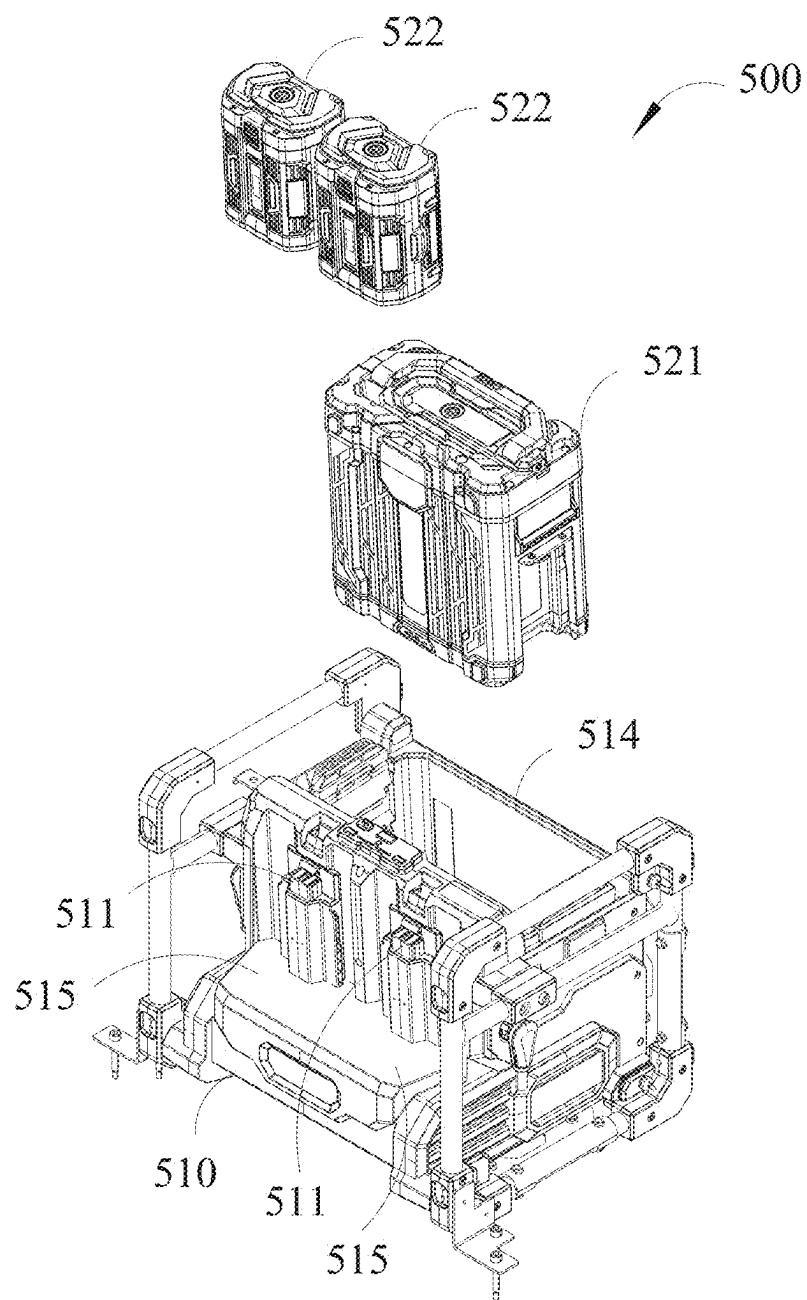


FIG. 37

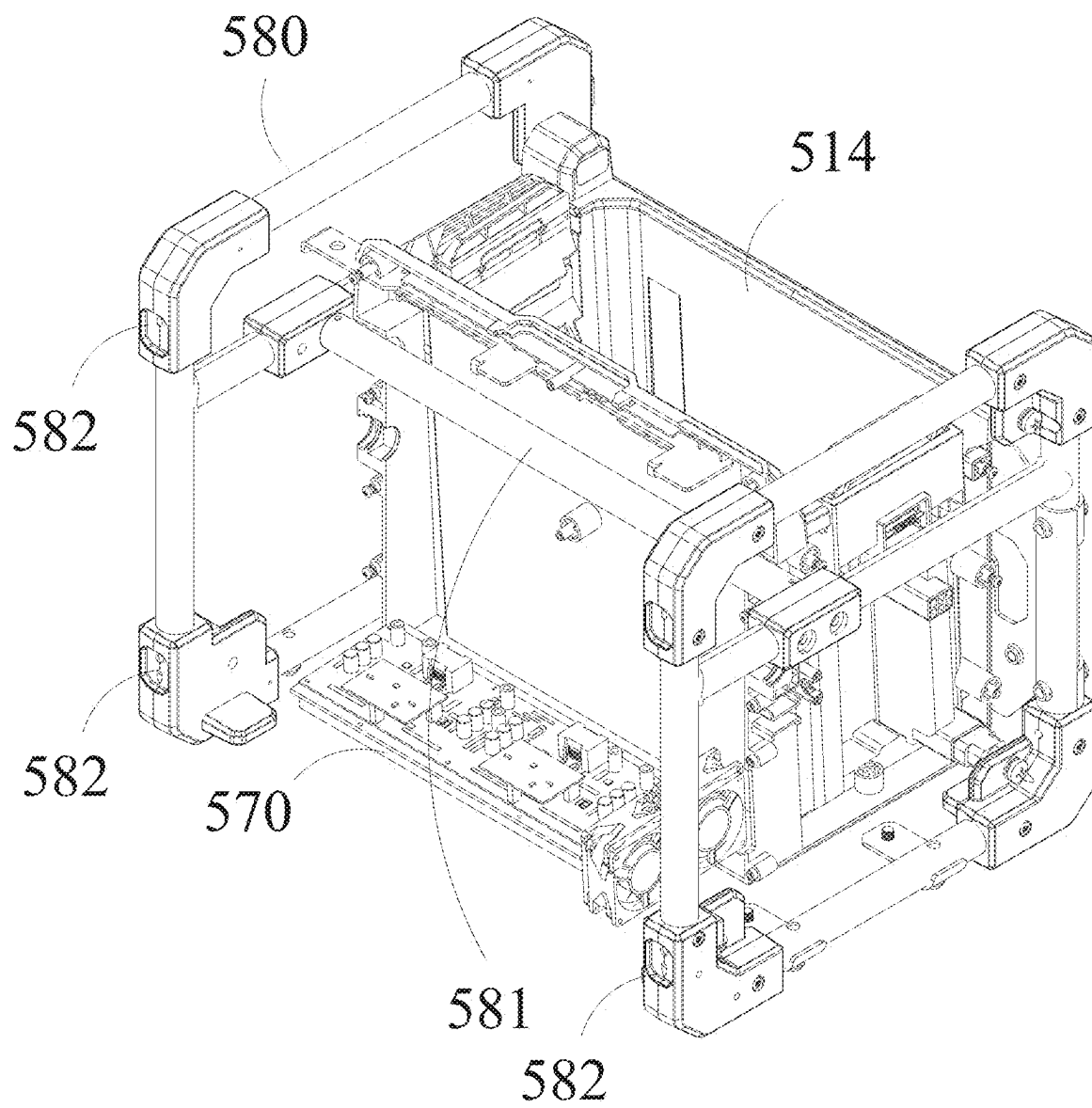


FIG. 38

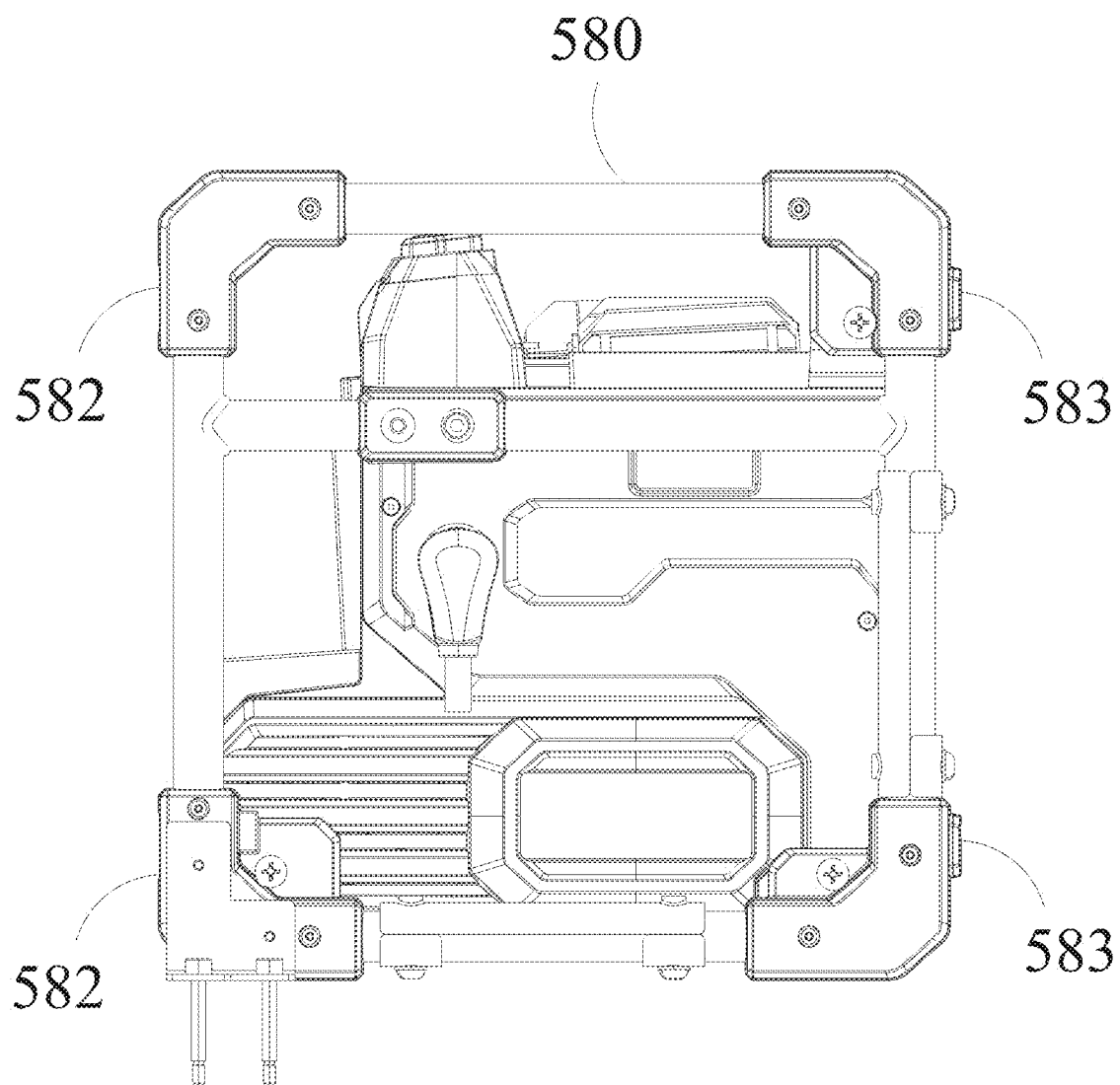


FIG. 39

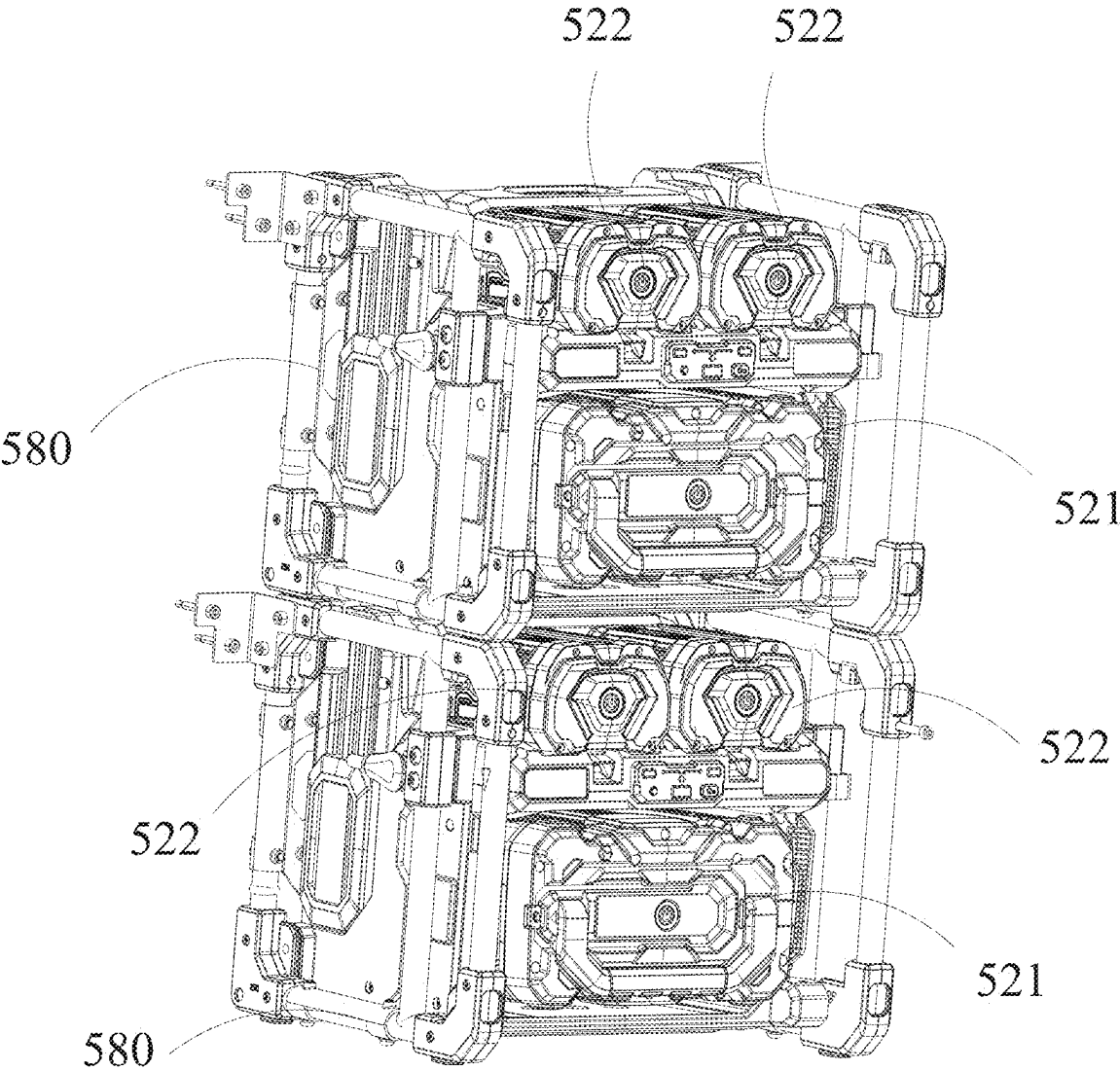


FIG. 40

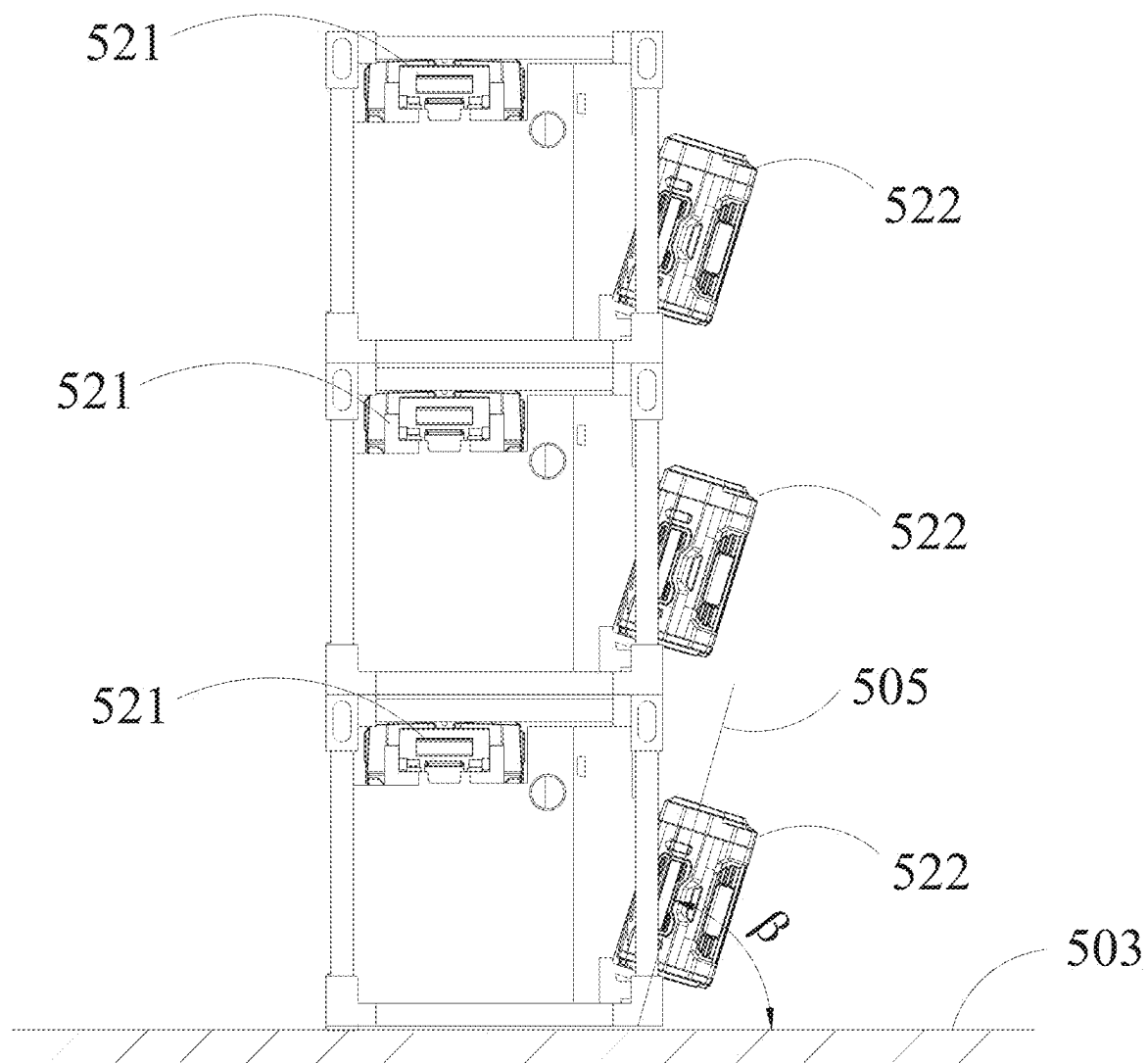


FIG. 41

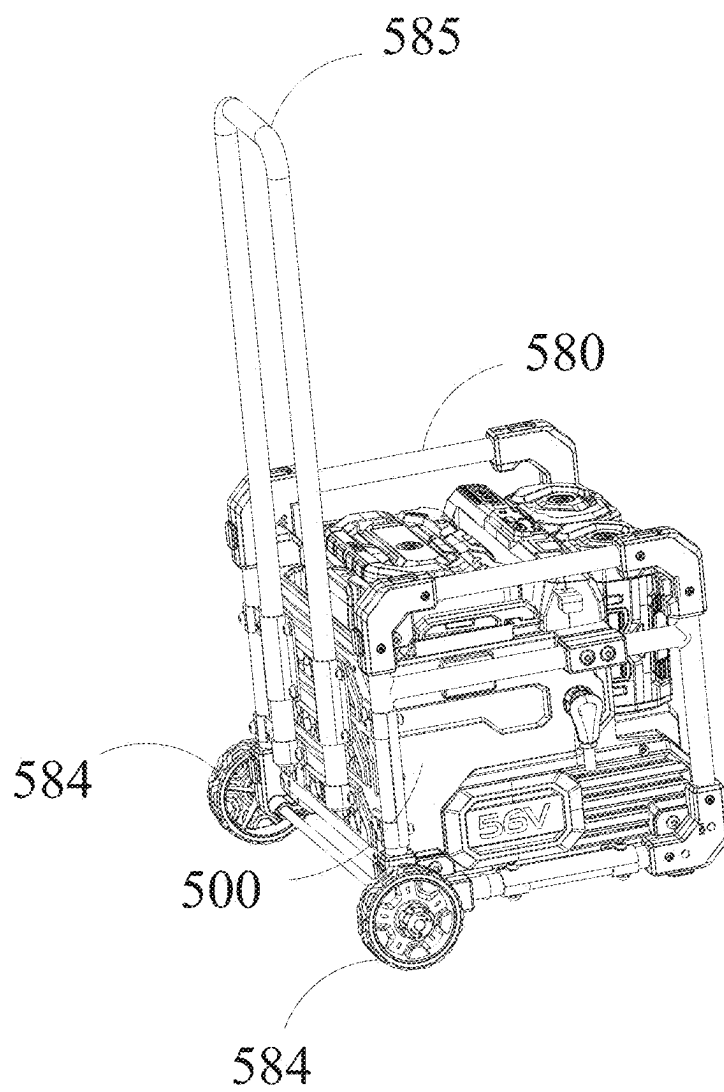


FIG. 42

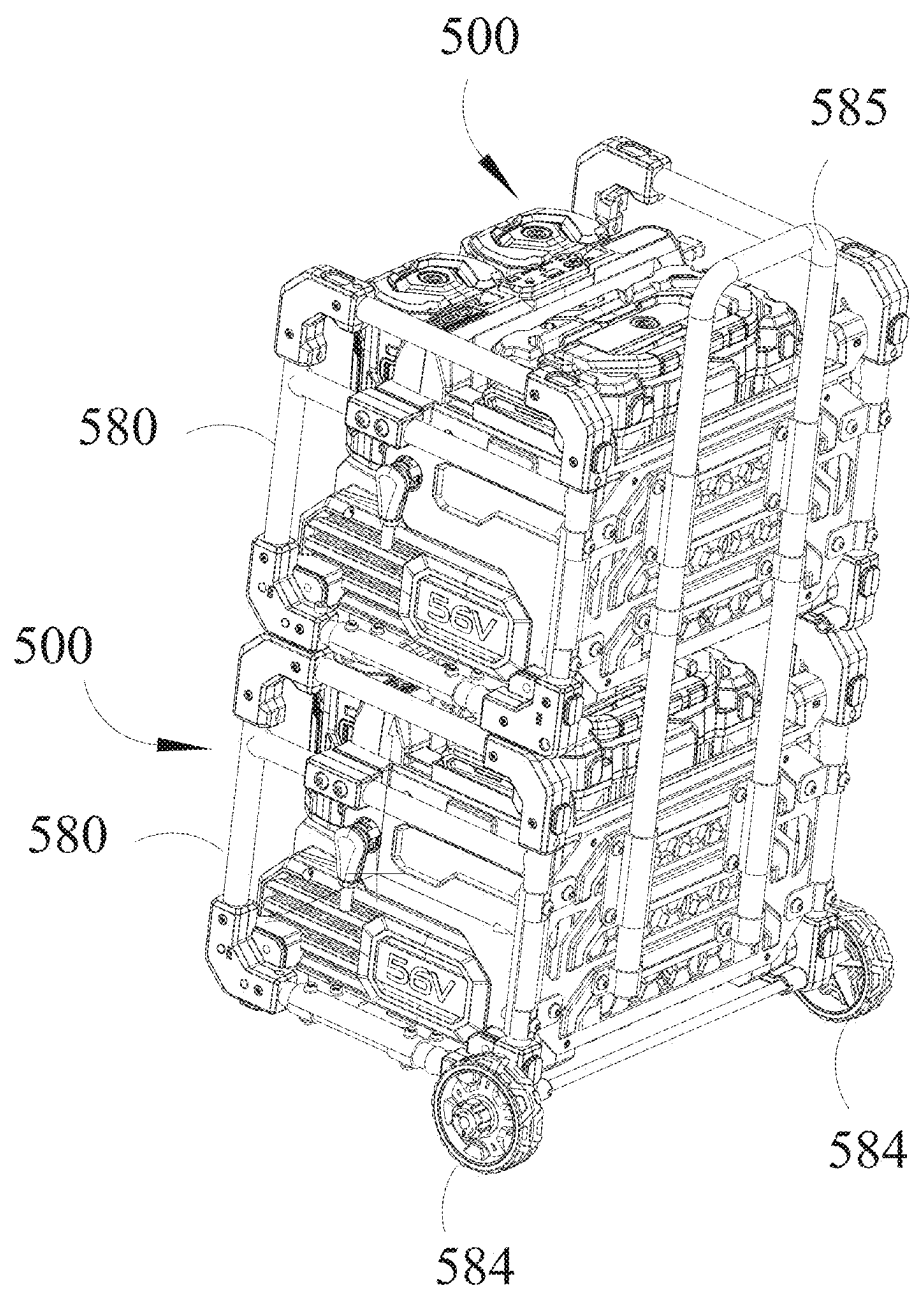


FIG. 43

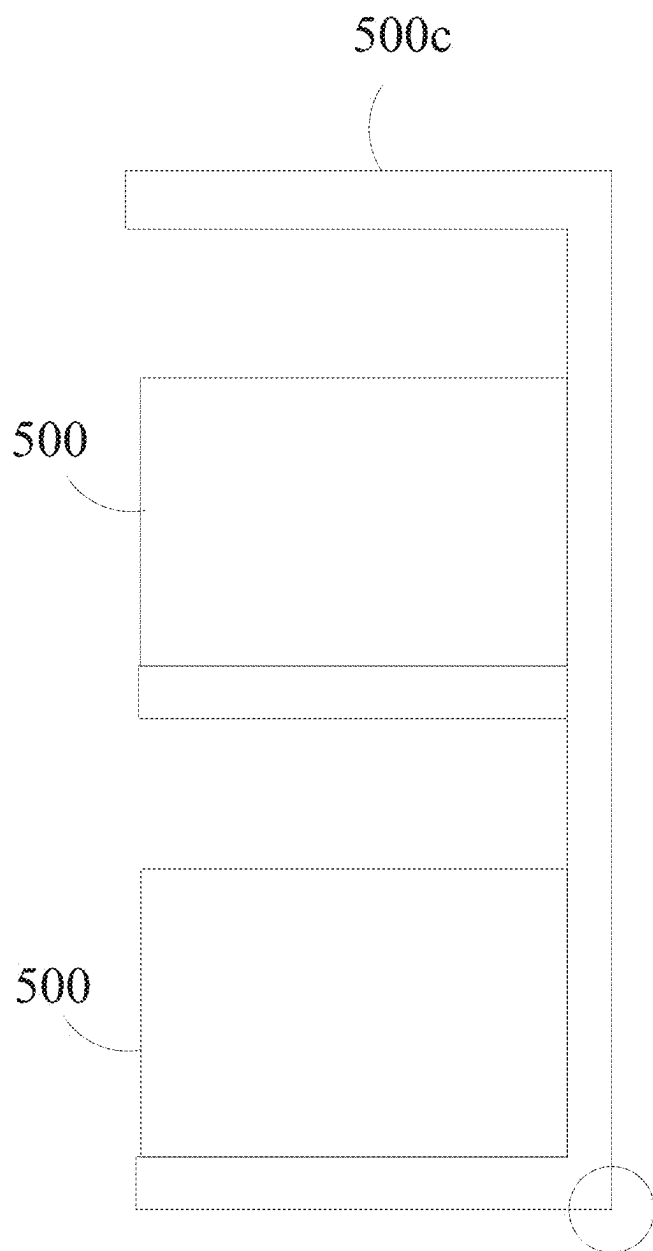


FIG. 44

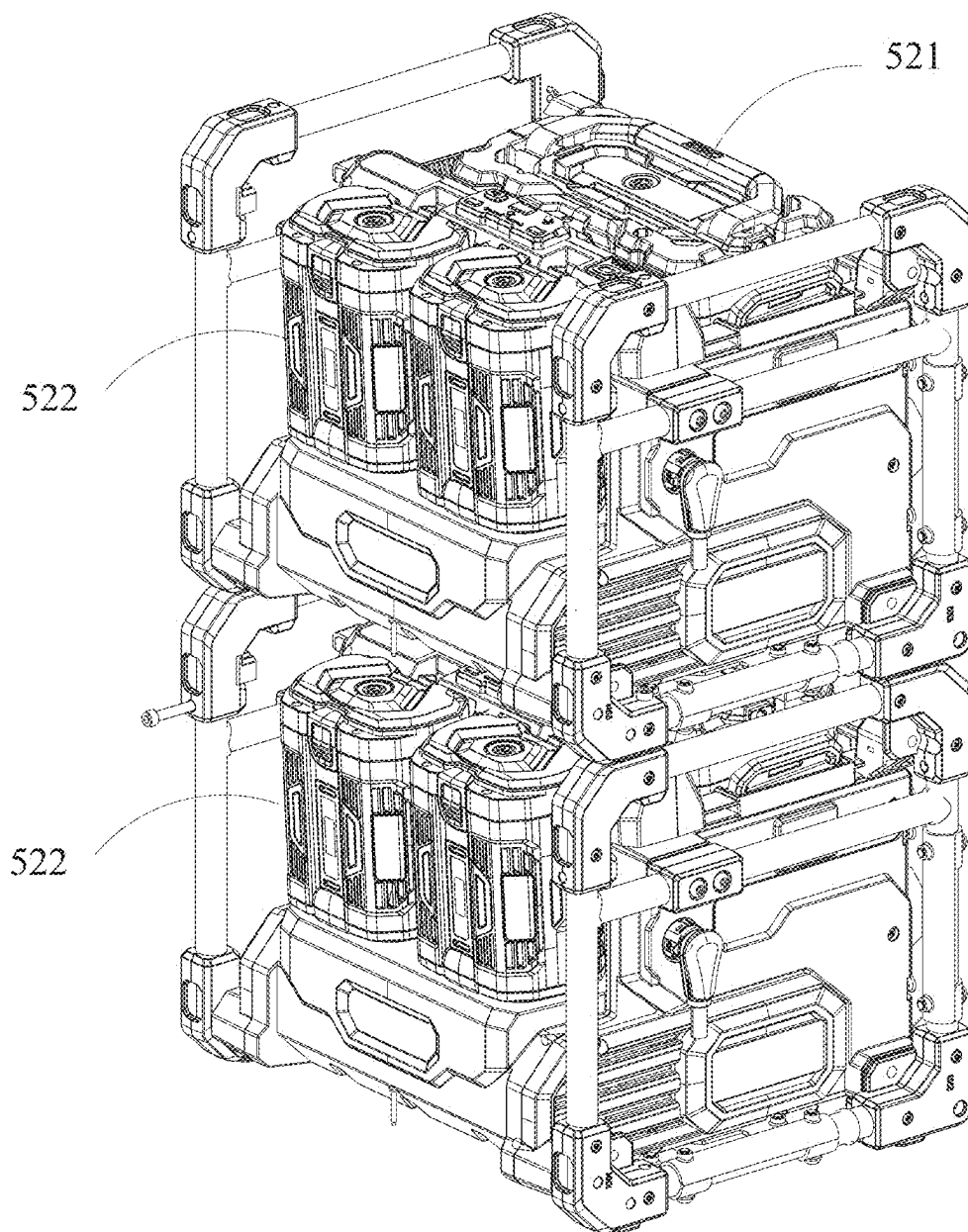


FIG. 45

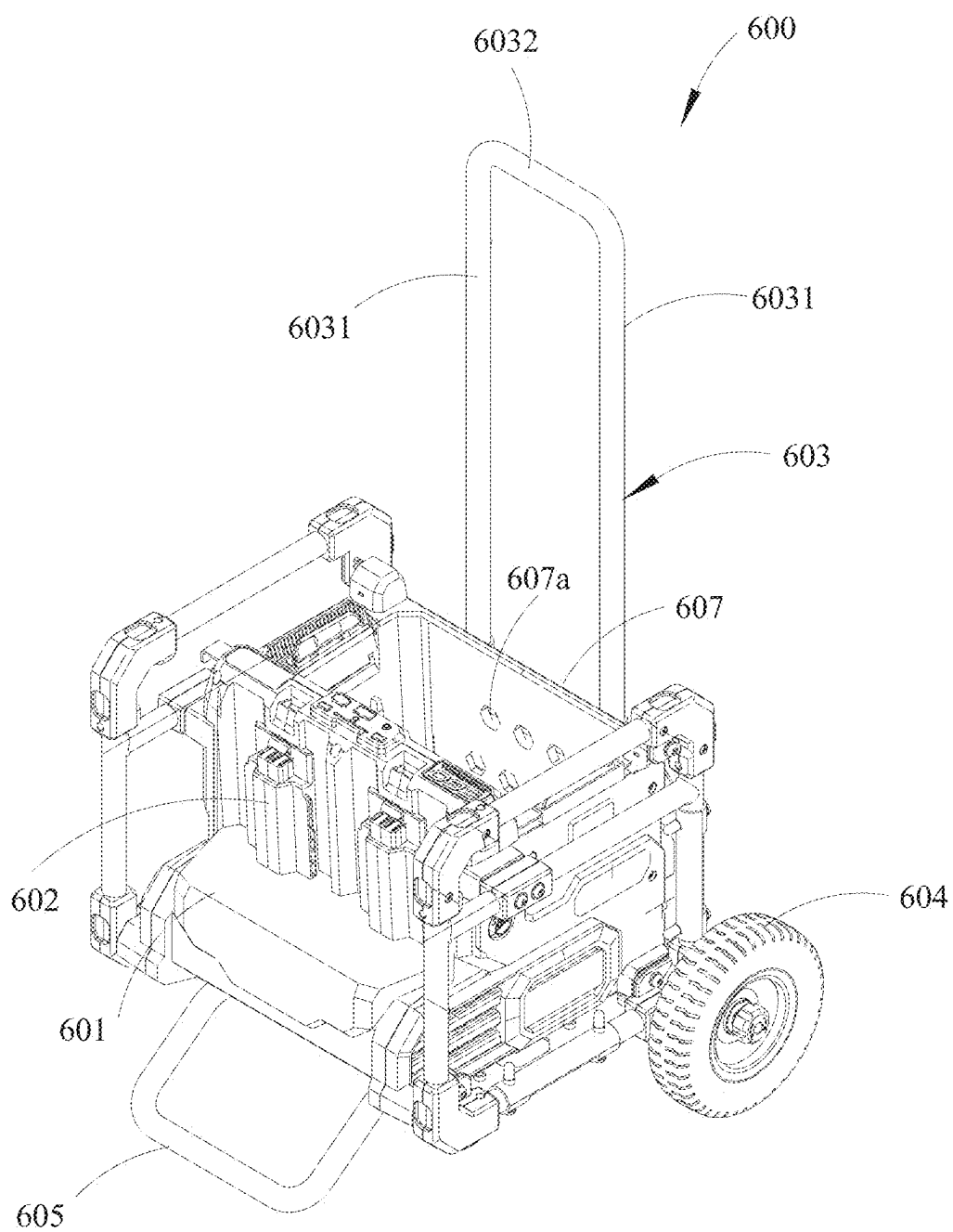


FIG. 46

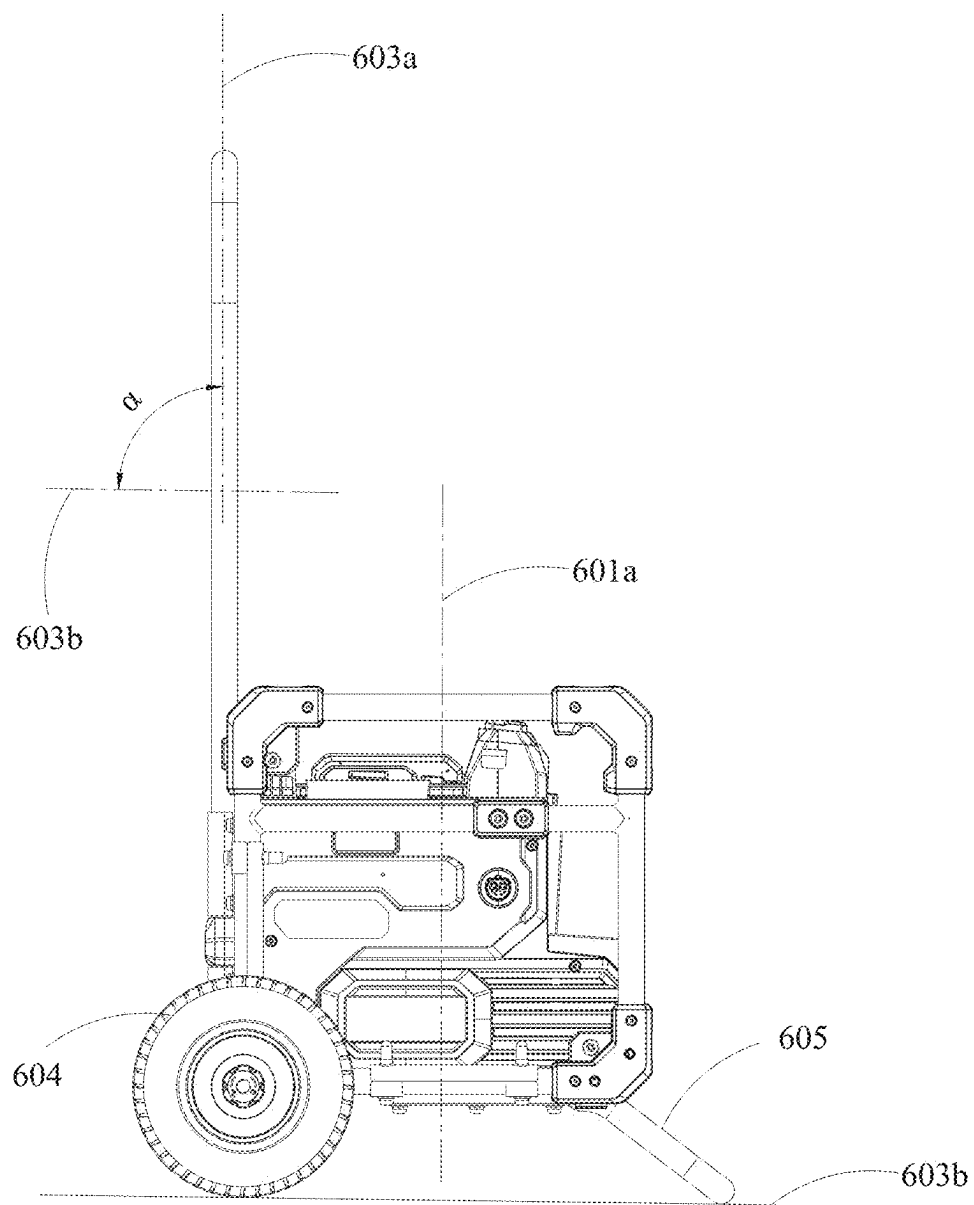


FIG. 47

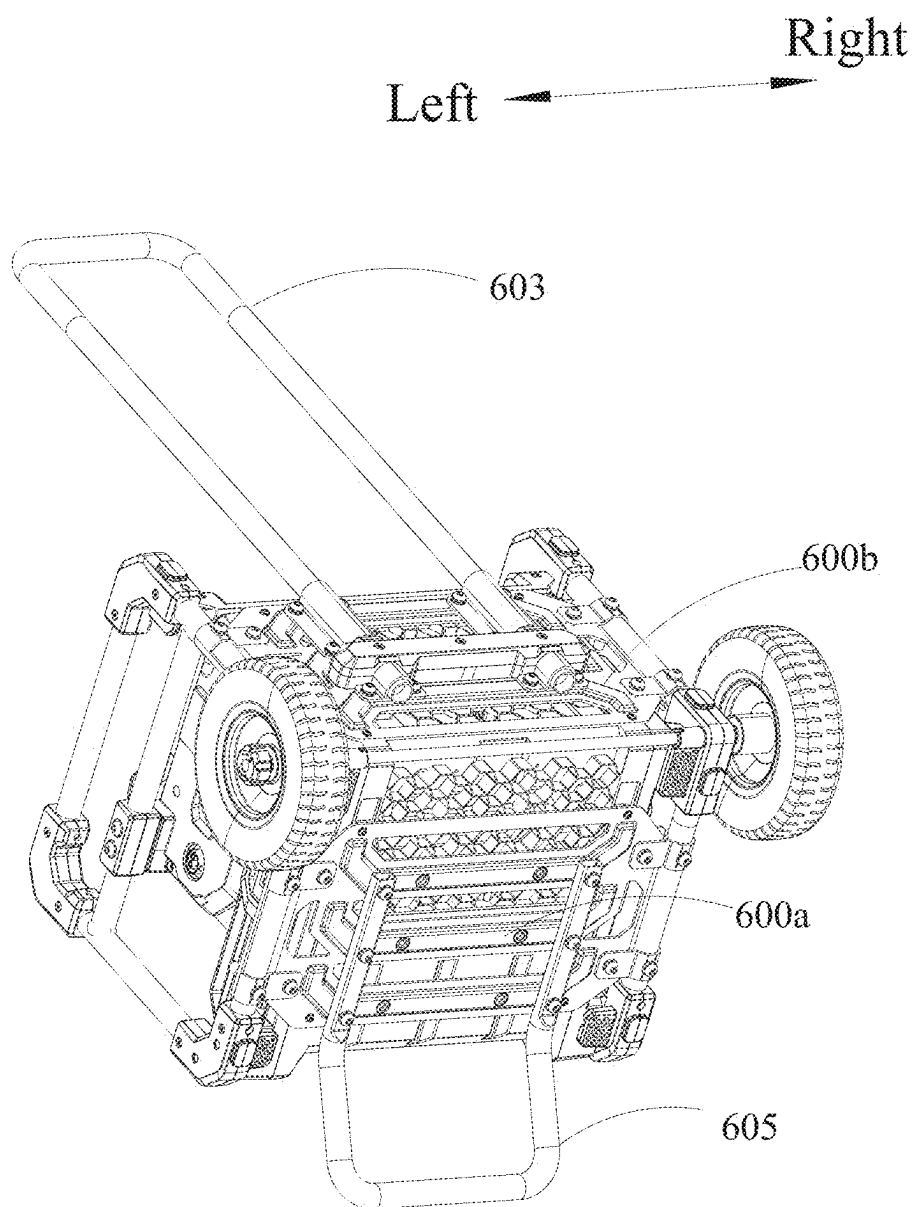


FIG. 48

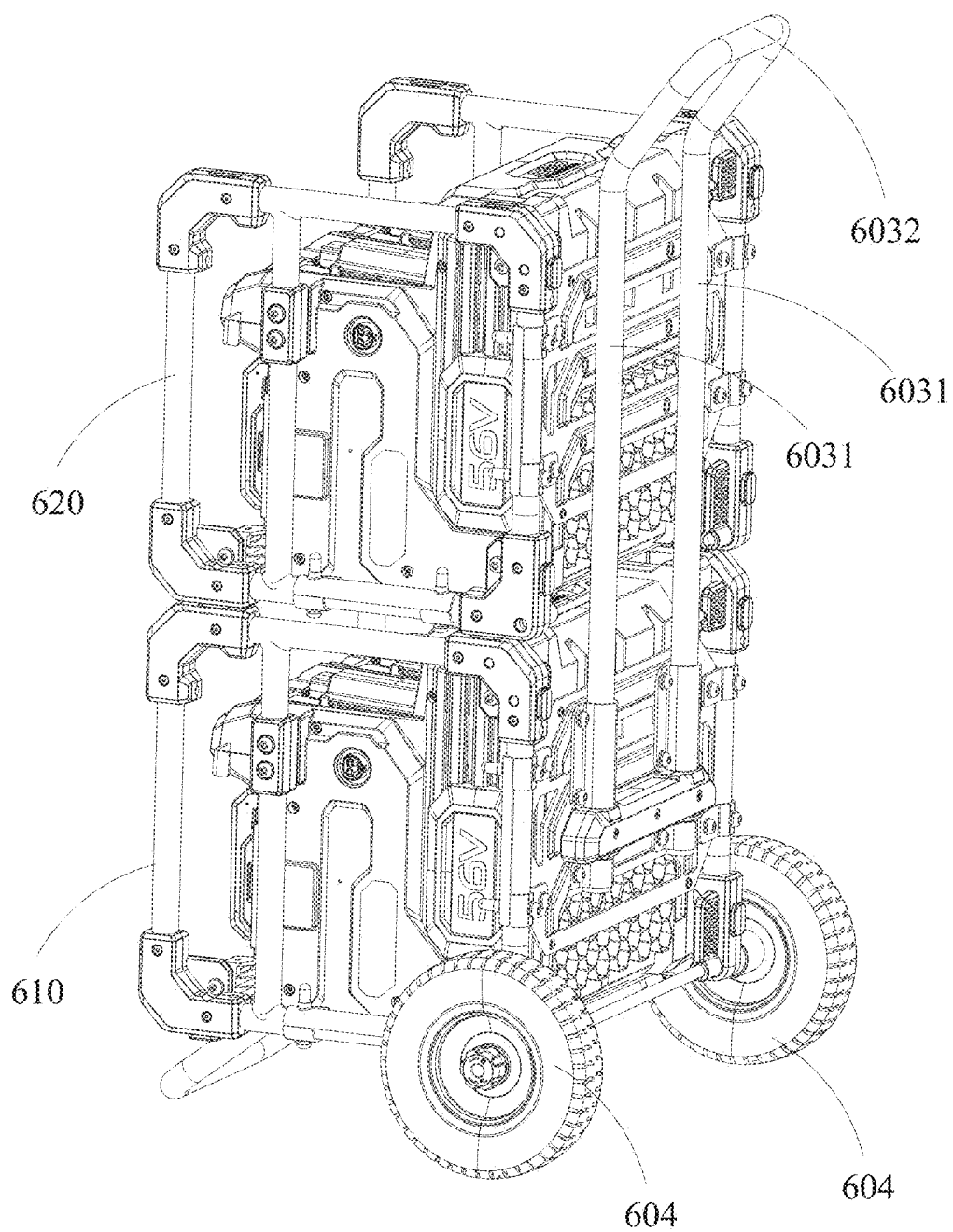


FIG. 49

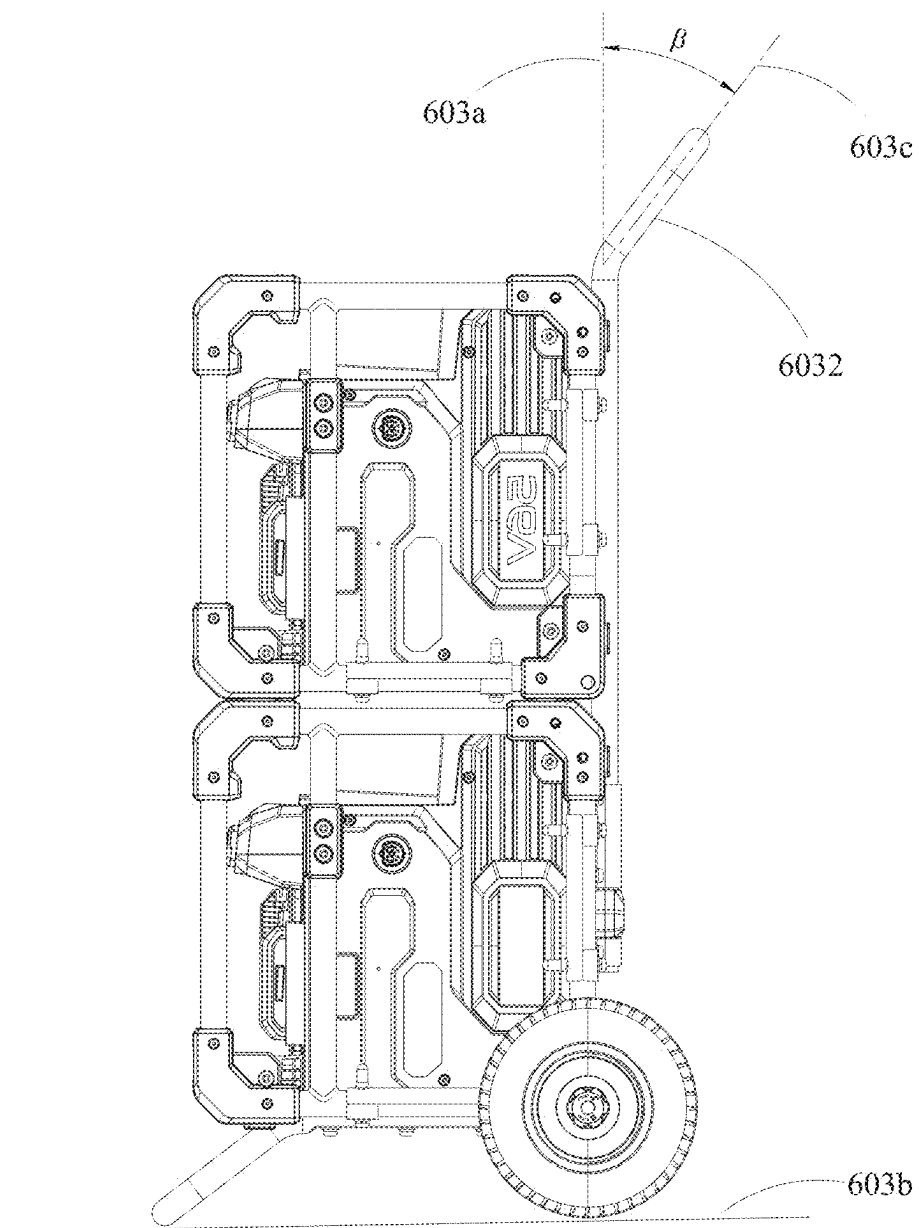


FIG. 50

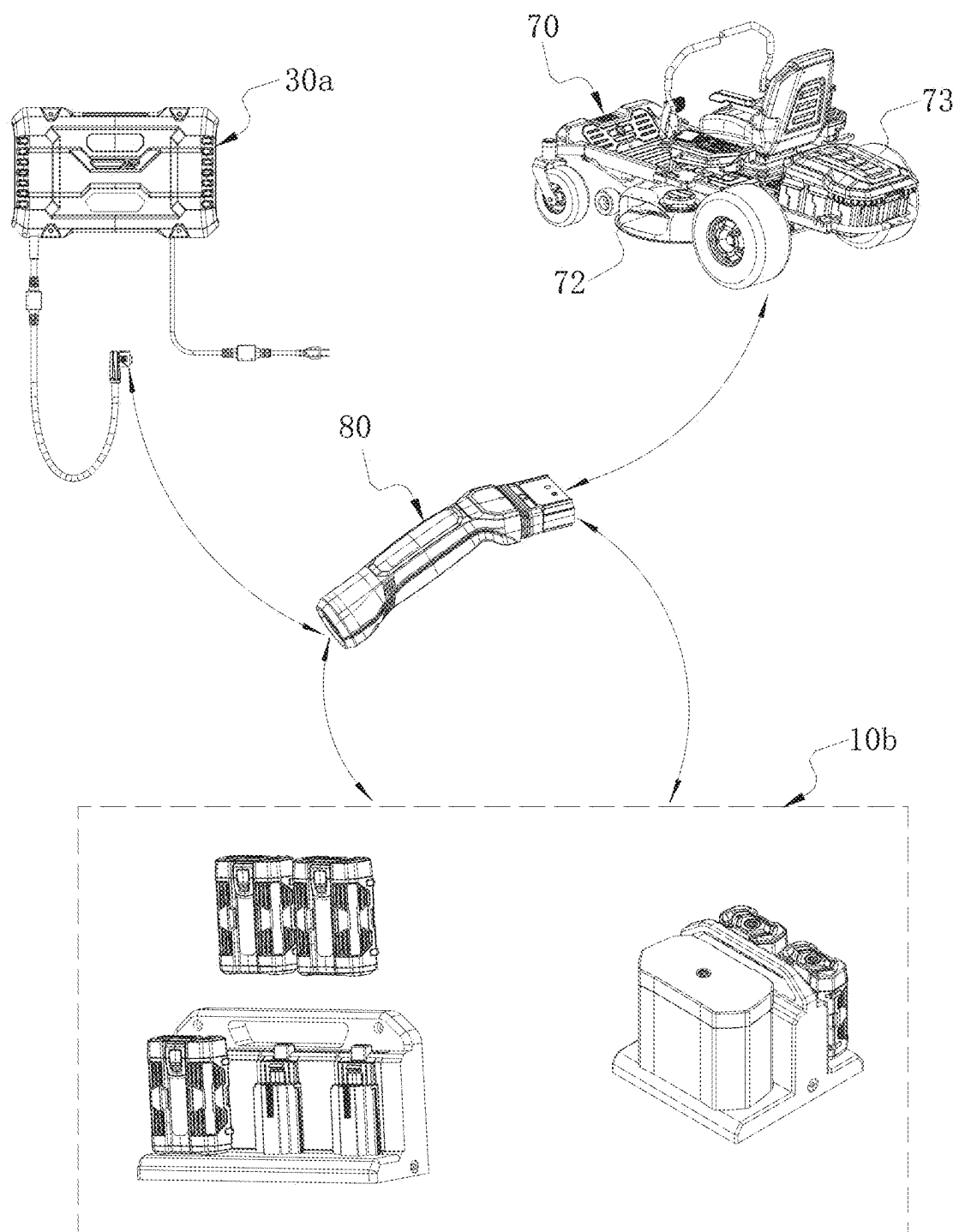


FIG. 51

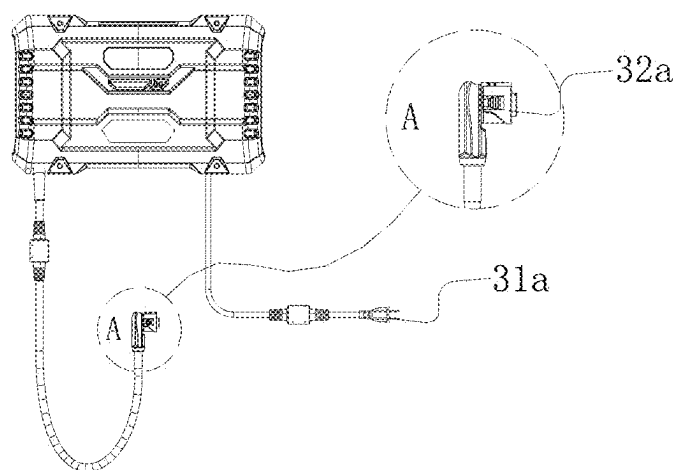


FIG. 52

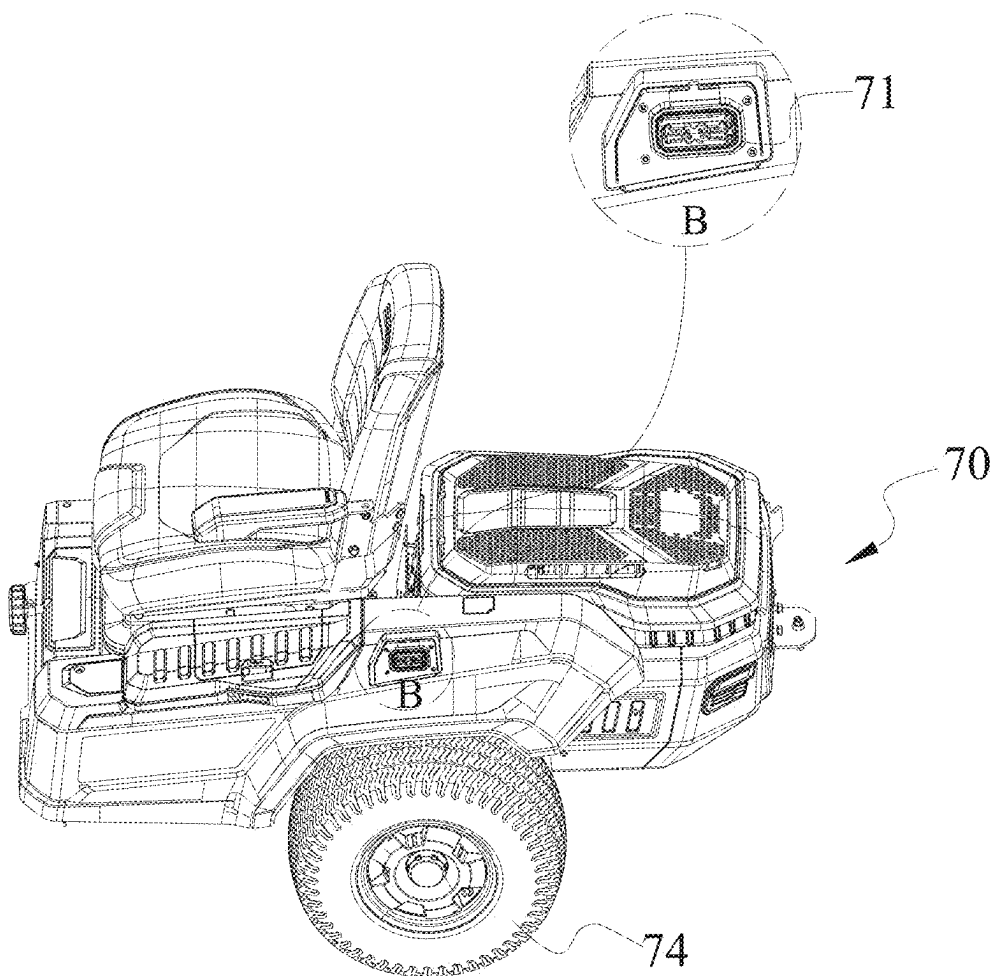


FIG. 53

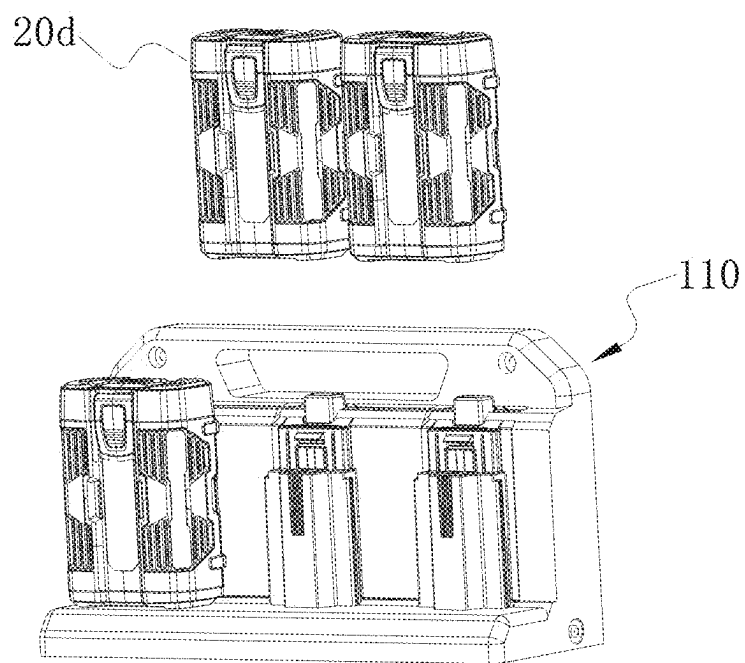


FIG. 54

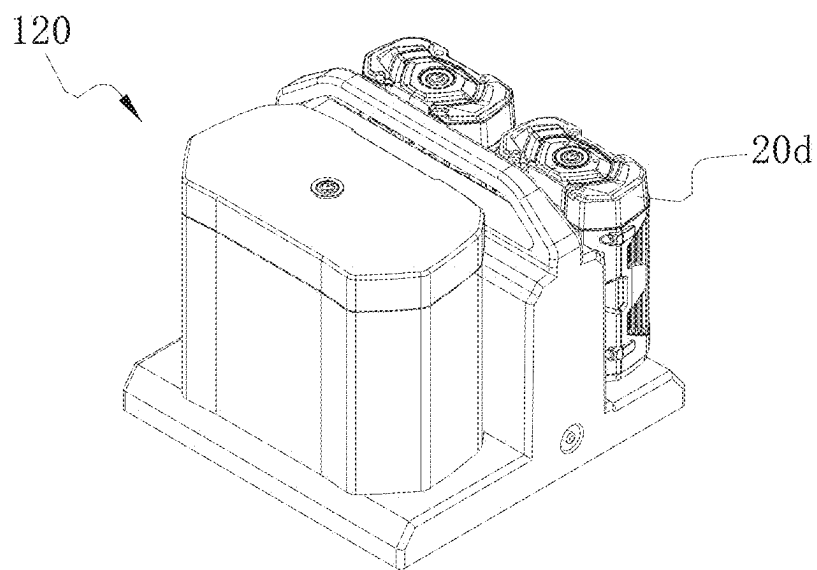


FIG. 55

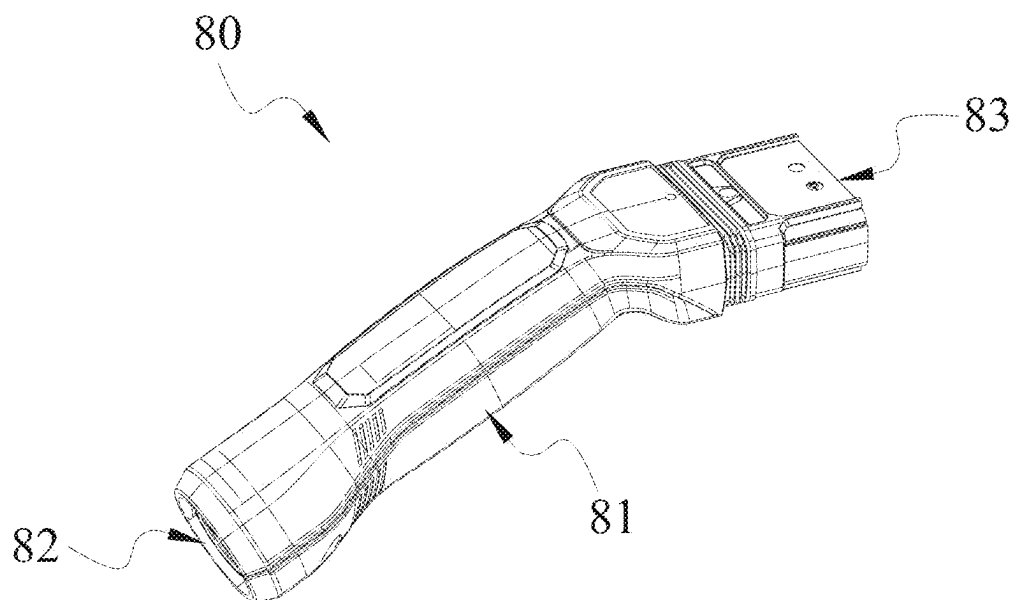


FIG. 56

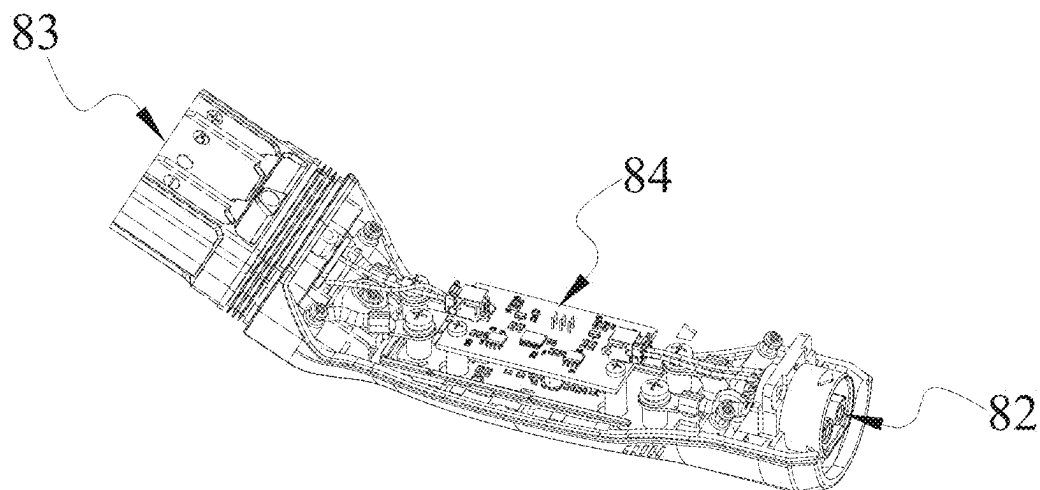


FIG. 57

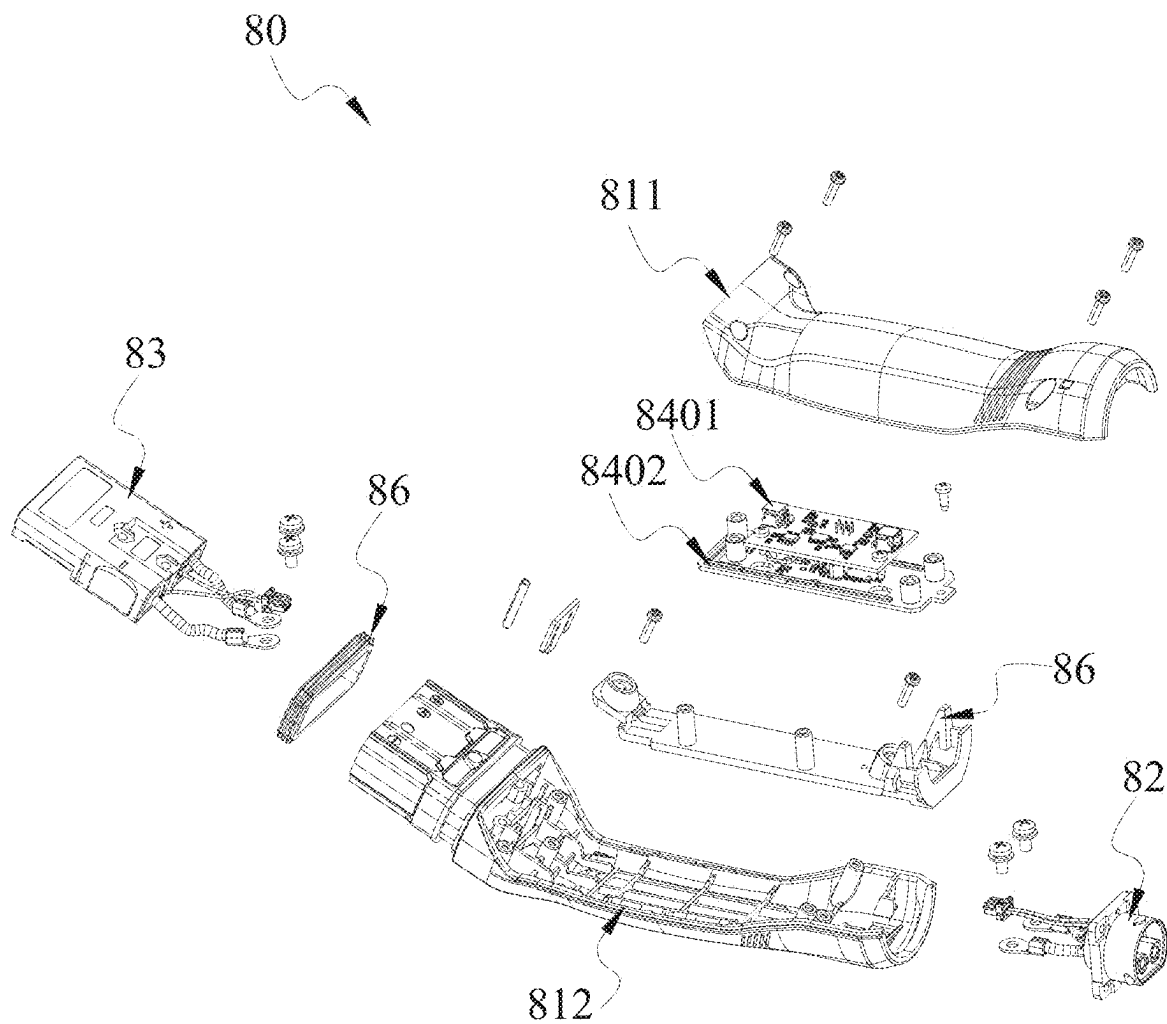


FIG. 58

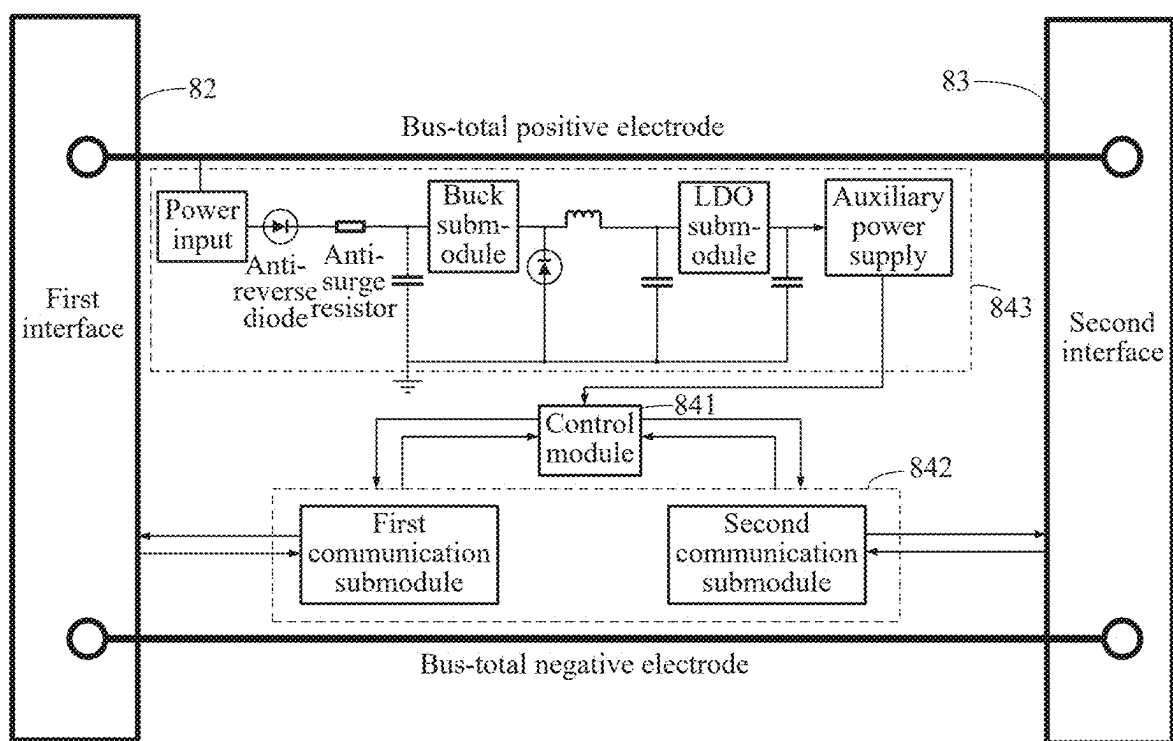


FIG. 59

CHARGING SYSTEM

RELATED APPLICATION INFORMATION

[0001] This application is a continuation-in-part of U.S. application Ser. No. 19/022,865, filed on Jan. 15, 2025, which application claims the benefit under 35 U.S.C. § 119 (a) of Chinese Patent Application No. 202410117985.X, filed on Jan. 26, 2024, Chinese Patent Application No. 202410117497.9, filed on Jan. 26, 2024, Chinese Patent Application No. 202420210815.1, filed on Jan. 26, 2024, and Chinese Patent Application No. 202410117667.3, filed on Jan. 26, 2024, which applications are incorporated herein by reference in their entireties.

[0002] This application also claims the benefit under 35 U.S.C. § 119 (a) of Chinese Patent Application No. 202410692020.3, filed on May 30, 2024, and Chinese Patent Application No. 202410692358.9, filed on May 30, 2024, which applications are incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0003] The present application relates to a charging system.

BACKGROUND

[0004] A charging system is disclosed in the related art. The charging system includes multiple battery packs and a charger for charging the multiple battery packs. The charger includes multiple battery interfaces, and each battery interface is connected to one battery pack so that the charger can charge the multiple battery packs.

[0005] This part provides background information related to the present application, and the background information is not necessarily the existing art.

SUMMARY

[0006] In some examples, a charging system includes an electrical energy conversion device and multiple electrical energy receiving devices connected in cascade. The charging system charges battery packs electrically connected to the multiple electrical energy receiving devices. The electrical energy conversion device includes a controller; a power interface for connecting an external power supply; a connecting port for connecting one of the multiple electrical energy receiving devices; and a conversion circuit electrically connected to the power interface and the connecting port separately. Each of the multiple electrical energy receiving devices includes an electrical energy input port for connecting the electrical energy conversion device or another of the multiple electrical energy receiving devices; and an electrical energy output port for connecting another of the multiple electrical energy receiving devices. The controller is configured to adjust an output of the connecting port based on the number of the multiple electrical energy receiving devices connected in the charging system.

[0007] In some examples, the output of the connecting port includes an output voltage of the connecting port and an output current of the connecting port.

[0008] In some examples, a maximum output voltage output from the connecting port is greater than or equal to 60 V and less than or equal to 65 V.

[0009] In some examples, a maximum output current output from the connecting port is greater than or equal to 22 A and less than or equal to 24 A.

[0010] In some examples, when the number of the multiple electrical energy receiving devices is less than or equal to a first preset value, the controller is configured to adjust a maximum output voltage of the connecting port to a first voltage threshold.

[0011] In some examples, when the number of the multiple electrical energy receiving devices is greater than or equal to a second preset value, the controller is configured to adjust the maximum output voltage of the connecting port to a second voltage threshold, where the first preset value is less than the second preset value, and the first voltage threshold is less than the second voltage threshold.

[0012] In some examples, the controller in the electrical energy conversion device acquires an input voltage of an electrical energy receiving device currently charged among the multiple electrical energy receiving devices, and when a difference between the input voltage and the output voltage of the electrical energy conversion device is less than a first threshold, the controller controls a maximum output voltage of the connecting port to be a third voltage threshold.

[0013] In some examples, the first threshold is set to approximately 2 V and the third voltage threshold is set to approximately 60 V.

[0014] In some examples, the controller in the electrical energy conversion device acquires an input voltage of an electrical energy receiving device currently charged among the multiple electrical energy receiving devices, and when a difference between the input voltage and the output voltage of the electrical energy conversion device is greater than a second threshold, the controller controls a maximum output voltage of the connecting port to be a fourth voltage threshold.

[0015] In some examples, the second threshold is set to approximately 4 V and the fourth voltage threshold is set to approximately 65 V.

[0016] In some examples, each of the battery packs has a rated voltage greater than or equal to 21 V and less than or equal to 65 V.

[0017] In some examples, the number of the multiple electrical energy receiving devices connected in the charging system is less than or equal to 30.

[0018] In some examples, the output of the connecting port includes output power of the connecting port, and the output power of the connecting port is greater than or equal to 500 W and less than or equal to 2000 W.

[0019] In some examples, the controller in the electrical energy conversion device communicates with the multiple electrical energy receiving devices using a controller area network (CAN) communication protocol.

[0020] In some examples, the electrical energy conversion device includes a wireless communication module configured to communicate with a remote device.

[0021] In some examples, the controller in the electrical energy conversion device communicates directly or indirectly with controllers in the multiple electrical energy receiving devices and records identities of the multiple electrical energy receiving devices.

[0022] In some examples, after recording the identities of the multiple electrical energy receiving devices, the electri-

cal energy conversion device controls the multiple electrical energy receiving devices to be charged in an order of the identities.

[0023] The present application further discloses a technical solution. A charging system includes an electrical energy conversion device and multiple electrical energy receiving devices connected in cascade. The charging system is configured to be able to charge battery packs electrically connected to the multiple electrical energy receiving devices. The electrical energy conversion device includes a controller; a power interface for connecting an external power supply; a connecting port for connecting one of the multiple electrical energy receiving devices; and a conversion circuit electrically connected to the power interface and the connecting port separately. Each of the multiple electrical energy receiving devices includes a controller; an electrical energy input port for connecting the electrical energy conversion device or another of the multiple electrical energy receiving devices; and an electrical energy output port for connecting another of the multiple electrical energy receiving devices. The controller in the electrical energy conversion device communicates directly or indirectly with controllers in the multiple electrical energy receiving devices and records identities of the multiple electrical energy receiving devices.

[0024] In some examples, after recording the identities of the multiple electrical energy receiving devices, the electrical energy conversion device controls the multiple electrical energy receiving devices to be charged in an order of the identities.

[0025] In some examples, the order of the identities is to start from an electrical energy receiving device closest to the electrical energy conversion device and number the multiple electrical energy receiving devices in sequence according to an order in which the multiple electrical energy receiving devices are physically connected.

[0026] In some examples, the identities are numbered in ascending order.

[0027] In some examples, the electrical energy conversion device is configured to give a level signal to trigger an electrical energy receiving device physically connected to the electrical energy conversion device to perform identification, and the electrical energy receiving device is configured to output the given level signal to the next physically connected electrical energy receiving device.

[0028] In some examples, the electrical energy conversion device is configured to, when at least one of the multiple electrical energy receiving devices fails in a process of recording the identities of the multiple electrical energy receiving devices, start from the last electrical energy receiving device to perform reverse numbering, identify an identity of the at least one electrical energy receiving device that fails, and report the identity to a remote device.

[0029] In some examples, the electrical energy conversion device is configured to have an information polling function; in a process of implementing the information polling function, the electrical energy conversion device is configured to collect information from an electrical energy receiving device corresponding to each identity in the order of the identities and report the information to the remote device.

[0030] In some examples, the electrical energy conversion device is configured to re-record identities of multiple electrical energy receiving devices **230** connected in cascade

when detecting in the polling process that a new electrical energy receiving device is connected to the charging system.

[0031] In some examples, a maximum number of the identities of the multiple electrical energy receiving devices is less than or equal to 32.

[0032] In some examples, the electrical energy conversion device is configured to adjust output power according to the maximum number of the identities.

[0033] In some examples, an output voltage of the connecting port is greater than or equal to 60 V and less than or equal to 65 V.

[0034] In some examples, a maximum output voltage of the connecting port is greater than or equal to 22 A and less than or equal to 24 A.

[0035] The present application further discloses a technical solution. A charging device includes a housing; a mounting portion for mounting or accommodating an energy storage device; a pull rod assembly mounted to the housing; and wheels mounted to the housing. When the pull rod assembly is pushed or pulled by a user, the wheels roll so that the user conveniently moves the charging device. When the charging device stands still on the ground, the angle between a plane formed by the pull rod assembly and the ground is less than or equal to 85°.

[0036] In some examples, the pull rod assembly includes two parallel extension rods and a grip rod connecting the two extension rods.

[0037] In some examples, the extension rods are retractable.

[0038] In some examples, the extension rods are foldable.

[0039] In some examples, the charging device further includes a bottom support detachably mounted to the housing.

[0040] In some examples, the height of the bottom support is greater than the radius of the wheels.

[0041] In some examples, the bottom support is detachably mounted to a first mounting portion of the housing, and the pull rod assembly is detachably mounted to a second mounting portion of the housing; and the first mounting portion is also able to be used for mounting the pull rod assembly, and the second mounting portion is also able to be used for mounting the bottom support.

[0042] In some examples, the bottom support is disposed on a side of the housing facing away from the wheels.

[0043] In some examples, the wheels and the pull rod assembly are disposed on the same side of the charging device.

[0044] In some examples, the charging device includes a first charging device and a second charging device. When the first charging device and the second charging device are stacked, the pull rod assembly is able to be pushed or pulled by the user to facilitate the movement of the first charging device and the second charging device.

[0045] The present application further discloses a technical solution. A charging system includes an electrical energy conversion device and multiple electrical energy receiving devices connected in cascade. The charging system is configured to charge battery packs electrically connected to the multiple electrical energy receiving devices. The electrical energy conversion device includes a controller; a power interface for connecting an external power supply; a connecting port for connecting one of the multiple electrical energy receiving devices; and a conversion circuit electrically connected to the power interface and the connecting

port separately. Each of the multiple electrical energy receiving devices includes a controller; an electrical energy input port for connecting the electrical energy conversion device or another of the multiple electrical energy receiving devices; and an electrical energy output port for connecting another of the multiple electrical energy receiving devices. The controller in the electrical energy conversion device communicates with controllers in the multiple electrical energy receiving devices using a CAN communication protocol.

[0046] In some examples, the electrical energy conversion device includes a wireless communication module configured to communicate with a remote device.

[0047] In some examples, the controller in the electrical energy conversion device is configured to record identities of the multiple electrical energy receiving devices.

[0048] In some examples, after recording the identities of the multiple electrical energy receiving devices, the electrical energy conversion device controls the multiple electrical energy receiving devices to be charged in an order of the identities.

[0049] In some examples, the remote device sends an instruction to the electrical energy conversion device through the wireless communication module, and the wireless communication module transmits the instruction to the multiple electrical energy receiving devices through the CAN communication protocol.

[0050] In some examples, after recording the identities of the multiple electrical energy receiving devices, the electrical energy conversion device controls the multiple electrical energy receiving devices to be charged based on an instruction from the remote device.

[0051] In some examples, the multiple electrical energy receiving devices transmit information to the electrical energy conversion device through the CAN communication protocol, and the electrical energy conversion device reports the information through the wireless communication module to display relevant information on the remote device.

[0052] In some examples, the charging system further includes a communication bus for implementing information exchange between the electrical energy conversion device and the multiple electrical energy receiving devices, and the electrical energy conversion device is configured to perform information polling on the multiple electrical energy receiving devices through the communication bus.

[0053] In some examples, the electrical energy conversion device is configured to re-record, when idle, identities of multiple electrical energy receiving devices connected in cascade when detecting in the polling process that a new electrical energy receiving device is connected to the charging system.

[0054] In some examples, the remote device is configured to be able to implement an over-the-air (OTA) upgrade function for the electrical energy conversion device and the multiple electrical energy receiving devices through the wireless communication module and the CAN communication protocol.

BRIEF DESCRIPTION OF THE DRAWINGS

[0055] FIG. 1 is a perspective view of a charging system as an example.

[0056] FIG. 2 is a perspective view of an adapter and energy storage devices in the charging system of FIG. 1.

[0057] FIG. 3 is a perspective view of an adapter and energy storage devices in the charging system of FIG. 1 from another angle.

[0058] FIG. 4 is a perspective view of an adapter in the charging system of FIG. 1.

[0059] FIG. 5 is a perspective view of the adapter of FIG. 4 with energy storage devices mounted.

[0060] FIG. 6 is a perspective view of a charging system according to another example of the present application.

[0061] FIG. 7 is a circuit schematic of the charging system of FIG. 1.

[0062] FIG. 8 shows another example of an adapter and energy storage devices in a charging system.

[0063] FIG. 9A is a perspective view of the adapter in FIG. 8.

[0064] FIG. 9B is a perspective view of the adapter in FIG. 8 from another angle.

[0065] FIG. 10 is a schematic view of another charging system as an example.

[0066] FIG. 11 is a perspective view of another charging system as an example.

[0067] FIG. 12 is a circuit schematic of the charging system of FIG. 11.

[0068] FIG. 13 is a circuit topology diagram of a direct current-direct current conversion circuit in the charging system of FIG. 12.

[0069] FIG. 14A is a circuit diagram of the direct current-direct current conversion circuit of FIG. 13 in mode A.

[0070] FIG. 14B is a circuit diagram of the direct current-direct current conversion circuit of FIG. 13 in mode B.

[0071] FIG. 14C is a circuit diagram of the direct current-direct current conversion circuit of FIG. 13 in mode C.

[0072] FIG. 14D is a circuit diagram of the direct current-direct current conversion circuit of FIG. 13 in mode D.

[0073] FIG. 15 is a control timing diagram of the direct current-direct current conversion circuit of FIG. 13 and a waveform graph of an inductor current and an output current.

[0074] FIG. 16 is a schematic diagram of a bidirectional charging system as an example.

[0075] FIG. 17 is a perspective view of an electrical energy conversion device as an example in FIG. 16.

[0076] FIG. 18 is a schematic diagram of charge and discharge principles of an outdoor traveling device as an example of the electrical energy conversion device of FIG. 17.

[0077] FIG. 19A is a schematic diagram of electrical connections when a bidirectional inverter module is mounted inside an outdoor traveling device.

[0078] FIG. 19B is a schematic diagram of electrical connections when a bidirectional inverter module is mounted outside an outdoor traveling device.

[0079] FIG. 20 is a schematic diagram of a power supply system where an electrical energy conversion device is an alternating current-direct current charger.

[0080] FIG. 21 is a schematic diagram of a power supply system where an electrical energy conversion device is a direct current-direct current charger.

[0081] FIG. 22 is a control schematic of a power supply system supplying power to a home grid.

[0082] FIG. 23 is a schematic diagram of a remote device communicatively connected to a power supply system and a home grid.

[0083] FIG. 24 is a schematic diagram of a charging system according to an example of the present application.

[0084] FIG. 25 is a schematic diagram of identification of electrical energy receiving devices in a charging system according to an example of the present application.

[0085] FIG. 26 is a perspective view of an alternating current-direct current charger according to an example of the present application.

[0086] FIG. 27 is an electrical schematic of an electrical energy conversion device according to an example of the present application.

[0087] FIG. 28 is an electrical schematic of an electrical energy receiving device according to an example of the present application.

[0088] FIG. 29 is a communication schematic of a charging system according to an example of the present application.

[0089] FIG. 30 is a schematic diagram of an adapter mounted to an electric driving device.

[0090] FIG. 31 is a perspective view of a charging device.

[0091] FIG. 32 is a perspective view of a fixing device and a locking device of the charging device of FIG. 31.

[0092] FIG. 33 is a perspective view of the fixing device in FIG. 32.

[0093] FIG. 34 is a perspective view of the charging device of FIG. 31 from another angle.

[0094] FIG. 35 is a view illustrating a connection relationship between multiple charging devices and cables.

[0095] FIG. 36 is a perspective view of a charging device as another example.

[0096] FIG. 37 is a schematic view illustrating an installation relationship between the charging device and energy storage devices in FIG. 36.

[0097] FIG. 38 is a schematic view of the position of a circuit board of the charging device of FIG. 36.

[0098] FIG. 39 is a perspective view of the charging device of FIG. 36 from another angle.

[0099] FIG. 40 is a schematic view of multiple charging devices stacked in a first manner.

[0100] FIG. 41 is a schematic view of multiple charging devices stacked in a second manner.

[0101] FIG. 42 is a schematic view of a charging device with pull rods and wheels.

[0102] FIG. 43 is a schematic view of multiple stacked charging devices with pull rods and wheels.

[0103] FIG. 44 is a schematic diagram of multiple charging devices placed in a movable cabinet.

[0104] FIG. 45 is a schematic view of multiple charging devices stacked in a third manner.

[0105] FIG. 46 is a perspective view of a charging device according to an example of the present application.

[0106] FIG. 47 is a perspective view of the charging device of FIG. 46 from another angle.

[0107] FIG. 48 is a perspective view of the charging device of FIG. 47 from another angle.

[0108] FIG. 49 is a perspective view of a first charging device and a second charging device stacked.

[0109] FIG. 50 is a perspective view of a first charging device and a second charging device from another angle, which are stacked.

[0110] FIG. 51 is a schematic view of a power-receiving system to which an adapter in the present application is applicable.

[0111] FIG. 52 is a plan view of a charger as an example of the present application.

[0112] FIG. 53 is a perspective view of an outdoor work vehicle as an example of the present application.

[0113] FIG. 54 is a perspective view of an electrical device as an example of the present application.

[0114] FIG. 55 is a perspective view of an electrical device as another example of the present application.

[0115] FIG. 56 is a perspective view of an adapter as an example of the present application.

[0116] FIG. 57 is a perspective view of part of the internal structure of the adapter shown in FIG. 56.

[0117] FIG. 58 is an exploded view of the adapter shown in FIG. 56.

[0118] FIG. 59 is a schematic diagram of electric control of an adapter as an example of the present application.

DETAILED DESCRIPTION

[0119] Before any examples of this application are explained in detail, it is to be understood that this application is not limited to its application to the structural details and the arrangement of components set forth in the following description or illustrated in the above drawings.

[0120] In this application, the terms “comprising”, “including”, “having” or any other variation thereof are intended to cover an inclusive inclusion such that a process, method, article or device comprising a series of elements includes not only those series of elements, but also other elements not expressly listed, or elements inherent in the process, method, article, or device. Without further limitations, an element defined by the phrase “comprising a . . .” does not preclude the presence of additional identical elements in the process, method, article, or device comprising that element.

[0121] In this application, the term “and/or” is a kind of association relationship describing the relationship between associated objects, which means that there can be three kinds of relationships. For example, A and/or B can indicate that A exists alone, A and B exist simultaneously, and B exists alone. In addition, the character “/” in this application generally indicates that the contextual associated objects belong to an “and/or” relationship.

[0122] In this application, the terms “connection”, “combination”, “coupling” and “installation” may be direct connection, combination, coupling or installation, and may also be indirect connection, combination, coupling or installation. Among them, for example, direct connection means that two members or assemblies are connected together without intermediaries, and indirect connection means that two members or assemblies are respectively connected with at least one intermediate members and the two members or assemblies are connected by the at least one intermediate members. In addition, “connection” and “coupling” are not limited to physical or mechanical connections or couplings, and may include electrical connections or couplings.

[0123] In this application, it is to be understood by those skilled in the art that a relative term (such as “about”, “approximately”, and “substantially”) used in conjunction with quantity or condition includes a stated value and has a meaning dictated by the context. For example, the relative term includes at least a degree of error associated with the measurement of a particular value, a tolerance caused by manufacturing, assembly, and use associated with the particular value, and the like. Such relative term should also be

considered as disclosing the range defined by the absolute values of the two endpoints. The relative term may refer to plus or minus of a certain percentage (such as 1%, 5%, 10%, or more) of an indicated value. A value that did not use the relative term should also be disclosed as a particular value with a tolerance. In addition, “substantially” when expressing a relative angular position relationship (for example, substantially parallel, substantially perpendicular), may refer to adding or subtracting a certain degree (such as 1 degree, 5 degrees, 10 degrees or more) to the indicated angle.

[0124] In this application, those skilled in the art will understand that a function performed by an assembly may be performed by one assembly, multiple assemblies, one member, or multiple members. Likewise, a function performed by a member may be performed by one member, an assembly, or a combination of members.

[0125] In this application, the terms “up”, “down”, “left”, “right”, “front”, and “rear” and other directional words are described based on the orientation or positional relationship shown in the drawings, and should not be understood as limitations to the examples of this application. In addition, in this context, it also needs to be understood that when it is mentioned that an element is connected “above” or “under” another element, it can not only be directly connected “above” or “under” the other element, but can also be indirectly connected “above” or “under” the other element through an intermediate element. It should also be understood that orientation words such as upper side, lower side, left side, right side, front side, and rear side do not only represent perfect orientations, but can also be understood as lateral orientations. For example, lower side may include directly below, bottom left, bottom right, front bottom, and rear bottom.

[0126] In this application, the terms “controller”, “processor”, “central processor”, “CPU” and “MCU” are interchangeable. Where a unit “controller”, “processor”, “central processing”, “CPU”, or “MCU” is used to perform a specific function, the specific function may be implemented by a single aforementioned unit or a plurality of the aforementioned unit.

[0127] In this application, the term “device”, “module” or “unit” may be implemented in the form of hardware or software to achieve specific functions.

[0128] In this application, the terms “computing”, “judging”, “controlling”, “determining”, “recognizing” and the like refer to the operations and processes of a computer system or similar electronic computing device (e.g., controller, processor, etc.).

[0129] As shown in FIGS. 1 and 2, a charging system 100 includes at least an adapter 10, energy storage devices 20, and a charger 30. The energy storage devices 20 are detachably connected to a power tool or the adapter 10 to provide the stored electrical energy to the power tool or receive and store the electrical energy outputted by the charger 30. In some examples, the charging system 100 is used for providing power to the power tool.

[0130] The charger 30 includes a housing 34, an alternating current input power cord 31, an alternating current-direct current conversion module (not shown in the figure) disposed in the housing 34, and an output port 33. The alternating current input power cord 31 is used for connecting the mains power or other forms of alternating current power. The alternating current-direct current conversion

module is used for converting the connected mains power or other forms of alternating current power into direct current power. In some examples, the alternating current-direct current conversion module includes an alternating current-direction current (AC-DC) conversion circuit, such as an inverter circuit. The AC-DC conversion circuit may be a bridge rectifier circuit. The output port 33 is electrically connected to the adapter 10 to provide a charge voltage for the energy storage devices 20 mounted on the adapter 10. In this example, the output power of the charger 30 is greater than or equal to 500 W and less than or equal to 2000 W.

[0131] In some examples, the charger 30 can also be used for charging the power tool. The power tool is a riding vehicle. Specifically, the riding power tool includes, but is not limited to, a riding mower, a riding snow thrower, an all-terrain vehicle, and an electric motorcycle. In some examples, since a cable header of the charger 30 does not match a charging interface of the vehicle, the charger 30 charges the riding vehicle through an adapter connector.

[0132] As shown in FIGS. 1 and 3, the adapter 10 includes a housing 11, at least one coupling portion 12, and a handle 111 for a user to hold. The at least one coupling portion 12 is disposed on the housing 11 and used for mounting multiple energy storage devices 20. The adapter 10 can charge the energy storage devices 20 mounted on the at least one coupling portion 12. Specifically, the energy storage device 20 includes a battery pack. In some examples, the multiple energy storage devices 20 may be the same type of battery packs or may include two or more different types of battery packs. For example, the battery packs have different capacities and different rated (nominal) voltages. The output power of the adapter 10 is greater than or equal to 500 W and less than or equal to 2000 W.

[0133] As shown in FIGS. 3 to 5, the directions such as up, down, left, right, front, and rear as shown in FIGS. 3 and 5 are defined. The left and right direction is defined as the extension direction of the housing 11 of the adapter 10. The projection of the adapter 10 on a plane perpendicular to the left and right direction is substantially L-shaped. The energy storage devices 20 can be detachably coupled to the coupling portions 12 of the adapter 10 along the up and down direction or the front and rear direction. When one of the energy storage devices 20 is completely charged, the user may remove the fully charged energy storage device 20 and use the energy storage device 20 immediately without waiting for the charging of other energy storage devices 20 to be completed.

[0134] The handle 111 is disposed on the upper portion of the housing 11 of the adapter 10 and extends along the left and right direction. In some examples, the length L1 of the handle 111 in the left and right direction is greater than or equal to 70 mm. In some examples, the length L1 of the handle 111 in the left and right direction is greater than or equal to 80 mm. In some examples, as shown in FIG. 4, when the user holds the handle 111 to lift the adapter 10, the projection of the center of gravity G1 of the adapter 10 on the horizontal plane falls within the projection of the adapter 10 on the horizontal plane. The horizontal plane may be understood as the ground or ground plane. Specifically, the center of gravity G1 of the adapter 10 is located below the handle 111 in the up and down direction. As shown in FIG. 5, when the energy storage device 20 is mounted to at least one coupling portion 12 of the adapter 10, the center of

gravity G2 of the adapter is located on the front side of the handle 1111 in the front and rear direction.

[0135] In this manner, regardless of whether the energy storage device 20 is mounted on the adapter 10, when the user lifts the handle 1111 on the adapter 10, the center of gravity of the adapter or the whole of the adapter and the energy storage device can be close to the handle 1111 so that the user can save more effort and have a better experience when lifting the adapter or the energy storage device.

[0136] The adapter 10 further includes a first port and a second port. The first port is used for electrically and communicatively connecting a first charging device, and the second port is used for electrically and communicatively connecting a second charging device. In some examples, the first charging device is the charger, and the second charging device is the adapter. In some examples, the first charging device is a first adapter, and the second charging device is a second adapter different from the first adapter. It is to be understood that the user may use one or more adapters 10 to mount and charge the multiple energy storage devices 20 at the same time. After charging, the user may optionally detach one or more of the energy storage devices 20 from the adapter 10. Thus, two or more energy storage devices can be charged easily and conveniently.

[0137] In some examples, the charging system includes multiple cascaded devices. The multiple cascaded devices include at least one first charging device and at least one second charging device. The first charging device includes a housing and multiple accommodation portions disposed at the housing and used for mounting energy storage devices. The first charging device can charge the energy storage devices mounted to the accommodation portions. The first charging device further includes a connecting port for connecting the second charging device or the first charging device. The second charging device includes a housing and multiple accommodation portions disposed at the housing and used for mounting energy storage devices. The second charging device can charge the energy storage devices mounted to the accommodation portions. The second charging device further includes a connecting port for connecting the first charging device or the second charging device. In some examples, the multiple accommodation portions of the second charging device include at least one first accommodation portion and at least one second accommodation portion. Specifically, the first accommodation portion and the second accommodation portion have different shapes or dimensions. The connecting port may be the first port or the second port. In some examples, the accommodation portions of the first charging device have the same dimension as the first accommodation portion or the second accommodation portion of the second charging device. The cascading may be understood as multiple charging devices connected end-to-end sequentially in structure.

[0138] In some examples, the charging system includes multiple cascaded devices, where the multiple cascaded devices include at least one first charging device and at least one second charging device. The first charging device includes a housing and at least one first accommodation portion disposed at the housing and used for mounting a first energy storage device. The first charging device can charge the first energy storage device mounted to the first accommodation portion. A connecting port is used for connecting the second charging device or the first charging device. The second charging device includes a housing and at least one

second accommodation portion disposed at the housing and used for mounting a second energy storage device. The second charging device can charge the second energy storage device mounted to the second accommodation portion. A connecting port is used for connecting the first charging device or the second charging device. The first accommodation portion and the second accommodation portion have different shapes or dimensions. The cascading may be understood as multiple charging devices connected end-to-end sequentially in structure.

[0139] In some examples, the first accommodation portion is configured to mount the first energy storage device, and the second accommodation portion is configured to mount the second energy storage device. The capacity of the first energy storage device is greater than the capacity of the second energy storage device. The first energy storage device includes multiple first energy storage units, and the second energy storage device includes multiple second energy storage units. In some examples, the first energy storage units and the second energy storage units have different chemical properties. In some examples, the first energy storage unit is a lithium iron phosphate cell. The second energy storage unit is a ternary lithium cell. In some examples, the first charging device and the second charging device are the same. In some examples, the first charging device and the second charging device are different. Optionally, the first charging device or the second charging device may be one of a power adapter and a direct current-direct current charger. Optionally, the second charging device may also be a riding mower with a charging function.

[0140] In some examples, the first port and the second port may be interchanged. It is to be understood that the first port may be used for connecting both the first charging device and the second charging device. Similarly, the second port may be used for connecting both the first charging device and the second charging device.

[0141] FIG. 6 shows the charging system 100 as an example. The adapter 10 includes at least one first adapter 10a and at least one second adapter 10b (for example, two second adapters 10b are provided in this example). The first adapter 10a and one or more second adapters 10b are connected in series. The charging system 100 can charge a larger number of energy storage devices 20 by connecting a larger number of first adapters 10a and second adapters 10b in series. In some examples, the second adapter 10b and the first adapter 10a are the same. In some examples, the second adapter 10b and the first adapter are different.

[0142] Specifically, the first adapter 10a has a first port 111 and a second port 112. The second adapter 10b has a first port 111 and a second port 112. The first port 111 of the first adapter 10a is used for connecting the output port 33 of the charger 30. In some examples, the output port 33 of the charger 30 is movable. The second port 112 of the first adapter 10a is used for connecting the first port 111 of the second adapter 10b. The second port 112 of the second adapter 10b is used for connecting the first port of the other second adapter connected in series. According to the preceding connection manner, the charger 30 is electrically connected to the first adapter 10a and the second adapter 10b in sequence to charge the energy storage device 20 mounted on the first adapter 10a or the second adapter 10b.

[0143] In this example, the first port 111 of the first adapter 10a is an electrical energy input port, and the second port 112 of the first adapter 10a is an electrical energy output

port. The first port **111** of the second adapter **10b** is an electrical energy input port, and the second port **112** of the second adapter **10b** is an electrical energy output port. The first adapter **10a** and the second adapter **10b** are electrically connected through the electrical energy output port and the electrical energy input port. The current outputted by the charger **30** flows through the first adapter **10a** and the second adapter **10b** so that the first adapter **10a** and the second adapter **10b** implement the charging function. In some examples, the charger **30** can also be used for charging the power tool. In some examples, the power tool includes the riding vehicle. Specifically, the riding vehicle includes the riding mower, a stand-on mower, the all-terrain vehicle, and the electric motorcycle. Of course, the power tool may also be a snow thrower or a table tool. When the second port **112** (the electrical energy output port) of the first adapter **10a** is electrically connected to the first port (the electrical energy input port) of the second adapter **10b**, the relative positions of the first adapter **10a** and the second adapter **10b** are adjustable. Specifically, the first port **111** and the second port **112** of the first adapter **10a** are disposed outside the housing **11** of the first adapter **10a**. The first port **111** and the second port **112** of the second adapter **10b** are disposed outside the housing **11** of the second adapter **10b**. In some examples, the second port **112** of the first adapter **10a** extends out of the housing **11** of the first adapter **10a** through a wire, and the second port **112** of the first adapter **10a** is movable relative to the housing **11** of the first adapter **10a**. The first port **111** and the second port **112** of the second adapter **10b** extend out of the housing **11** of the second adapter **10b** through the wire, and the first port **111** and the second port **112** of the second adapter **10b** are movable relative to the housing **11** of the second adapter **10b**. The wire may be understood as a cable. The above describes the electrical connection between the first adapter **10a** and the second adapter **10b**. It is to be noted that a similar connection manner is adopted between multiple second adapters **10b** or multiple first adapters **10a** for electric power transmission.

[0144] It should be noted that the substantially same or similar structures in different examples or devices may use the same labels. For example, the first adapter **10a** and the second adapter **10b** may be different devices, which have adopted different labels. The first port in the first adapter **10a** and the first port in the second adapter **10b** have realized similar functions, and both may be labeled with **111** for now. Other examples also use similar rules for labeling, which will not be described in detail below.

[0145] In this manner, the first adapter **10a** and the second adapter **10b** are electrically connected and connected through a signal via the cable so that the flexibility of the first adapter and the second adapter during installation can be improved. On the other hand, the heat dissipation of the adapter and the energy storage device during a charging process can be facilitated so that the service life and charging performance of the adapter and the energy storage device can be improved.

[0146] In some examples, the adapter **10** further includes a fixing assembly. The fixing assembly is used for fixing the adapter **10** to another object. In some examples, the adapter **10** can be fixed on the wall by the fixing assembly. Specifically, the fixing assembly in this example may be a fixing structure that is removable without a tool, such as a snap-fit structure. Of course, the fixing assembly may be a fixing structure that requires a tool to remove, such as a screw and

a bolt. In this manner, the fixing assembly is provided to fix the adapter **10** on the wall or a work surface, which is convenient for the user to store and manage the adapter and saves space when the adapter is in use or not in use.

[0147] In some examples, the adapter **10** further includes a support assembly so that multiple adapters are stacked. Specifically, the support assembly is used for stacking the first adapter **10a** and the second adapter **10b** or stacking multiple second adapters **10b**. In this manner, the support assembly is provided to stack multiple adapters so that when the user uses more adapters to charge the energy storage devices, or when the adapters are idle, space can be saved and storage and management can be facilitated.

[0148] Next, representative circuits of various functions of the charging system **100** are described below.

[0149] As shown in FIG. 7, in this example, the energy storage device **20** is configured to be the battery pack. The battery pack has multiple cell units **21**, at least one battery controller **22**, at least one temperature sensor, and at least one battery memory **24**. The cell unit **21** in this example is a lithium-ion battery. The multiple cell units **21** are connected to battery positive terminals **20a** and battery negative terminals **20b**. The battery controllers **22** are connected to the cell units **21** and can detect the electrical parameters of the multiple cell units **21**. For example, the battery controllers **22** can detect the voltage and current of the cell units. In addition, the battery controller **22** may estimate the charging level and internal resistance of the cell unit **21** based on the detected voltage of the cell unit **21**. The temperature sensor is disposed near the multiple cell units **21** and detects the temperature of the multiple cell units **21**. The temperature sensor is connected to the battery controller **22**, and the battery controller **22** acquires the temperature of the cell unit **21** through the temperature sensor. The battery controller **22** is connected to a battery communication terminal **20c**.

[0150] The battery memory **24** is used for storing battery information of the energy storage device. The battery information includes, but is not limited to, at least one or any combination of two or more of the following: a single identification code of the energy storage device **20**, a model code of the energy storage device **20**, the rated voltage of the energy storage device **20**, the rated current of the energy storage device **20**, the maximum allowable temperature of the energy storage device **20**, the maximum current experienced by the energy storage device **20**, the highest temperature experienced by the energy storage device **20**, the start date of use of the energy storage device **20**, the total charge count of the energy storage device **20**, the total discharge count of the energy storage device **20**, the total discharge time of the energy storage device **20**, and the administrator of the energy storage device **20**. The battery memory **24** is connected to the battery controller **22**. The battery controller **22** can read, update, overwrite, and delete the battery information stored in the battery memory **24**.

[0151] The first adapter **10a** and the second adapter **10b** each include charging circuits **14** and at least one control circuit **13**. The charger **30** is connected to an external alternating current power supply through the alternating current input power cord **31**. The alternating current-direct current conversion module in the charger **30** converts the alternating current power of the external alternating current power supply into the direct current power. The direct current outputted by the charger **30** is supplied to each

charging circuit **14** of the first adapter **10a** and the second adapter **10b**. Each charging circuit **14** is connected to a corresponding battery interface **15** and controls the charge current supplied from the battery interface **15** to the corresponding energy storage device **20**. Specifically, a switch **141** is disposed between the first port **111** of the first adapter **10a** and each charging circuit **14**. Each switch **141** is controlled by a corresponding control circuit **13**. When one of the energy storage devices **20** is connected to the corresponding battery interface **15**, the control circuit **13** controls the corresponding switch **141** to be turned on, so as to charge the energy storage device **20**. When the energy storage device **20** is separated from the battery interface **15**, or when the energy storage device **20** is completely charged, the control circuit **13** turns off the switch **141**. It is to be noted that the first port and the second port are configured to transmit an electrical signal or a communication signal.

[0152] When the energy storage device **20** is electrically connected to the battery interface **15**, the control circuit **13** in the adapter is connected to the corresponding battery controller **22** to enable communication between the control circuit **13** and the battery controller **22**. The control circuit **13** acquires the battery information stored in the corresponding battery memory **24** from each battery controller **22**. In addition, the control circuit **13** may acquire the state indication of the corresponding energy storage device **20** from each battery controller **22**. For example, the state indication of each energy storage device **20** includes at least one or any combination of two or more of the following: the charging level of the energy storage device **20**, the output voltage of the energy storage device **20**, the internal resistance of the energy storage device **20**, the temperature of the energy storage device **20**, and the charge time of the energy storage device **20**. The control circuit **13** stores the acquired battery information and the state indication of each energy storage device **20** in the memory. The battery information and the state indication of each energy storage device **20** are stored in the corresponding memory together with the interface identification information of the battery interface **15** connected to the energy storage device **20**.

[0153] In some examples, the first adapter **10a** further includes a communication circuit (not shown in the figure). Specifically, the communication circuit is connected to the control circuit **13**. The communication circuit may be connected to a remote device in a wireless or wired manner to enable communication between the communication circuit and the remote device. The control circuit **13** may send and receive information or signals to and from the remote device via the communication circuit. The remote device referred to herein may be any of the remote devices described above, such as, but not limited to, a mobile phone, a smartphone, a tablet computer, or some other (for example, portable) computer devices.

[0154] Next, a charging control method of the charging system **100** is described below.

[0155] In this example, according to the information received by at least one of the first port and the second port of the adapter, the control circuit **13** controls whether the charging circuit **14** charges the energy storage device **20** mounted on the adapter **10**. Specifically, the control circuit is at least used for controlling the charge current between the first port and the second port to be turned on and off.

[0156] In some examples, the control circuit **13** controls the charging circuit **14** to simultaneously charge multiple

energy storage devices accommodated in multiple coupling portions. Specifically, when multiple energy storage devices are mounted on the adapter and need to be charged, the control circuit **13** controls the corresponding switch **141** to be turned on so that the direct current outputted by the charger flows through the charging circuit **14**, thereby charging the energy storage devices. It is to be understood that when the charger is connected to multiple first adapters and second adapters at the same time, multiple energy storage devices mounted on the adapters may be charged in the preceding control method.

[0157] In some examples, the control circuit **13** controls the charging circuit **14** to preferentially charge an energy storage device with the maximum remaining power among multiple energy storage devices accommodated in multiple coupling portions. Specifically, when multiple energy storage devices are mounted to the adapter, the battery communication terminal **20c** of the energy storage device is connected to a communication terminal in the battery interface **15** of the adapter, and the control circuit **13** is connected to the battery controller **22** to acquire the current remaining power of the multiple energy storage devices and control the switch **141** corresponding to the energy storage device with the maximum remaining power and is not fully charged to be turned on, thereby preferentially charging the corresponding energy storage device. After the energy storage device is completely charged, the control circuit **13** controls the charging circuit **14** to charge the energy storage device with the maximum current remaining power and is not fully charged again. This cycle continues until all the energy storage devices mounted to the adapter are completely charged. The advantage is that when the user urgently needs a fully charged energy storage device, the adapter can maximize the power allocated to the energy storage device with the maximum current remaining power for charging to satisfy the requirement of the user as soon as possible. It is to be understood that when the charger is connected to the first adapters and multiple second adapters at the same time, multiple energy storage devices mounted on the adapters may be charged in the preceding control method.

[0158] In some examples, the charging system further includes an operating member, where the control circuit **13** controls the charging circuit **14** to charge multiple energy storage devices coupled to multiple coupling portions according to a sequence set by the operating member. Specifically, the operating member may be configured to be a switch mounted on the housing of the adapter. The operating member may also be configured to be a wireless device. When the user needs to charge multiple energy storage devices, the operating member is controlled so as to charge the designated energy storage devices. The advantage is that the adapter is featured with strong operability and high charging flexibility, and according to the requirement of the user, the user can set the energy storage device that needs to be charged preferentially. It is to be understood that when the charger is connected to the first adapters and multiple second adapters at the same time, multiple energy storage devices mounted on the adapters may be charged in the preceding control method.

[0159] Specifically, the adapter may be configured to receive an operation instruction signal from the operating member. The operation instruction signal may be a charge start signal or a charge stop signal for instructing the adapter to perform various operations. In this case, the operation

instruction signal is preferably received together with the interface identification information of the battery interface **15** which is the target of the operation instruction signal. According to this configuration, the user may designate (select) a specific battery interface **15** and start or stop charging of the battery interface **15**.

[0160] In some examples, when the charger is connected to at least one first adapter and at least one second adapter at the same time, charging may be performed according to a preset sequence. Specifically, the energy storage device on the first adapter is charged preferentially. Specifically, the energy storage device on the second adapter is charged preferentially. The preset sequence may be set by the user. Of course, the preset sequence may be a priority among the adapters preset at the factory.

[0161] In some examples, the current outputted by the charger flows through the first adapter and the second adapter so that the first adapter and the second adapter implement the charging function. When the first adapter is partially damaged, the control circuit controls the current outputted by the charger to flow through the second adapter so that the second adapter implements the charging function.

[0162] The charging system **100** discussed above includes at least one first adapter **10a** and at least one second adapter **10b** and is configured such that the first adapter **10a** and the second adapter **10b** are electrically connected. However, in the present application, the adapters to which the charging system **100** is applicable are not limited to this configuration. For example, FIG. **8** shows another configuration of the adapter in the present application. The adapter in this example may be used as both a charger and an adapter.

[0163] The adapter of another configuration is now described with reference to FIGS. **8** to **13**.

[0164] The charging system **100** further includes an adapter **10c**, where the adapter **10c** includes the housing **11**, multiple coupling portions **12**, and the handle **1111** for the user to hold. The multiple coupling portions **12** are disposed on the housing **11** and used for mounting multiple energy storage devices. The coupling portions **12** include at least one first coupling portion **121** and at least one second coupling portion **122**. The handle **1111** is integrally formed with the housing **11**. Of course, the handle **1111** is also configured to be fixedly mounted to the housing **11** by means of assembly. In this example, the output power of the adapter **10c** is greater than or equal to 500 W and less than or equal to 2000 W.

[0165] In some examples, the adapter **10c** includes at least two second coupling portions **122**. Multiple second coupling portions **122** are arranged in a straight line parallel to the extension direction of the handle **1111**. The first coupling portion **121** and the second coupling portion **122** have the same interface. In some examples, the first coupling portion **121** and the second coupling portion **122** are different. Specifically, the first coupling portion **121** and the second coupling portion **122** have different interfaces.

[0166] In some examples, the energy storage devices mounted to the adapter **10c** may have different characteristics. Specifically, the energy storage devices include at least one first energy storage device **20a** and at least one second energy storage device **20b**. The capacity of the first energy storage device **20a** is greater than or equal to twice the capacity of the second energy storage device **20b**. The first energy storage device includes at least one first cell unit, the second energy storage device includes at least one second

cell unit, and the capacity of the first cell unit is at least four times the capacity of the second cell unit. The first energy storage device **20a** is mounted to the first coupling portion **121**, and the second energy storage device **20b** is mounted to the second coupling portion **122**. In this example, the first energy storage device **20a** has a first characteristic, and the second energy storage device **20b** has a second characteristic different from the first characteristic. The first characteristic or the second characteristic is at least one of the physical dimension, the shape, the chemical property, and the operational characteristic. The first energy storage device **20a** includes a lithium iron phosphate cell, and the second energy storage device **20b** includes a ternary lithium cell. Types of the first energy storage device include, but are not limited to, lithium ferrophosphate (LFP), Na-ion, solid/semi-solid, the 21700 battery, the 40135 battery, the 4680 battery, the 46950 battery, the 46120 battery, and 35184.

[0167] In some examples, the height of the first energy storage device **20a** is greater than the height of the second energy storage device **20b**. The height of the first energy storage device **20a** mounted on the adapter **10c** is greater than the height of the second energy storage device **20b** mounted on the adapter **10c**.

[0168] In this example, the first coupling portion **121** for mounting the first energy storage device **20a** and the second coupling portion **122** for mounting the second energy storage device **20b** are disposed on the front and rear sides of the adapter **10c**. It is to be understood that the first coupling portion **121** and the second coupling portion **122** are arranged back to back. The advantage of this design is that when the first energy storage device **20a** and the second energy storage device **20b** mounted on the adapter **10c** have different weights and volumes, the adapter on which the energy storage devices are mounted can be placed more stably. On the other hand, with such a design, the overall center of gravity can be closer to the handle, and the user can save more effort when lifting the handle **1111**.

[0169] In this example, the directions such as up, down, left, right, front, and rear as shown in FIG. **8** are defined. The left and right direction is defined as the extension direction of the housing **11** of the adapter **10c**. The projection of the adapter **10c** on a plane perpendicular to the left and right direction is substantially T-shaped. Referring to FIGS. **8** to **9B**, the first energy storage device **20a** and the second energy storage device **20b** can be detachably coupled to the first coupling portion **121** or the second coupling portion **122** of the adapter **10c** along the up and down direction or the front and rear direction. The handle **1111** is disposed on the upper portion of the housing **11** of the adapter **10c** and extends along the left and right direction. In some examples, as shown in FIG. **9A**, the ratio of the length **L2** of the handle **1111** in the left and right direction to the length **L3** of the adapter **10c** in the left and right direction is greater than or equal to 0.7 and less than or equal to 1.

[0170] In some examples, the projection of the handle **1111** on the bottom surface of the adapter **10c** is located between the projection of the first coupling portion **121** on the bottom surface of the adapter **10c** and the projection of the second coupling portion **122** on the bottom surface of the adapter. Specifically, the area of the projection of the second coupling portion **122** on the bottom surface of the adapter **10c** is greater than the area of the projection of the first coupling portion **121** on the bottom surface of the adapter **10c**.

[0171] In this example, the adapter 10c has the first port 111 and the second port 112. The first port 111 is used as the electrical energy input port, and the second port 112 is used as the electrical energy output port for connecting the second charging device. Specifically, when the first port 111 is not connected to an external power supply, the first energy storage device 20a charges the second energy storage device 20b. When the first port 111 is connected to the external power supply, the charging circuit charges the first energy storage device 20a or the second energy storage device 20b using the external power supply. When the first port 111 is connected to the external power supply, the charging circuit simultaneously charges the first energy storage device 20a and the second energy storage device 20b using the external power supply. In some examples, the charging circuit is further configured to charge the second energy storage device with larger remaining power among two second energy storage devices using the first energy storage device. It is to be noted that the external power supply may be the charger or other forms of external direct current power. In some examples, the external power supply is an alternating current power supply. The charger further includes the inverter circuit for converting the alternating current power supply into a direct current power supply. In some examples, the external power supply is a direct current power supply outputted by a high-power charger. The high-power charger may be understood as the charger 30 in the charging system 100. In some examples, the adapter 10c includes the fixing assembly and the support assembly. The structures and functions of the fixing assembly and the support assembly are described above and are not repeated here.

[0172] Next, the working principle when the adapter 10c is electrically connected to the charger 30 is described.

[0173] In some examples, as shown in FIG. 10, the charging system 100 includes the charger 30, the first adapter 10a, the second adapter 10b, and an adapter 10c. Specifically, the first port 111 of the adapter 10c is connected to the second port 112 of the second adapter 10b, so as to receive the direct current outputted from the charger 30 to charge the first energy storage device 20a and the second energy storage device 20b simultaneously. It is to be noted that the adapter 10c may also be connected between the first adapter 10a and the second adapter 10b. It is to be understood that the connection sequence of the first adapter 10a, the second adapter 10b, and the adapter 10c is not fixed.

[0174] In some examples, as shown in FIG. 11, the charging system 100 includes the charger 30 and the adapter 10c. The first port 111 of the adapter 10c is electrically connected to the output port 33 of the charger 30, so as to receive the direct current outputted from the charger 30 to charge the energy storage device 20a or 20b mounted on the adapter 10c.

[0175] In the preceding examples, the adapter 10c is electrically connected to the charger 30 and used for providing a charge voltage to the energy storage device mounted on the adapter 10c. In this example, the energy storage devices on the first adapter 10a, the second adapter 10b, and the adapter 10c are charged according to the preset sequence.

[0176] In some examples, the adapter 10c is used as the charger. The difference from the preceding two examples is that the adapter 10c can charge the second energy storage device 20b without being electrically connected to the charger 30. It is to be understood that the first energy storage

device 20a mounted on the adapter 10c charges the second energy storage device 20b mounted on the adapter 10c. The control circuit is configured to control the first energy storage device 20a to charge the second energy storage device 20b when an operating parameter of the first energy storage device 20a is greater than or equal to a preset value. The operating parameter includes, but is not limited to, the state of charge and the state of health of the first energy storage device 20a.

[0177] In some examples, the first energy storage device 20a simultaneously charges multiple second energy storage devices 20b mounted on the adapter 10c. In some examples, the first energy storage device 20a preferentially charges the second energy storage device 20b with the maximum remaining power among the multiple second energy storage devices 20b. In some examples, the first energy storage device 20a charges the multiple second energy storage devices 20b according to the preset sequence.

[0178] The difference between a second charging circuit 14c of the adapter 10c in this example and the charging circuits 14 in the first adapter 10a and the second adapter 10b is that the second charging circuit 14c further includes a direct current-direct current conversion circuit 142. In this example, the output power of the charging circuit 14 is greater than or equal to 500 W and less than or equal to 2000 W.

[0179] Next, the working manner of the adapter 10c in this example is described in conjunction with FIG. 12.

[0180] After the adapter 10c is electrically connected to the charger 30, the control circuit 13 controls the switches 141 to be turned on. Part of the direct current outputted by the charger 30 charges the first energy storage device 20a through the charging circuit 14. The other part of the direct current outputted by the charger 30 charges the second energy storage device 20b through the second charging circuit 14c.

[0181] When the adapter 10c is not connected to the charger 30, the control circuit 13 controls the switches 141 to be turned on, and the direct current outputted by the first energy storage device 20a charges the second energy storage device 20b through the second charging circuit 14c.

[0182] In some examples, when the adapter 10c is not connected to the charger 30, while charging the second energy storage device 20b, the first energy storage device 20a can also charge the energy storage devices on other adapters electrically connected to the charger 30. Specifically, as shown in FIG. 10, when the adapter 10c is not connected to the charger 30, the first energy storage device 20a can also charge the energy storage device mounted on the first adapter 10a or the second adapter 10b.

[0183] In some examples, part of the direct current outputted by the charger 30 charges the first energy storage device 20a through the charging circuit 14. The other part of the direct current outputted by the charger 30 charges the second energy storage device 20b through the second charging circuit 14c. It is to be noted that the preceding direct current does not flow through the direct current-direct current conversion circuit 142. When the adapter 10c is not connected to the charger 30, the direct current outputted by the first energy storage device 20a charges the second energy storage device 20b through the second charging circuit 14c including the direct current-direct current conversion circuit 142.

[0184] In some examples, the direct current-direct current conversion circuit 142 adopts the topology as shown in FIG. 13 and converts the direct current outputted by the charger 30 or the direct current U1 outputted by the first energy storage device 20a into the direct current U2, so as to provide a charge voltage to the second energy storage device 20b. The direct current-direct current conversion circuit 142 includes at least a switching tube Q1, a switching tube Q2, a switching tube Q3, a switching tube Q4, a capacitor C1, a capacitor C2, and an inductor L. The control circuit 13 is separately connected to the gates of the switching tube Q1, the switching tube Q2, the switching tube Q3, and the switching tube Q4 and is used for controlling the switching tube Q1, the switching tube Q2, the switching tube Q3, and the switching tube Q4 to be turned on and off, thereby implementing a voltage conversion function of the direct current-direct current conversion circuit. The direct current-direct current conversion circuit 142 in this example may be understood as a four-switch buck-boost (FSBB) circuit. The FSBB circuit is adopted so that soft switching is achieved, thereby improving the conversion efficiency of the direct current-direct current conversion circuit.

[0185] Next, the topology and control method of the FSBB circuit are described in detail in conjunction with FIGS. 13 to 15.

[0186] Specifically, as shown in FIGS. 14A and 15, the control circuit controls the switching tubes Q1 and Q4 to be turned on and the switching tubes Q2 and Q3 to be turned off during the T1 period, the inductor voltage is U1, the rising slope of the inductor current i_L is $U1/L$, and the output current i_o is 0. The preceding process is referred to as the FSBB circuit in working mode A. As shown in FIGS. 14B and 15, the control device controls the switching tubes Q1 and Q2 to be turned on and the switching tubes Q3 and Q4 to be turned off during the T2 period, the inductor voltage is $U1-U2$, the rising slope of the inductor current i_L is $U1-U2/L$, and the output current i_o is equal to the inductor current i_L . The preceding process is referred to as the FSBB circuit in working mode B. As shown in FIGS. 14C and 15, the control device controls the switching tubes Q2 and Q3 to be turned on and the switching tubes Q1 and Q4 to be turned off during the T3 period, the inductor voltage is U2, the rising slope of the inductor current i_L is $-U1/L$, and the output current i_o is equal to the inductor current i_L . The preceding process is referred to as the FSBB circuit in working mode C. As shown in FIGS. 14D and 15, the control device controls the switching tubes Q3 and Q4 to be turned on and the switching tubes Q1 and Q2 to be turned off during the T4 period, the inductor voltage is 0, the slope of the inductor current i_L is 0, and the output current i_o is equal to 0. The preceding process is referred to as the FSBB circuit in working mode D. In one cycle, the direction of the inductor current changes twice, and the negative current I_d exists to achieve the zero-voltage turn-on of the switching tubes Q1 and Q4.

[0187] The preceding four working modes may form different working modes of the FSBB circuit. For example, mode A and mode B may be equivalent to the boost working mode, and mode B and mode C may be equivalent to the buck working mode. The inductor voltage of mode D is 0, and the operating frequency of the circuit may be maintained by increasing or decreasing the duration of mode D.

[0188] Referring to FIGS. 16 to 18, the present application further discloses a power supply system 200, specifically an

alternating current and direct current bidirectional power supply system. The power supply system 200 includes a power supply 210, an electrical energy conversion device 220, and an electrical energy receiving device 230. The electrical energy conversion device 220 is configured to be electrically connected to the power supply 210 and convert the characteristic of the current drawn from the power supply 210. Multiple electrical energy receiving devices 230 are electrically connected to the electrical energy conversion device 220 to acquire electrical energy from the power supply 210.

[0189] In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 10 W and less than or equal to 10 KW. In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 10 W and less than or equal to 250 W. In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 250 W and less than or equal to 550 W. In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 500 W and less than or equal to 10 kW. In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 1000 W and less than or equal to 10 kW. In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 1 kW and less than or equal to 3 kW. In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 3 KW and less than or equal to 5 kW. In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 5 kW and less than or equal to 7 kW. In some examples, the output power of the electrical energy conversion device 220 is greater than or equal to 7 kW and less than or equal to 10 KW. In some examples, the output power of the electrical energy conversion device 220 may further include 10 W, 50 W, 100 W, 200 W, 500 W, 800 W, 1 KW, 1.5 KW, 2 KW, 2.5 KW, 3 KW, 3.5 KW, 4 KW, 4.5 kW, 5 KW, 5.5 kW, 6 KW, 6.5 kW, 7 KW, 7.5 kW, 8 KW, 8.5 kW, 9 KW, 9.5 kW, and 10 kW.

[0190] The source of the power supply 210 includes at least one of a solar panel 211, a charging pile for an electric vehicle 212, a car cigarette lighter 213, a mains socket 214, or a battery pack 215. The output power of the solar panel 211 is greater than or equal to 100 W. Of course, the solar panel 211 may have a high power output capability and the output power of the solar panel 211 is greater than or equal to 1000 W. The output of the charging pile for an electric vehicle 212 supplies power to the electrical energy receiving device 230 in the power supply mMm system 200 through the electrical energy conversion device 220. In some examples, the output power of the charging pile for an electric vehicle 212 is greater than or equal to 1000 W. The output power of the car cigarette lighter 213 is greater than or equal to 100 W. Of course, the car cigarette lighter 213 may also have a high power output capability and the output power of the car cigarette lighter 213 is greater than or equal to 1000 W. In some examples, the car cigarette lighter 213 may be connected to the direct current-direct current charger, so as to output electrical energy in an in-vehicle power supply to supply power to the battery pack or the direct current appliance electrically connected to the direct current-direct current charger. The battery pack 215 may be an external battery pack and, of course, the battery pack 215 may also be a built-in battery pack of the riding vehicle.

Types of the battery pack include, but are not limited to, LFP, Na-ion, solid/semi-solid, the 21700 battery, the 40135 battery, the 4680 battery, the 46950 battery, the 46120 battery, and 35184.

[0191] The electrical energy conversion device 220 includes at least one of the alternating current-direct current charger, the bidirectional power supply, the direct current-direct current charger, and the riding vehicle. Specifically, the alternating current-direct current charger is at least used for converting the alternating current connected to the mains socket into the direct current for the electrical energy receiving device 230 to use. In some examples, the output power of the alternating current-direct current charger is greater than or equal to 100 W. In some examples, the output power of the alternating current-direct current charger is greater than or equal to 210 W. In some examples, the alternating current-direct current charger also has a relatively high power output capability, and the output power of the alternating current-direct current charger is greater than or equal to 1000 W. The direct current-direct current charger is at least used for converting the direct current from the solar panel, the electric vehicle, the car cigarette lighter, or the battery pack into the direct current adapted to the electrical energy receiving device. When the riding vehicle is used as the electrical energy conversion device 220, the built-in battery pack may be used as the power supply to convert the direct current in the energy storage device into the alternating current for output.

[0192] The electrical energy receiving device 230 includes at least one of a device equipped with a battery pack and an alternating current appliance 235. The alternating current appliance may be a household appliance such as a television or a grill, and the power supply 210 outputs the alternating current through the electrical energy conversion device 220 to supply electrical energy to the television or the grill. In some examples, the device equipped with the battery pack includes, but is not limited to, at least one of the direct current-direct current charger, the power adapter, the riding vehicle, or an outdoor traveling device. The direct current-direct current charger is connected to the riding vehicle. In some examples, the electrical energy receiving device 230 may include multiple devices equipped with battery packs, where the devices are connected in series. Multiple energy storage devices are detachably mounted on the electrical energy receiving device. The multiple energy storage devices include at least the first energy storage device and the second energy storage device. When the power supply is powered on, the power supply charges the first energy storage device and the second energy storage device; and when the power supply is cut off, the first energy storage device charges the second energy storage device.

[0193] In some examples, the first energy storage device includes at least one battery pack with a large capacity, where the energy of the battery pack is greater than or equal to 1 kWh and less than or equal to 30 kWh. In some examples, the energy of the battery pack is equal to 2 kWh. In some examples, the energy of the battery pack is greater than or equal to 3 kWh. In some examples, the energy of the battery pack is greater than or equal to 4 kWh. In some examples, the energy of the battery pack is greater than or equal to 5 kWh. In some examples, the energy of the battery pack is greater than or equal to 6 kWh. In some examples, the second energy storage device includes at least one battery pack with a small capacity, where the energy of the

battery pack is greater than or equal to 100 Wh and less than or equal to 2 kWh. The energy of the battery pack is greater than or equal to 0.1 kWh. In some examples, the energy of the battery pack is greater than or equal to 0.4 kWh. In some examples, the energy of the battery pack is greater than or equal to 0.6 kWh. In some examples, the energy of the battery pack is greater than or equal to 1 kWh.

[0194] In some examples, the device equipped with the battery pack includes a first power adapter 231, a second power adapter 232, a direct current-direct current charger 233, and a riding vehicle 234 connected in sequence. Of course, it is also feasible that the first power adapter 231, the direct current-direct current charger 233, the second power adapter 232, and the riding vehicle 234 are connected in sequence. The electrical energy conversion device 220 exchanges control information and other information with the first power adapter 231, the second power adapter 232, and the direct current-direct current charger 233 through power carrier communication or wireless communication.

[0195] In some examples, the battery pack includes at least a cell unit, and at least one electrical energy receiving device 230 is capable of achieving the charging rate of the cell unit greater than or equal to 8 C. In some examples, the at least one electrical energy receiving device 230 is capable of achieving the charging rate of the cell unit greater than or equal to 9 C. In some examples, the at least one electrical energy receiving device 230 is capable of achieving the charging rate of the cell unit greater than or equal to 10 C.

[0196] Specifically, the first power adapter 231 is used for mounting a first battery pack 231a. The second power adapter 232 is used for mounting a second battery pack 232a. One or more first battery packs 231a or second battery packs 232a are provided. Multiple first battery packs 231a or second battery packs 232a may have the same or different types, chemical parameters, or characteristic parameters. One or more direct current-direct current chargers 233 may be provided. The direct current-direct current charger 233 is used for mounting a first energy storage device 233a or a second energy storage device 233b. When the direct current-direct current charger 233 is electrically connected to the power supply system 200, the electrical energy outputted by the electrical energy conversion device 220 can be received so as to charge the first energy storage device 233a or the second energy storage device 233b connected on the direct current-direct current charger 233. When the direct current-direct current charger 233 cannot acquire electrical energy from the power supply system 200, the first energy storage device 233a may charge the second energy storage device 233b. In some examples, the first energy storage device 233a can simultaneously charge multiple second energy storage devices 233b mounted on the direct current-direct current charger 233. In some examples, the first energy storage device is mounted to a first direct current-direct current charger, and the second energy storage device is mounted to a second direct current-direct current charger. The first energy storage device can charge the second energy storage device. In some examples, the first energy storage device is mounted to the riding vehicle, and the second energy storage device is mounted to the direct current-direct current charger. The first energy storage device charges the second energy storage device.

[0197] A bidirectional charging system further includes a wireless communication module communicatively connected to the remote device. The wireless communication

module is configured to receive a signal outputted by the remote device to control the first energy storage device to charge the second energy storage device. In some examples, the wireless communication module is configured to receive the signal outputted by the remote device to control the first energy storage device on the direct current-direct current charger or the first energy storage device on the riding vehicle to charge the second energy storage device. In some examples, the remote device is configured to acquire a signal outputted by the wireless communication module to display remaining charge time or charging failure information.

[0198] In some examples, the riding vehicle or the outdoor traveling device can receive the direct current outputted by the electrical energy conversion device **220** to charge the battery pack built in the riding vehicle or the outdoor traveling device. In some examples, the riding vehicle can also output the direct current to supply power to other electrical energy receiving devices **230** connected to the electrical energy conversion device **220**. The riding vehicle includes a first interface, where the first interface is configured to be electrically connected to the electrical energy output port of the first adapter or the electrical energy output port of the second adapter. The connection manner between the riding vehicle and the electrical energy output port includes, but is not limited to, a wired connection, a wireless connection, and a hard connection.

[0199] In some examples, when the electrical energy conversion device **220** in the power supply system is configured to be the riding vehicle, the power supply **210** is the battery pack disposed on the riding vehicle. In this example, the electrical energy conversion device **220** includes a bidirectional inverter module. Through the bidirectional inverter module, the electrical energy in the battery pack may output the alternating current for the television and a barbecue grill to use and output the direct current for the battery packs mounted on the first adapter or the second adapter and the direct current-direct current charger to use. The riding vehicle can also output the alternating current to directly supply power to the household appliance such as the television or the grill. In some examples, the riding vehicle can also output the direct current adapted to the mobile phone or a 4G gateway to charge the mobile phone or the 4G gateway.

[0200] Specifically, the riding vehicle or the outdoor traveling device includes a body; a traveling wheel set including traveling wheels supporting the body; a drive assembly mounted to the body and including an electric motor driving the traveling wheels to rotate; and an energy storage device detachably mounted to the body and configured to supply power to at least the electric motor. The outdoor traveling device further includes a bidirectional inverter module, where the bidirectional inverter module is configured to be capable of being electrically connected to the energy storage device and outputting the electrical energy of the energy storage device in the form of the alternating current. The energy storage device in the riding vehicle or outdoor traveling device may be detachably installed on the body, or may be fixedly installed on the body.

[0201] Referring to FIGS. **19A** and **19B**, the riding vehicle **234** is connected to a bidirectional inverter module **234a** for converting the electrical energy or has a built-in bidirectional inverter module **234a** for converting the electrical energy. The bidirectional inverter module is disposed in the body, or the bidirectional inverter module is independent of the body and detachably connected to an energy storage

device **234b**. The riding vehicle **234** includes a first interface **2341**, a second interface **2342**, and a third interface **2343** electrically connected to the bidirectional inverter module. The first interface **2341** is used for receiving the electrical energy from the power supply system or outputting the electrical energy to the power supply system. Specifically, the first interface **2341** is used for receiving the direct current outputted by the electrical energy conversion device **220**, so as to charge the energy storage device. Of course, the first interface **2341** can also output the direct current to supply electrical energy to other electrical energy receiving devices **230** in the power supply system **200**. The second interface **2342** is used for outputting the electrical energy in the energy storage device in the form of the alternating current, so as to supply power to the household appliance such as the television or the barbecue grill. The third interface **2343** is used for outputting the direct current so as to charge the mobile phone or the 4G gateway. The third interface may be a Universal Serial Bus (USB) interface, specifically Type-A or Type-C. The energy storage device in the riding vehicle may include multiple battery packs of the same type or may include battery packs of different types. For example, the energy storage device includes a battery pack with a lithium iron phosphate cell and a battery pack with a ternary lithium cell.

[0202] In some examples, as shown in FIG. **20**, when the electrical energy conversion device **220** in the power supply system is configured to be the alternating current-direct current charger, the electrical energy in the power supply **210** can be transmitted to the first power adapter **231**, the second power adapter **232**, the direct current-direct current charger **233**, and the riding vehicle **234**. Of course, the electrical energy in the power supply **210** can also be transmitted to the household appliance such as the television or the barbecue grill. The power supply **210** may be the solar panel **211**, the charging pile for an electric vehicle **212**, the car cigarette lighter **213**, or the mains socket **214**. It is to be noted that the alternating current-direct current charger also has a direct current input port for receiving the direct current, performing voltage conversion, and outputting the direct current adapted to the electrical energy receiving device. Of course, the alternating current-direct current charger may not have the direct current input port, and the power supply that can be connected is the mains socket **214**.

[0203] In some examples, as shown in FIG. **21**, when the electrical energy conversion device **220** in the power supply system is configured to be the direct current-direct current charger, the electrical energy in the power supply **210** can be transmitted to the first energy storage device and the second energy storage device electrically connected to the direct current-direct current charger and the riding vehicle **234**. The power supply **210** may be the solar panel **211**, the charging pile for an electric vehicle **212**, or the car cigarette lighter **213**. It is to be noted that the power supply **210** may also specifically be the battery pack. The battery pack may be the first energy storage device, the second energy storage device, or the battery pack on the riding vehicle.

[0204] In some examples, when the electrical energy conversion device **220** in the power supply system is configured to be the bidirectional power supply, the electrical energy in the power supply **210** can be transmitted to the first power adapter **231**, the second power adapter **232**, the direct current-direct current charger **233**, and the riding vehicle **234**. The power supply **210** may be the solar panel **211**, the

charging pile for an electric vehicle **212**, the car cigarette lighter **213**, or the mains socket **214**. It is to be noted that the battery pack may also be used as the power supply, and the electrical energy of the battery pack passes through the bidirectional power supply and is converted into the alternating current for the alternating current appliance to use. The battery pack may be mounted on the first adapter, the second adapter, the direct current-direct current charger, or the riding vehicle.

[0205] As shown in FIG. 17, in some examples, the electrical energy conversion device **220** further includes an alternating current output port **221**. When the power supply is connected, the electrical energy can supply power to the alternating current appliance **235** connected to the alternating current output port **221** through the alternating current output port **221**. When the power supply is cut off, the electrical energy of the multiple energy storage devices mounted on the electrical energy receiving device is outputted in the form of the alternating current through the alternating current output port **221**. The alternating current output port **221** includes at least one of a mains 2 pin socket and a mains 3 pin socket.

[0206] In some examples, the power supply system supplies electrical energy to the alternating current appliance to ensure the normal operation of the alternating current appliance. When the mains interface cannot be connected to the mains power, the battery pack in the charging system is discharged to ensure the normal use of the alternating current appliance. The battery pack includes the battery pack mounted on the first adapter, the second adapter, the direct current-direct current charger, or the riding vehicle. Specifically, the bidirectional power supply converts the direct current outputted by the battery pack into the alternating current adapted to the alternating current appliance. Of course, the discharge power of the power supply system depends on the types and number of battery packs. Through the remote device, the user may set the discharge sequence of multiple battery packs in the charging system and provide a reminder of the discharge countdown, that is, the remaining battery life. When the mains interface can be connected to the mains power, the charging system performs discharging on the alternating current appliance through the bidirectional power supply and the charging pile for an electric vehicle to ensure the normal use of the alternating current appliance. In some examples, when more alternating current appliances exist and the required power is relatively large, the battery pack in the electrical energy receiving device may supply power to the alternating current appliances through the bidirectional power supply.

[0207] The charging system in the present application also has the function of discharging while charging. For example, the electrical energy conversion device receives the mains power connected to the mains interface and transmits the received electrical energy to the adapter, the direct current-direct current charger, and the riding vehicle, so as to charge the battery packs on the adapter, the direct current-direct current charger, and the riding vehicle. At the same time, the electrical energy conversion device also outputs the received mains power for the alternating current appliance to use. For example, the electrical energy conversion device receives the electrical energy outputted by the solar panel or the charging pile for an electric vehicle and transmits the received electrical energy to the adapter, the direct current-direct current charger, and the riding vehicle, so as to charge

the battery packs on the adapter, the direct current-direct current charger, and the riding vehicle. At the same time, the battery packs are discharged to the peripheral device. For example, the electrical energy conversion device receives the electrical energy outputted by the charging pile for an electric vehicle and transmits the electrical energy to the battery packs mounted on the adapter, the direct current-direct current charger, and the riding vehicle, so as to charge the battery packs. At the same time, the battery packs are discharged to the peripheral device. For example, the solar panel outputs the electrical energy to the adapter or the direct current-direct current charger to charge multiple battery packs electrically connected to the adapter or the direct current-direct current charger. At the same time, the first energy storage device electrically connected to the direct current-direct current charger can charge the second energy storage device.

[0208] As shown in FIG. 22, the alternating current output port **221** can be connected to a home grid **300** to supply power to the home grid **300**. The home grid **300** has an alternating current input interface **310** for connecting the alternating current output port **221**, a 56V power bus **22**, an alternating current device interface **13** for connecting an alternating current device **131**, and a direct current device interface **14** for connecting a direct current device **143** or an energy storage device **142**. The direct current device interface **14** includes a power converter or a bidirectional direct current converter **141**. The home grid **300** further includes a bidirectional inverter device **15** electrically connected to a mains interface **11** and used for processing the mains power connected to the mains interface **11**. Specifically, the bidirectional inverter device **15** is a bidirectional inverter for converting the inputted alternating current into the direct current or converting the inputted direct current into the alternating current. The home grid **300** further includes a solar interface **16** electrically connected to the power bus **22**.

[0209] As shown in FIG. 23, the power supply system **200** further includes a wireless communication module **240** communicatively connected to a remote device **400**, where the wireless communication module is configured to be communicatively connected to the home grid. The remote device is capable of recording the number of times and the duration that the home grid acquires the electrical energy through the alternating current input interface. The remote device is configured for the user to set which circuit in the home grid is prioritized for power supply. The remote device **400** may be a mobile phone, a computer, a tablet computer, or another device.

[0210] In some examples, as shown in FIG. 24, the charging system **200** includes the electrical energy conversion device **220** and multiple electrical energy receiving devices **230** connected in cascade, and the charging system **200** is configured to charge battery packs electrically connected to the electrical energy receiving devices **230**. The electrical energy conversion device **220** includes a controller **222**, a power interface **223** for connecting an external power supply, a connecting port **224** for connecting the electrical energy receiving device **230**, and a conversion circuit **225** electrically connected to the power interface **223** and the connecting port **224** separately. The electrical energy receiving device **230** includes a controller **236**, an electrical energy input port **237a** for connecting the electrical energy conversion device **220** or another electrical energy receiving device and an electrical energy output port **237b** for connecting

another electrical energy receiving device. The controller **222** in the electrical energy conversion device **220** communicates directly or indirectly with controllers **236** in the multiple electrical energy receiving devices **230** and records identities of the multiple electrical energy receiving devices **230**. Specifically, the electrical energy conversion device **220** controls, based on the recorded identities, the corresponding electrical energy receiving devices **230** on a link to be charged in an order of the identities. In some examples, the order of the identities is to start from an electrical energy receiving device **230** closest to the electrical energy conversion device **220** and number the electrical energy receiving devices **230** in sequence according to an order in which they are physically connected. Numbers may be Arabic numerals in ascending order, for example, 1, 2, and 3. Of course, the numbers may be 26 English letters, for example, a, b, and c. The electrical energy conversion device **220** includes at least one of the alternating current-direct current charger, the bidirectional power supply, the direct current-direct current charger, and the riding vehicle. The electrical energy receiving device **230** includes, but is not limited to, at least one of the direct current-direct current charger, the power adapter, the riding vehicle, or the outdoor traveling device. The riding vehicle includes the riding mower, the stand-on mower, and the all-terrain vehicle.

[0211] In some examples, the multiple electrical energy receiving devices may be connected in parallel.

[0212] Specifically, as shown in FIG. 25, after powered on, the electrical energy conversion device **220** provides voltages for a communication bus **250** and a power bus **260** to power on the multiple electrical energy receiving devices **230** connected in cascade. Specifically, the electrical energy conversion device **220** gives a level signal (for example, a pulse-width modulation (PWM) wave) to trigger an electrical energy receiving device **230a** physically connected thereto to perform identification. After receiving a PWM signal, the electrical energy receiving device **230a** completes the identification as No. 01, reports a message including an identity to the communication bus **250**, and outputs the PWM wave to the next physically connected electrical energy receiving device **230b**. After receiving the PWM signal from the electrical energy receiving device **230a**, the electrical energy receiving device **230b** completes the identification as No. 02, reports an identity to the communication bus, and outputs the PWM wave to the next physically connected electrical energy receiving device **230c**. The multiple electrical energy receiving devices connected in cascade perform the identification in sequence. In some examples, a maximum number of the preceding identities may be 20, 30, 40, or 50. Of course, the maximum number may be a larger or smaller value. It is to be noted that in a process of recording the identities, if no fault occurs, the maximum number of the identities may be understood as the number of electrical energy receiving devices in the charging system. In addition to the identity, the message reported by each electrical energy receiving device further includes information indicating whether it is the last electrical energy receiving device. Specifically, each electrical energy receiving device **230** includes an End Detect pin. Whether an electrical energy receiving device is inserted at the next stage depends on whether the End Detect pin of the electrical energy receiving device **230** at this stage is triggered.

[0213] In some examples, the communication bus may be a controller area network (CAN) bus. The electrical energy

conversion device **220** may communicate with the electrical energy receiving devices **230** via the CAN bus. As for downlink information data, the remote device sends an instruction to the electrical energy conversion device through the wireless communication module, and the wireless communication module transmits the instruction to the electrical energy receiving devices through a CAN communication protocol. As for downlink information data, the electrical energy receiving devices transmit information to the electrical energy conversion device for summary through the CAN communication protocol, and the electrical energy conversion device reports the information through the wireless communication module to display relevant information on the remote device. In some examples, the communication bus may be a 485 bus or a USB-PD bus.

[0214] In addition to the numbering of the identities according to the order in which the electrical energy receiving devices are physically connected in the preceding example, the charging system may perform the identification in other forms. In some examples, the charging system charges according to the most recent charging order. In some examples, the charging system charges based on an optimal charging scheme. The preceding optimal charging scheme may be a scheme ensuring that the user can obtain relatively many fully charged battery packs in a shortest time, that is, preferentially charging battery packs with relatively high power. The preceding optimal charging scheme may be a scheme ensuring that the user can obtain relatively many partially charged battery packs in the shortest time, that is, preferentially charging battery packs with relatively low power. In some examples, the charging system charges the battery packs in the system based on a random principle.

[0215] In some examples, when one or some of the electrical energy receiving devices **230** fail in the process of the electrical energy conversion device **220** recording the identities of the multiple electrical energy receiving devices **230**, the electrical energy conversion device **220** starts from the last electrical energy receiving device **230** to perform reverse numbering, identifies a number of an identity of the electrical energy receiving device that fails, and reports the identity to the remote device. Specifically, in the process of the electrical energy conversion device **220** performing the identification on the multiple electrical energy receiving devices **230**, when one or some of the electrical energy receiving devices **230** are damaged and cannot transmit the PWM signal, the reverse numbering is performed through a terminal detection function until an abnormal electrical energy receiving device **230** is encountered so that other normal electrical energy receiving devices **230** can be prevented from being affected.

[0216] In some examples, the power-on of the power interface **223** triggers the electrical energy conversion device **220** to re-perform the operation of recording the identities of multiple electrical energy receiving devices **230**. Of course, after powered on, the electrical energy conversion device **220** identifies whether the electrical energy receiving devices **230** and the battery packs thereon are allowed to access and starts to number the electrical energy receiving devices **230** on the link after completing a handshake protocol. In some examples, when a new electrical energy receiving device **230** is added to the charging system **200**, the electrical energy conversion device **220** is triggered to re-perform the operation of recording the identities of multiple electrical energy receiving devices **230**. In

some examples, when the charging system 200 has a fault, the electrical energy conversion device 220 is triggered to re-perform the operation of recording the identities of multiple electrical energy receiving devices 230. Of course, the electrical energy conversion device 220 initiates a renumbering operation after completing the current round of charging.

[0217] In some examples, the charging system 200 also has an information polling function. After recording the identities of the multiple electrical energy receiving devices 230, the electrical energy conversion device 220 collects information from the electrical energy receiving device 230 corresponding to each identity in sequence, summarizes the information or records the information of a single electrical energy receiving device 230, and reports the information to an Internet of things (IoT) module to display the relevant information on the remote device. In the polling process, the electrical energy conversion device 220 queries whether the bus is connected to a new electrical energy receiving device 230. If so, the electrical energy conversion device re-records the identities of multiple electrical energy receiving devices 230 connected in cascade when being idle. When the power supply system 200 is in a standby state, the electrical energy conversion device 220 identifies, in the polling process, that an electrical energy receiving device 230 needs to be charged and charges the electrical energy receiving device 230 to be charged. Specifically, the electrical energy conversion device 220 periodically inquires whether the electrical energy receiving devices 230 with the identities need to be charged. When a battery pack is inserted into the electrical energy receiving device 230, the electrical energy receiving device 230 replies that it needs to be charged. The electrical energy receiving device 230 detects a voltage of the battery pack and transmits information such as the voltage of the battery pack and a required current to the electrical energy conversion device 220. The electrical energy conversion device 220 charges the electrical energy receiving device 230 according to the transmitted information.

[0218] In some examples, when the multiple electrical energy receiving devices 230 are connected in the charging system 200, the harness is long, and a relatively large voltage drop on the harness causes an electrical energy receiving device at the rear end to fail to be charged with full power, affecting the charging efficiency. Therefore, the electrical energy conversion device 220 improves the charging efficiency by increasing an output voltage. The electrical energy conversion device 220 adjusts an output voltage and current according to information about the electrical energy receiving device 230. For an electrical energy receiving device 230 with a long harness, the output voltage is increased and the current is appropriately reduced to prevent rated power from being exceeded.

[0219] In some examples, the controller in the electrical energy conversion device adjusts the output of the connecting port based on the number of the electrical energy receiving devices connected in the charging system. The output includes, but is not limited to, the output voltage, the output current, and output power.

[0220] In some examples, the electrical energy conversion device 220 adjusts the output power according to the maximum number. The maximum number of the identities of the electrical energy receiving devices is less than or equal to 32. Specifically, the charging system 200 charges battery packs

238 electrically connected to the electrical energy receiving devices 230. The controller 222 in the electrical energy conversion device 220 adjusts the output voltage of the connecting port 224 based on the number of the electrical energy receiving devices 230 connected in the charging system 200. In some examples, a maximum output voltage of the connecting port 224 is greater than or equal to 60 V and less than or equal to 65 V. A maximum output current is greater than or equal to 22 A and less than or equal to 24 A. In some examples, the number of the electrical energy receiving devices 230 is less than or equal to 30. In some examples, a maximum number of the electrical energy receiving devices 230 is less than or equal to 25. In some examples, the maximum number of the electrical energy receiving devices 230 is less than or equal to 15. In some examples, the maximum number of the electrical energy receiving devices 230 may be 5, 10, 15, 35, 40, 45, 50, or larger. In some examples, the rated voltage of a battery pack 238 is greater than or equal to 21 V and less than or equal to 65 V. In some examples, the rated voltage of the battery pack 238 is greater than or equal to 40 V and less than or equal to 60 V. In some examples, the rated voltage of the battery pack 238 is greater than or equal to 42 V and less than or equal to 56 V. In some examples, the rated voltage of the battery pack 238 is greater than or equal to 48 V and less than or equal to 56 V.

[0221] In some examples, when the number of the electrical energy receiving devices is less than or equal to a first preset value, the controller is configured to adjust the maximum output voltage of the connecting port to a first voltage threshold. When the number of the electrical energy receiving devices is greater than or equal to a second preset value, the controller is configured to adjust the maximum output voltage of the connecting port to a second voltage threshold. The first preset value is less than the second preset value, and the first voltage threshold is less than the second voltage threshold. Specifically, the first preset value is set to 8 and the second preset value is set to 12. The first voltage threshold is set to 60 V and the second voltage threshold is set to 65 V. For the first eight electrical energy receiving devices, the maximum output voltage and the maximum output current of the electrical energy conversion device are set to 60 V and 24 A. For the ninth to twelfth electrical energy receiving devices, the maximum output voltage and the maximum output current of the electrical energy conversion device are set to 62 V and 23 A. For the 13th to 30th electrical energy receiving devices, the maximum output voltage and the maximum output current of the electrical energy conversion device are set to 65 V and 22 A.

[0222] In some examples, the controller in the electrical energy conversion device acquires an input voltage of an electrical energy receiving device currently charged, and when a difference between the input voltage and the output voltage of the electrical energy conversion device is less than a first threshold, the controller controls the maximum output voltage of the connecting port to be a third voltage threshold. The controller in the electrical energy conversion device acquires the input voltage of the electrical energy receiving device currently charged, and when the difference between the input voltage and the output voltage of the electrical energy conversion device is greater than a second threshold, the controller controls the maximum output voltage of the connecting port to be a fourth voltage threshold. Specifically, the first threshold is set to 2 V and the second

threshold is set to 4 V. The third voltage threshold is set to 60 V and the fourth voltage threshold is set to 65 V. For example, the electrical energy conversion device **220** detects the output voltage of the electrical energy conversion device **220** and collects the input voltage of the electrical energy receiving device **230** currently charged. When the difference between the input voltage and the output voltage is less than 2 V, the maximum output voltage and the maximum output current of the connecting port are 60 V and 24 A. When the difference between the input voltage and the output voltage is greater than or equal to 2 V and less than or equal to 4 V, the maximum output voltage and the maximum output current of the connecting port are 62 V and 23 A. When the difference between the input voltage and the output voltage is greater than 4 V, the maximum output voltage and the maximum output current of the connecting port of the electrical energy conversion device **220** are set to 65 V and 22 A.

[0223] In some examples, when the charging system **200** charges one or more battery packs **238** connected to the electrical energy receiving device **230**, a battery pack with lowest power is charged first. When all the battery packs have substantially the same power, the charging system **200** charges all the battery packs **238**. In some examples, when a battery pack with low power is newly inserted into the electrical energy receiving device **230**, the battery pack with low power is charged first, and the charging of a battery pack with high power is suspended. In some examples, when a battery pack with the same power is newly inserted, the battery pack is directly added and charged together. In some examples, when a battery pack with high power is newly inserted, the battery pack enters a waiting state and is added to be charged when a voltage is charged to the same level. Charging strategies here are for an independent electrical energy receiving device **230**. For all the electrical energy receiving devices **230** in the charging system **200**, after recording the identities of the multiple electrical energy receiving devices **230** connected in cascade, the electrical energy conversion device **220** controls the multiple electrical energy receiving devices **230** connected in cascade to be charged in the order in which they are physically connected. In some examples, the electrical energy conversion device **220** specifies any charging order based on external input (the IoT module or other buttons).

[0224] In some examples, the electrical energy conversion device **220** identifies a fault of an electrical energy receiving device **230** to be charged, reports fault information to the IoT module, and continues to charge a battery pack on the next electrical energy receiving device **230** physically connected to the currently faulty electrical energy receiving device **230**.

[0225] In some examples, in the polling process, if the electrical energy conversion device **220** identifies that an abnormal electrical energy receiving device **230** waits to be charged, the electrical energy receiving device **230** currently charged is not affected, an identity of the faulty electrical energy receiving device **230** is reported, and the abnormal electrical energy receiving device **230** is skipped during sequential charging. After all the electrical energy receiving devices **230** are charged, the electrical energy conversion device **220** re-records the identities of multiple electrical energy receiving devices connected in cascade and reports fault information.

[0226] In some examples, when identifying that the electrical energy receiving device **230** being charged is faulty,

the electrical energy conversion device **220** stops charging the electrical energy receiving device **230**, charges the next physically connected electrical energy receiving device **230**, and reports the identity of the faulty electrical energy receiving device **230**. After all the electrical energy receiving devices **230** are charged, the electrical energy conversion device **220** re-records the identities of multiple electrical energy receiving devices connected in cascade and reports the fault information to the IoT module. In a process of the charging system implementing the charging function, the electrical energy conversion device is configured to, after identifying a fault of the power adapter or the direct current-direct current charger, report the fault information to the wireless communication module for transmission to the remote device and continue to charge a battery pack on the next physically connected power adapter or direct current-direct current charger.

[0227] In some examples, when identifying that a battery pack on a corresponding electrical energy receiving device **230** is faulty, the electrical energy conversion device **220** reports the fault, forbids the battery pack from being charged, and normally charges a remaining battery pack.

[0228] In some examples, when the charging system **200** is in an idle state, the electrical energy conversion device **220** performs the polling operation on the multiple electrical energy receiving devices **230** connected in cascade. In the polling process, after identifying the abnormal electrical energy receiving device **230**, the electrical energy conversion device **220** reports the fault information to the IoT module and re-records the identities of multiple electrical energy receiving devices connected in cascade.

[0229] In some examples, the charging system **200** also has a low power consumption mode. Specifically, the electrical energy conversion device has the low power consumption mode. In the low power consumption mode, the electrical energy conversion device is configured to exit the low power consumption mode based on information obtained by the wireless communication module. In some examples, the electrical energy conversion device **220** automatically enters the low power consumption mode in the case where it has not charged for two consecutive hours (except a scheduled charging mode). In the low power consumption mode, an auxiliary power circuit and a relay in the electrical energy conversion device **220** are turned off. When detecting a short press on a button, the IoT module sends information to the electrical energy conversion device **220**, and the electrical energy conversion device **220** exits the low power consumption mode. In some examples, replugging an alternating current plug of the electrical energy conversion device **220** may cause the electrical energy conversion device **220** to exit the low power consumption mode. The button may be disposed on the remote device or may be disposed on the electrical energy conversion device, of course.

[0230] In some examples, the electrical energy conversion device **220** includes the wireless communication module **240** configured to communicate with the remote device **400**. The wireless communication module **240** includes a Bluetooth module, a Wireless Fidelity (Wi-Fi) module, or a Zigbee module. In some examples, the electrical energy conversion device **220** specifies any charging order based on external input from the remote device **400**.

[0231] In some examples, the charger summarizes or records information about adapters and battery packs and reports the information to the wireless communication mod-

ule 240 to display the relevant information on the remote device 400. In some examples, the cloud sends an upgrade package to the charging system 200 through the wireless communication module 240 so that an OTA upgrade function of the electrical energy conversion device 220, the electrical energy receiving devices 230, and the battery packs thereon can be implemented.

[0232] In some examples, the present application further discloses an electronic combination including a primary node device and a secondary node device. The primary node device includes the wireless communication module configured to communicate with the remote device. The primary node device includes the connecting port electrically and communicatively connected to the secondary node device. Each secondary node device includes the electrical energy input port and the electrical energy output port so that secondary node devices are electrically and communicatively connected to each other in sequence. The electrical energy input port of the first secondary node device is configured to be connected to the connecting port of the primary node device.

[0233] In some examples, the electronic combination further includes a tertiary node device. The secondary node device includes multiple tertiary interfaces, and each tertiary interface is configured to be electrically connected to one tertiary node device. The tertiary node device is communicatively connected to the secondary node device. The tertiary node device implements data transmission with the primary node device through the secondary node device. Specifically, the primary node device may be the electrical energy conversion device, the secondary node device may be the electrical energy receiving device, and the tertiary node device may be the battery pack, where the electrical energy conversion device charges the battery pack through the electrical energy receiving device. In some examples, the electrical energy conversion device 220 includes at least one of the alternating current-direct current charger, the bidirectional power supply, the direct current-direct current charger, and the riding vehicle. The electrical energy receiving device 230 includes at least one of the device mounted with the battery pack and the alternating current appliance. The device mounted with the battery pack includes, but is not limited to, at least one of the direct current-direct current charger, the power adapter, the riding vehicle, or the outdoor traveling device. Specifically, when the primary node device is the riding vehicle, the riding vehicle includes the wireless communication module for wireless communication with the remote device or the cloud. The riding vehicle further includes a communication module for communication with the electrical energy receiving devices physically connected thereto. The riding vehicle further includes an energy storage device for charging the electrical energy receiving devices physically connected thereto.

[0234] Specifically, the electrical energy conversion device 220 is the alternating current-direct current charger, and the electrical energy receiving device 230 is the direct current-direct current charger, the power adapter, or the riding vehicle. As for uplink communication data, thing model data of the battery pack is sent to the direct current-direct current charger or the power adapter, and the direct current-direct current charger, the power adapter, or the riding vehicle transparently transmits the thing model data as it originally is to the alternating current-direct current charger. The controller in the alternating current-direct cur-

rent charger transparently transmits the thing model data to the wireless communication module. As for downlink communication data, the wireless communication module sends data to the alternating current-direct current charger, the alternating current-direct current charger transparently transmits the data to the direct current-direct current charger or the power adapter, and the direct current-direct current charger, the power adapter, or the riding vehicle transparently transmits the data to the corresponding battery pack. In this manner, data or control commands can be delivered, devices at all levels can be upgraded, and other functions can be implemented.

[0235] In some examples, as shown in FIGS. 26 and 27, the electrical energy conversion device 220 is specifically the alternating current-direct current charger, and the power interface 223 of the alternating current-direct current charger is connected to the external power supply. The connecting port 224 of the alternating current-direct current charger is connected to the electrical energy receiving device 230. Specifically, an alternating current connected to the power interface 223 passes through an electromagnetic noise (EMI) filtering circuit 228a and is rectified by a rectifier bridge 228b. The voltage is increased by a power-factor correction (PFC) circuit 228c and decreased by being input into a full-bridge inductor-inductor-capacitor (LLC) resonant conversion circuit 228d. The controller 222 controls a relay 228e to output the voltage to an output terminal 228f, where the output terminal 228f is connected to the battery pack to charge the battery pack. The voltage passing through the PFC circuit 228c is input into an auxiliary power circuit 238g for voltage conversion, so as to supply power to a fan circuit and light display circuit 238h, the controller 222, and the wireless communication module 240. The controller 222 is configured to control an output voltage and an output current of the full-bridge LLC resonant conversion circuit 228d, the communication module, a voltage detection module 228m, a current detection module 228n, the fan circuit and light display circuit 238h, and the wireless communication module 240.

[0236] In some examples, as shown in FIG. 28, the connecting port 224 of the alternating current-direct current charger is connected to the electrical energy receiving device 230. Specifically, the electrical energy receiving device 230 is the power adapter. The power adapter includes the controller 236, a buck circuit 239a, a display circuit 239b, a sampling circuit 239c, and a switch circuit 239d. The buck circuit is configured to decrease the voltage output from the connecting port 224 to supply power to the controller 236 and the display circuit 239b. The display circuit 239b is configured to display a charging or discharging state, the remaining power, and other information of the battery pack 238. The sampling circuit 239c is configured to collect a port voltage and an output current of the battery pack 238. The switch circuit 239d is connected to the controller 236 and triggered by the controller 236 to switch between a closed state or an open state. When the switch circuit 239d is in the closed state, the electrical energy output from the connecting port 224 flows into the battery pack 238 to charge the battery pack 238. In some examples, the power adapter further includes a communication module 239e and a PWM signal transceiver module 239f. The communication module 239e communicates with the electrical energy conversion device 220 via the communication bus 250. The PWM signal transceiver module 239f is connected to the controller 236

and also connected to the PWM signal transceiver module in the next power adapter to transmit the PWM signal to the next physically connected power adapter.

[0237] In some examples, as shown in FIG. 29, the direct current-direct current charger 233 is used for mounting the first energy storage device 233a or the second energy storage device 233b. After electrically connected to the charging system 200, the direct current-direct current charger 233 can receive the electrical energy output from the electrical energy conversion device 220 to charge the first energy storage device 233a or the second energy storage device 233b connected to the direct current-direct current charger 233. When the direct current-direct current charger 233 cannot acquire electrical energy from the power supply system 200, the first energy storage device 233a may charge the second energy storage device 233b. After switched on, the direct current-direct current charger 233 automatically enters a first working mode. In the first working mode, the first energy storage device 233a charges the second energy storage device 233b. In the first working mode, the direct current-direct current charger 233 switches from the first working mode to a second working mode after connected to the charging system. In the second working mode, the first energy storage device 233a switches from a discharging state to a charged state. In some examples, the direct current-direct current charger 233 has a switch button, and the switch button is forbidden from a response in the second working mode. When the direct current-direct current charger exits the second working mode, the user can operate the switch button to make the direct current-direct current charger 233 enter the first mode. The case where the direct current-direct current charger exits the second working mode may include no output from the electrical energy conversion device 220 or no input into the electrical energy conversion device 220 in an AC mode. In this manner, the first working mode and the second working mode of the direct current-direct current charger 233 are managed to ensure the safe charging of the charging system and the battery pack.

[0238] In some examples, as shown in FIG. 29, the direct current-direct current charger 233 includes a Bluetooth module for transmitting a working state and data to the remote device 400. The data may include states and data of the first energy storage device 233a and the second energy storage device 233b connected to the direct current-direct current charger. In some examples, the remote device 400 may remotely upgrade the controller and the Bluetooth module in the direct current-direct current charger 233. In some examples, the first energy storage device 233a includes a Bluetooth module, and the remote device 400 may remotely upgrade the first energy storage battery pack 233a. In some examples, the Bluetooth module in the direct current-direct current charger 233 can also be used for warehouse management.

[0239] In some examples, the switch button on the direct current-direct current charger 233 has multiple functions. Specifically, a long press on the switch button can enable the Bluetooth function of the direct current-direct current charger 233, and the direct current-direct current charger 233 can respond to a short press on the switch button to enter the first working mode.

[0240] In the present application, the power supply system may also be controlled by the remote device. The user can control the operation of the power supply system through the

remote device. Specifically, the power supply system further includes the wireless communication module configured to be communicatively connected to the first adapter and the second adapter, where when the wireless communication module receives a signal from the remote device, the first adapter and the second adapter charge the energy storage device mounted to the first adapter and the energy storage device mounted to the second adapter according to a sequence set by the signal.

[0241] In some examples, through the remote device, the user can control the sequence of charging and discharging of the electrical energy receiving devices in the power supply system. Specifically, the electrical energy receiving devices include multiple sequentially arranged adapters and the direct current-direct current charger. The battery packs are mounted on the adapters and the direct current-direct current charger. For example, in the charging system, three sequentially connected adapters are provided, and two direct current-direct current chargers are provided. To illustrate the charging strategy more clearly, the numbers of the adapters are set to No. 1, No. 2, and No. 3 in sequence. The numbers of the direct current-direct current chargers are No. 1 and No. 2. After the power supply 10 is connected, the user may select different charging modes on the remote device, such as a sequential charging mode and an alternate charging mode.

[0242] In the sequential charging mode, the user may choose a single sequential charging mode or a combined sequential charging mode. After the user selects the single sequential charging mode, the electrical energy provided by the power supply 10 first charges the battery pack on the No. 1 adapter, charges the battery pack on the No. 2 adapter after the battery pack on the No. 1 adapter is fully charged, charges the battery pack on the No. 3 adapter after the battery pack on the No. 2 adapter is fully charged, charges the battery pack on the No. 1 direct current-direct current charger after the battery pack on the No. 3 adapter is fully charged, and charges the battery pack on the No. 2 direct current-direct current charger after the battery pack on the No. 1 direct current-direct current charger is fully charged until the battery pack on the No. 2 direct current-direct current charger is fully charged. After the user selects the combined sequential charging mode, the electrical energy provided by the power supply 10 first simultaneously charges the battery packs on the No. 1 adapter and the No. 2 adapter, then charges the battery packs on the No. 3 adapter and the No. 1 direct current-direct current charger, and finally, charges the battery pack on the No. 2 direct current-direct current charger. Of course, the user may also set the preceding charging combination, for example, the battery packs on the No. 1 adapter, the No. 2 adapter, and the No. 3 adapter are first charged, and then the battery packs on the No. 1 direct current-direct current charger and the No. 2 direct current-direct current charger are charged.

[0243] In the alternate charging mode, the user may choose a single alternate charging mode or a combined alternate charging mode. After the user selects the single alternate charging mode, the electrical energy in the power supply 10 stops charging after charging the battery pack on the No. 1 adapter for a preset time, stops charging after charging the battery pack on the No. 2 adapter for a preset time, stops charging after charging the battery pack on No. 3 adapter for a preset time, stops charging after charging the battery pack on the No. 1 direct current-direct current

charger for a preset time, stops charging after charging the battery pack on the No. 2 direct current-direct current charger for a preset time, stops charging after charging the battery pack on the No. 1 adapter for a preset time, stops charging after charging the battery pack on the No. 2 adapter for a preset time, and cycles in sequence until all the battery packs in the power supply system are fully charged. After the user selects the combined alternate charging mode, the electrical energy in the power supply 10 first stops charging after charging the battery packs on the No. 1 adapter, the No. 2 adapter, and the No. 3 adapter for a preset time, stops charging after charging the battery packs on the No. 1 direct current-direct current charger and the No. 2 direct current-direct current charger for a preset time, stops charging after charging the battery packs on the No. 1 adapter, the No. 2 adapter, and the No. 3 adapter for a preset time, and cycles in sequence until all the battery packs are fully charged. Of course, the user may also set the preceding charging combination, for example, the battery packs on the No. 1 adapter and the No. 2 adapter are first charged, and then the battery packs on the No. 3 adapter, the No. 1 direct current-direct current charger, and the No. 2 direct current-direct current charger are charged. The battery packs are charged by adopting the preceding combined alternate charging mode so that the charging efficiency of the power supply system can be improved.

[0244] In some examples, the user may set the mode of the charging power through the remote device. Specifically, a fast charging mode or a normal charging mode may be selected. Of course, the charging power may also be set according to the requirement. For example, the user may set the gear of the charging power. For example, the user may select the first gear, the second gear, the third gear, or other gears, and different gears correspond to different charging power.

[0245] In some examples, the user may also set the life mode of the battery pack through the remote device. Specifically, the user may set the life mode to 100%, 90%, or 80%. It is to be understood that when the life mode of the battery pack is set to 80%, the battery pack no longer receives the electrical energy input when the power reaches 80%. Of course, the life mode may also include other gears, such as 70% or others.

[0246] In some examples, the user may set a reminder of the charging percentage of the battery pack through the remote device. For example, the user wants to receive a reminder from the remote device when the battery pack reaches 50% charge. Of course, the preceding percentage is only set to 50% schematically, and the user may reasonably set the percentage on the remote device according to actual requirements.

[0247] In some examples, the user may also perform scheduled charging settings or charging timing settings through the remote device. For example, the user may charge the preset battery pack or the electrical energy receiving device in the power supply system after an hour reservation on the remote device. Of course, the user may also perform charging timing settings of the power supply system on the remote device so that the repeated operations of the user can be avoided and the convenience and intelligence of the entire power supply system can be improved.

[0248] In some examples, a controller and a memory are built in the remote device and used for storing and calculating the total charging power of the power supply system

every day, every week, every month, or every year. The user can view the total charging power through the remote device to know the working state of the entire power supply system. In addition, the remote device can also display the proportion of the charging power outputted by the solar panel in the total charging power, so as to analyze the utilization rate of solar energy.

[0249] In some examples, the user may also select the electrical energy receiving device that is charged preferentially through the remote device. For example, through the remote device, the user may set the battery pack with the maximum power to be charged preferentially. Of course, the user may also set the battery pack that is the most accessible for the user in the power supply system to be charged preferentially.

[0250] Of course, in some examples, through the remote device, the user may customize the charging sequence of the electrical energy receiving devices or choose and charge part of the battery packs.

[0251] To improve the user experience and the intelligence of the power supply system, the power supply system also has the functions of a full charge reminder and a failure reminder. Specifically, when the electrical energy receiving device in the power supply system is fully charged, the user can receive a full charge reminder of the corresponding electrical energy receiving device through the remote device. When the power supply system fails, the remote device displays the abnormal problem on a display interface and sends a message, makes a call, or generates an alarm to remind the user.

[0252] Referring to FIG. 30, an adapter 234c is detachably mounted to an electric driving device 234 on which the energy storage device 234b is disposed. Specifically, the adapter 234c includes an alternating current input interface 2341c, an alternating current output interface 2342c, an energy storage device interface 2343c, and the bidirectional inverter module. The alternating current input interface 2341c is configured to input the alternating current. The alternating current output interface 2342c is configured to output the alternating current. The energy storage device interface 2343c is configured to be electrically connected to the energy storage device 234b. The bidirectional inverter module is electrically connected to the alternating current input interface 2341c, the alternating current output interface 2342c, and the energy storage device interface 2343c. The bidirectional inverter module 234a is configured to charge the energy storage device 234b using the alternating current inputted from the alternating current input interface 2341c and output the electrical energy of the energy storage device 234b using the alternating current output interface 2342c. In some examples, the alternating current input interface 2341c is 3 prong outlet. In some examples, the alternating current output interface 2342c is a 3 pin socket or a 2 pin socket. In some examples, the adapter 234c further includes a USB interface for supplying power to the mobile phone or the 4G gateway. In some examples, the energy storage device 234b includes at least one battery pack detachably connected to the electric driving device. In this example, the output power of the adapter 234c is greater than or equal to 500 W and less than or equal to 2000 W. The total energy of the energy storage device is greater than or equal to 0.5 kW·h and less than or equal to 6 kW·h. The electric driving device may be the riding vehicle, such as the riding mower or the all-terrain vehicle. Of course, the

electric driving device may also be a push snow thrower or a push mower. In some examples, the electric driving device may also be an electric motorcycle, a utility task vehicle (UTV), a motorboat, a drone, or the like.

[0253] In some examples, two electric driving devices can charge each other. The electric driving device includes a body; a traveling wheel set including traveling wheels supporting the body; a drive assembly mounted to the body and including an electric motor for driving the traveling wheels to rotate; an energy storage device detachably mounted to the body and configured to supply power to at least the electric motor; and a direct current interface electrically connected to the energy storage device and configured to charge the energy storage device when the direct current interface is connected to an external power supply. The direct current interface is further configured to output the direct current and charge another electric driving device using the electrical energy of the energy storage device. In some examples, the external power supply is another electric driving device. In some examples, the external power supply is the charger. In some examples, the direct current interface can also charge other devices. Other devices are the mobile phone, a lighting device, a fan, and the like.

[0254] The charging device in the present application has a relatively strong loading capacity, and at the same time, anti-theft design and monitoring are performed on the battery pack mounted on the charging device.

[0255] In some examples, a charging device includes a housing; and a battery pack interface disposed on the housing and configured to be capable of being coupled with a battery pack. The charging device further includes an anti-theft device, where the anti-theft device includes a locked state and an unlocked state. When the anti-theft device is in the locked state, the battery pack is constrained within a first distance from the charging device; and when the anti-theft device is in the unlocked state, the battery pack is removable from the charging device. In some examples, the anti-theft device is a chain. An end of the anti-theft device is connected to the battery pack, and the other end of the anti-theft device is connected to the charging device. The first distance is less than or equal to 1 meter. In some examples, the first distance is less than or equal to 0.5 meters. An unlocking manner of the anti-theft device may be at least one of fingerprint recognition, a mechanical lock, voice control, face recognition, and remote unlocking. The anti-theft device is configured to unlock or lock multiple battery packs mounted to the battery pack interface one by one. The anti-theft device is configured to simultaneously unlock or lock multiple battery packs mounted to the battery pack interface.

[0256] In some examples, as shown in FIGS. 31 to 33, a charging device 500 includes a housing 510 and battery pack interfaces 511 disposed on the housing 510 and configured to be capable of being coupled with battery packs 520. Fixing devices 530 are configured to be operated by the user to switch between a first state and a second state. When the fixing devices 530 are in the first state, the battery packs 520 are fixed on the charging device 500; and when the fixing devices 530 are in the second state, the battery packs 520 are movable freely relative to the charging device 500. The charging device 500 further includes a locking device 540, where the locking device 540 includes a locked state and an unlocked state. When the locking device 540 is in the locked state, the fixing device 530 is constrained in the first state;

and when the locking device 540 is in the unlocked state, the fixing device 530 is capable of being operated by the user to switch to the second state.

[0257] Specifically, a limiting portion 531 is formed on the fixing device 530, and ratchets 541 are disposed on the locking device 540. When the locking device 540 is in the locked state, the limiting portion 531 abuts against the ratchet 541, and the fixing device 530 cannot switch from the first state to the second state. When the user operates the locking device 540 to move along the direction indicated by arrow a, the limiting portion 531 is not in contact with the ratchet 541, and the fixing device 530 can switch from the first state to the second state. When the locking device 540 is in the unlocked state and the user operates the locking device 540 to move along the direction indicated by arrow b, the limiting portion 531 abuts against the ratchet 541, and the fixing device 530 cannot switch from the first state to the second state.

[0258] In some examples, the locking device 540 is a latch. In some examples, the fixing device 530 is a snap. In some examples, an unlocking manner of the locking device may be at least one of fingerprint recognition, a mechanical lock, voice control, face recognition, and remote unlocking.

[0259] In some examples, the locking device or the fixing device is configured to unlock or lock multiple battery packs mounted to the battery pack interface one by one. In some examples, the locking device or the fixing device is configured to simultaneously unlock or lock multiple battery packs mounted to the battery pack interface.

[0260] In some examples, as shown in FIGS. 34 and 35, the charging device 500 includes a first port 512 disposed on the housing 510 and used for inputting the current or outputting the current; and a second port 513 disposed on the housing 510 and used for outputting the current or inputting the current. The first port 512 and the second port 513 are capable of being used interchangeably. When the first port 512 is configured to input the current, the second port 513 is configured to output the current; and when the second port 513 is configured to input the current, the first port 512 is configured to output the current. Specifically, the first port 512 and the second port 513 have the same shape. In some examples, the first port 512 and the second port 513 are symmetrically disposed on two sides of the charging device. In some examples, the first port 512 and the second port 513 are disposed on the upper portion of the housing 510. In some examples, the first port is disposed on the upper portion of the housing, and the second port is disposed on the lower portion of the housing. In some examples, the first port and the second port are disposed on the lower portion of the housing. In some embodiments, the first port 512 and the second port 513 have basically the same voltage. In some examples, the first port 512 is used for connecting a first charging device 500a. The second port 513 is used for connecting a second charging device 500b. The first port 512 and the second port 513 have the same structure and are capable of separately receiving a current input at different times. The first port 512 and the second port 513 are symmetrical about a middle plane 501 of the charging device 500. The first port 512 and the second port 513 are located on the upper portion of the charging device 500. The first port 512 and the second port 513 are connected to other devices through cables. Specifically, a cable 550 includes quarter-turn connectors 551 and a wire 552. Specifically, the cable 550 includes two quarter-turn connectors connected at

two ends of the wire. The cables **550** are adopted so that the installation space when two charging devices are connected can be saved. In some examples, the charging device **500** further includes fixing assemblies **560** for fixing the charging device **500** to another object. In some examples, the fixing assemblies **560** are detachably mounted to the housing **510**. In some examples, the fixing assemblies **560** are snap clips.

[0261] In some examples, a charging device includes a first adapter and a second adapter. The first adapter includes a housing; at least one accommodation portion for mounting an energy storage device, where the first adapter is capable of charging the energy storage device mounted to one of the at least one accommodation portion; and a first port disposed on the housing and used for connecting the second adapter. The second adapter includes a housing; at least one accommodation portion for mounting an energy storage device, where the second adapter is capable of charging the energy storage device mounted to one of the at least one accommodation portion; and a second port disposed on the housing and used for connecting the first adapter. The first port and the second port have the same structure and are connected through a cable. The cable is configurable in different lengths. At least one end of the cable is configured to be the quarter-turn connector. An end of the cable is detachable from the first port. An end of the cable is detachable from the second port. In some examples, the first adapter and the second adapter are the same. In some examples, the first adapter and the second adapter are different. The first adapter further includes the support assembly capable of supporting the second adapter. The first adapter further includes the fixing assembly for fixing the first adapter to another object. The first port is disposed on the upper portion of the housing of the first adapter.

[0262] In this manner, the first port and the second port of the charging device have the same appearance and can be used interchangeably so that the user experience can be improved, the charging device can be adapted to more space application scenarios, and it is convenient for the user to perform installation and wiring.

[0263] In some examples, as shown in FIGS. 36 and 37, the charging device **500** includes the housing **510**; a first accommodation portion **514** disposed at the housing **510** and configured to mount a first energy storage device **521**; and at least two second accommodation portions **515** disposed at the housing **510** and configured to mount second energy storage devices **522**. The volume of the first energy storage device **521** is greater than the volume of the second energy storage device **522**. The first accommodation portion **514** and the at least two second accommodation portions **515** are separately disposed on two sides of a partition surface **502**. When the second energy storage devices are mounted to the second mounting portions and the first energy storage device is mounted to the first accommodation portion, the width of the projection of the first energy storage device on the partition surface is basically equal to the width of the projection of all the second energy storage devices on the partition surface.

[0264] In some examples, the weight of the first energy storage device **521** is greater than or equal to 9 kg. In an example, the weight of the first energy storage device **521** is greater than or equal to 10 kg, greater than or equal to 11 kg, greater than or equal to 12 kg, greater than or equal to 13 kg, greater than or equal to 14 kg, or greater than or equal to 15

kg. For example, the weight of the battery pack is 9 kg, 10 kg, or 15 kg. The capacity of the first energy storage device **521** is greater than or equal to 20 Ah. In an example, the capacity of the first energy storage device **521** is greater than or equal to 30 Ah, the capacity of the first energy storage device **521** is greater than or equal to 40 Ah, or the capacity of the first energy storage device **521** is greater than or equal to 50 Ah. For example, the capacity of the first energy storage device **521** may be 20 Ah, 30 Ah, 40 Ah, or 50 Ah. The ratio of the capacity to the weight of the first energy storage device **521** is greater than or equal to 2 Ah/kg, for example, 2 Ah/kg, 4 Ah/kg, or 5 Ah/kg. An average discharge current of the first energy storage device **521** is greater than or equal to 30 A, which may be, for example, 30 A, 35 A, or 40 A. The energy of the battery pack **100** is greater than or equal to 2 kWh. In an example, the energy of the first energy storage device **521** is greater than or equal to 3 kWh, the energy of the first energy storage device **521** is greater than or equal to 4 kWh, or the energy of the first energy storage device **521** is greater than or equal to 5 kWh. For example, the energy of the first energy storage device **521** may be 2 kWh, 3 kWh, 4 kWh, or 5 kWh. In an example, the first energy storage device **521** is basically in the shape of a rectangular cuboid, and the height of the first energy storage device **521** is greater than or equal to 300 mm and less than or equal to 350 mm. The length of the first energy storage device **521** is greater than or equal to 300 mm and less than or equal to 350 mm. In some examples, the length of the first energy storage device **521** is greater than or equal to 320 mm and less than or equal to 330 mm. In some examples, the width of the first energy storage device **521** is greater than or equal to 150 mm and less than or equal to 200 mm. In some examples, the width of the first energy storage device **521** is greater than or equal to 170 mm and less than or equal to 180 mm. It is to be noted that the height dimension $\geq$ the length dimension $\geq$ the width dimension for the first energy storage device **521**.

[0265] In some examples, the charging device **500** includes the housing **510**; the first accommodation portion **514** disposed at the housing **510** and used for mounting the first energy storage device **521**; and the second accommodation portions **515** disposed at the housing **510** and used for mounting the second energy storage devices **522**. The weight of the first energy storage device **521** is greater than the weight of the second energy storage device **522**. When the charging device **500** is placed on a plane **503**, the first energy storage device **521** is inserted into the first accommodation portion **514** along the direction of a first straight line **504**, and the included angle formed by the first straight line **504** and the plane **503** is greater than or equal to 0 degrees and less than or equal to 60 degrees. In some examples, the included angle formed by the first straight line **504** and the plane **503** is greater than or equal to 10 degrees and less than or equal to 45 degrees. In some examples, the included angle formed by the first straight line **504** and the plane **503** is equal to 15 degrees. When the charging device **500** is placed on the plane **503**, the first energy storage device **521** is located below the second energy storage device **522** in the up and down direction.

[0266] In some examples, the battery pack is connected to the charging device via a cable. Specifically, the charging device includes a socket adapted to a plug connected to the cable, and the battery pack is provided with a socket adapted to a plug connected to the cable. In this manner, in one

aspect, the charging device can move relative to the battery pack, making its application scenario more flexible. In the other aspect, the charging device is smaller in volume, more convenient to carry, store, and transport, and lower in cost.

[0267] In some examples, as shown in FIG. 38, the charging device 500 further includes a circuit board 570 disposed in the housing 510, where the circuit board 570 is basically parallel to the bottom surface of the charging device 500. Specifically, the circuit board 570 is at least partially disposed below the first accommodation portion 514 and the second accommodation portions 515 in the up and down direction. A heat dissipation airflow is formed in the housing 510 and basically flows along a long end of the circuit board 570.

[0268] In some examples, the charging device 500 may be further provided with a fan, and an airflow generated by the fan is used for dissipating heat for the charging device or the battery pack. In some examples, the charging device 500 may use a coolant for heat dissipation. In some examples, the charging device 500 may be made of a material with a relatively good heat dissipation effect, such as a phase change material. In some examples, the charging device may be provided with a hydrogel material with a relatively good heat dissipation effect.

[0269] In some examples, as shown in FIG. 38, the charging device 500 further includes a circuit board 570 disposed in the housing 510, where the circuit board 570 is basically parallel to the bottom surface of the charging device 510. Specifically, the circuit board 570 is at least partially disposed below the first accommodation portion 514 and the second accommodation portions 515 in the up and down direction. A heat dissipation airflow is formed in the housing 510 and basically flows along the long end of the circuit board 570.

[0270] In some examples, the charging device 500 includes the housing 510; and at least one battery pack interface 511 disposed on the housing 510 and configured to be capable of being coupled with the battery pack. As shown in FIGS. 38 and 39, the charging device 500 further includes a frame 580, and the housing is accommodated within a cuboid defined by the frame 580. Specifically, the frame includes or is formed with a handle for the user to lift. In some examples, the frame is made of plastic. In some examples, the frame is made of metal. In some examples, the battery pack is configured to be insertable from the outside of the cuboid defined by the frame 580 into the inside of the cuboid to be coupled to the battery pack interface 511. Specifically, the frame includes a transverse bar 581 disposed near the partition surface 502. It is to be noted that the frame may be understood as a specific example of the support assembly in the present application.

[0271] In some examples, the charging device 500 includes the housing 510; and at least one battery pack interface 511 disposed on the housing 510 and configured to be coupled with the battery pack. Multiple charging devices 500 are capable of being stacked in a first manner and a second manner, where the placement direction of the multiple charging devices in the first manner is different from the placement direction of the multiple charging devices in the second manner. The charging device further includes the frame, and the housing is accommodated within the cuboid defined by the frame 580. In some examples, the frame includes or is formed with the handle for the user to lift. In some examples, the top corners of the frame 580 form

protrusions 583 or recesses 582; and when frames 580 are stacked, protrusions and recesses 582 are capable of mating with each other and being fixed. Specifically, when the charging devices are stacked in the first direction as shown in FIG. 40, all the battery packs are accessible. When the charging devices are stacked in the second direction as shown in FIG. 41, at least part of the battery packs are inaccessible. In some examples, when the charging devices are stacked in the third direction as shown in FIG. 45, only the battery packs on the uppermost charging device are accessible.

[0272] In some examples, the charging device 500 further has a power display unit for displaying a power state of the battery pack mounted on the charging device 500.

[0273] In some examples, the charging device 500 includes the housing 510; the first accommodation portion 514 disposed at the housing 510 and used for mounting the first energy storage device 521, where the first energy storage device 521 includes multiple first cells; and the second accommodation portions 515 disposed at the housing 510 and used for mounting the second energy storage devices 522, where the second energy storage device 522 includes multiple second cells. The volume of the first cell is greater than the volume of the second cell. As shown in FIGS. 41, multiple charging devices are capable of being stacked, and when the multiple charging devices are stacked on the plane 503, the first cells are basically parallel to the plane 503. In some examples, the second energy storage device 522 is inserted into the second accommodation portion 515 along the extension direction of a second straight line 505, and the included angle β formed by the second straight line 505 and the plane 503 is greater than or equal to 45 degrees and less than or equal to 80 degrees. In some examples, the included angle β formed by the second straight line 505 and the plane 503 is greater than or equal to 50 degrees and less than or equal to 70 degrees. In some examples, the included angle β formed by the second straight line 505 and the plane 503 may be 45°, 50°, 55°, 60°, 65°, 70°, 75°, or 80°.

[0274] In some examples, as shown in FIGS. 42 and 43, the charging system includes multiple charging devices, where the multiple charging devices 500 can be stacked through the frames 580. Specifically, wheels 584 and a pull rod 585 are capable of being detachably mounted on the frame 580. When the multiple charging devices 500 need to be moved, the wheels and pull rods are mounted on the frame 580 of the lowest charging device so that the charging system can be easily moved. In some examples, as shown in FIG. 44, the charging system includes the multiple charging devices 500 and a cabinet 500c on which the multiple charging devices are placed, where the cabinet 500c has wheels for easy movement.

[0275] In some examples, as shown in FIGS. 46 to 48, a charging device 600 includes a housing 601; a mounting portion 602 for mounting or accommodating an energy storage device; a pull rod assembly 603 mounted to the housing 601; and wheels 604 mounted to the housing 601. When the pull rod assembly 603 is pushed or pulled by a user, the wheels 604 roll, making it convenient for the user to move the charging device 600.

[0276] In some examples, the housing 601 is formed with a compartment 607 for mounting the energy storage device, and the compartment 607 surrounds a space 607a for accommodating the energy storage device. When the energy

storage device is disposed in the space 607a and charged or discharged, a large amount of heat is generated. To dissipate the heat generated by the energy storage device as quickly as possible, multiple heat dissipation holes 607a are provided on the compartment 607. The multiple heat dissipation holes 607a are configured to transfer the heat in the compartment 607 to the outside of the compartment 607.

[0277] In some examples, the pull rod assembly 603 includes two parallel extension rods 6031 and a grip rod 6032 connecting the two extension rods. The extension rods 6031 extend obliquely in a direction away from the energy storage device. In some examples, the extension rods 6031 are retractable or foldable so that the requirement of the user for storage can be satisfied.

[0278] In some examples, the charging device 600 further includes a bottom support 605. The bottom support 605 is disposed on a side of the housing facing away from the wheels. When the charging device 600 is placed on the ground plane, the bottom support 605 and the wheels 604 jointly support the charging device 600. When the charging device 600 is pushed or pulled by the user to move on the ground plane, the bottom support 605 is suspended and the wheels 604 support the charging device 600. Specifically, the height of the bottom support 605 is greater than the radius of the wheels 604.

[0279] In some examples, the housing 601 is symmetrically distributed about a middle plane 601a, the pull rod assembly 603 and the wheels 604 are located on the same side of the middle plane 601a, and the bottom support 605 is located on the other side of the middle plane 601a.

[0280] In some examples, as shown in FIG. 47, when the charging device 600 stands still on the ground 603b, the angle α between a first plane 603a formed by the pull rod assembly 603 and the ground 603b is greater than or equal to 75° and less than or equal to 90°. In some examples, the angle α between the first plane 603a formed by the pull rod assembly 603 and the ground 603b is greater than or equal to 75° and less than or equal to 85°. In some examples, the angle α between the first plane 603a formed by the pull rod assembly 603 and the ground 603b may be 80°. In this manner, the charging device is more convenient and labor-saving for the user to push or pull.

[0281] In some examples, the distance between two wheels 604 in a left and right direction is greater than or equal to half of the width of the housing 601 in the left and right direction. In some examples, the distance between the two wheels 604 in the left and right direction is greater than or equal to three-quarters of the width of the housing 601 in the left and right direction. In some examples, the distance between the two wheels 604 in the left and right direction is greater than or equal to the width of the housing 601 in the left and right direction.

[0282] In some examples, as shown in FIGS. 49 and 50, the pull rod assembly 603 includes the two parallel extension rods 6031 and the grip rod 6032 connecting the two extension rods. The extension rods 6031 extend along the first plane 603a, the grip rod 6032 extends along a second plane 603c, and the angle β between the second plane 603c and the first plane 603a is greater than or equal to 0° and less than or equal to 45°.

[0283] In some examples, the bottom support 605 is detachably mounted to a first mounting portion 600a of the housing 601, and the pull rod assembly 603 is detachably mounted to a second mounting portion 600b of the housing

601. The first mounting portion 600a can also be used for mounting the pull rod assembly 603, and the second mounting portion 600b can also be used for mounting the bottom support 605. In this manner, when the user needs to change a direction in which the charging device 600 is placed to push or pull the charging device 600, the bottom support 605 and the pull rod assembly 603 can be conveniently detached or mounted and their mounting positions can be interchanged so that a quick change can be achieved without removing the battery pack or disassembling the housing.

[0284] In some examples, the charging device includes a first charging device 610 and a second charging device 620. When the first charging device 610 and the second charging device 620 are stacked, the pull rod assembly 603 can be pushed or pulled by the user to facilitate the movement of the first charging device 610 and the second charging device 620. In this manner, the requirement of the user for placement in different directions can be satisfied. In some examples, the pull rod assembly 603, the wheels 604, and the bottom support 605 are all detachably mounted to the housing 601. In some examples, the charging device includes the first charging device 610 and the second charging device 620. When the first charging device 610 and the second charging device 620 are stacked, the pull rod assembly can be pushed or pulled by the user to facilitate the movement of the first charging device 610 and the second charging device 620.

[0285] Referring to FIG. 51, FIG. 51 shows a power-receiving system with an adapter 80 described in the present application as a core in conjunction with a charger 30a, an outdoor work vehicle 70, and an electrical device 10b that may be involved in different application scenarios. The power-receiving system mainly provides services for the outdoor work vehicle 70, the power tool, and the related electrical device 10b, and various indicators should satisfy the design requirements of the preceding scenarios.

[0286] FIG. 52 shows the charger 30a as an example of the present application. The main function of the charger 30a in the preceding power-receiving system is to achieve electrical energy conversion such as alternating current-direct current conversion. The charger 30a has at least one alternating current port (AC port) and one direct current port (DC port). The AC port is an input port 31a that can be connected to the mains power or other forms of alternating current, and the DC port is an output port 32a that outputs the direct current that satisfies the requirements of the current scenario. In some examples, a voltage conversion module is provided in the charger 30a, and the voltage conversion module can convert the alternating current inputted into the charger 30a into the direct current for output and may include relevant circuits such as a rectifier circuit, a filter circuit, an isolator circuit, and a transformer circuit. In the present application, the output power of the output port 32a of the charger 30a is greater than or equal to 550 W and less than or equal to 5000 W to satisfy the design requirements in the application scenario of outdoor power equipment. It is to be understood that the preceding description is the power-receiving scenario mainly described below and the cases where the charger 30a does not perform alternating current-direct current conversion are not excluded. For example, boost conversion, buck conversion, or other monitoring and management tasks may be performed without changing the alternating current/direct current form.

[0287] FIG. 53 shows the outdoor work vehicle 70 as an example of the present application. Specifically, FIG. 53 is a perspective view of some structures of a riding mower. The outdoor work vehicle 70 in the present application may include a manned mower. For example, the outdoor work vehicle 70 may include at least a riding mower and a stand-on mower. In addition, the outdoor work vehicle 70 may further include a robotic mower, a snow thrower, a cleaning machine, and other devices that can perform other functional tasks while traveling. Furthermore, the outdoor work vehicle 70 further includes some devices that only perform outdoor traveling, for example, a utility task vehicle (UTV)/farmer's vehicle, an all-terrain vehicle (ATV)/dune buggy, and a golf cart. In the present application, the outdoor work vehicle 70 is equipped with a power supply device, and the power supply device is used for supplying the electrical energy required for the outdoor work vehicle 70 to travel and implement other functions. In some examples, the preceding power supply device includes at least one battery pack. In some examples, the power supply device may also include multiple battery packs, which may be identical battery packs or battery packs with parameters such as the capacity, the voltage, and the current that are not completely identical. In some examples, the battery pack in the power supply device of the outdoor work vehicle 70 may supply power to other power tools in a detachable or non-detachable manner. In some examples, the power supply device includes a power supply circuit and related monitoring and management circuits in addition to the battery pack. As shown in FIG. 53, the outdoor work vehicle 70 has an electrical interface for connecting an external device, and the electrical interface is directly or indirectly connected to the power supply device.

[0288] FIG. 54 shows the electrical device 10b as an example of the present application. FIG. 55 shows the electrical device 10b as another example of the present application. The electrical device 10b in the present application can at least be used for mounting and charging the battery pack of the outdoor power vehicle or power tool. The electrical device 10b shown in FIG. 54 is denoted as a first electrical device 110. The first electrical device 110 can charge at least two identical battery packs. FIG. 54 is used as an example, and the first electrical device 110 can simultaneously charge three identical battery packs with consistent charging and discharging requirements. The electrical device 10b shown in FIG. 55 is denoted as a second electrical device 120. The second electrical device 120 can charge at least two different battery packs. FIG. 55 is used as an example. The second electrical device 120 can charge two identical battery packs with consistent charging and discharging requirements and can also charge another type of battery packs different from the preceding battery packs in size, model, capacity, charge voltage, charge current, charge power, or the like. In addition to using external power to charge the battery packs mounted on the electrical device 10b, the electrical device 10b can use the electrical energy outputted by the battery packs mounted on the electrical device 10b to power/charge other connected external devices.

[0289] Referring to FIGS. 56 to 59, the adapter 80 as an example of the present application is shown. The adapter 80 is connected between any two of the charger 30a, the outdoor work vehicle 70, and the electrical device 10b in the preceding power-receiving system and can serve as an intermediary for power transmission between the two. It is

to be noted here that, in some cases, the electrical device 10b described above may also be described as the adapter 80, while the adapter 80 described here may be described as a transfer adapter or an adapter connector, and those skilled in the art can distinguish such differences in description.

[0290] As shown in FIGS. 56 to 58, the adapter 80 includes at least a housing 81, a first interface 82, and a second interface 83. The housing 81 forms the body of the adapter 80, and an accommodation space is formed inside the housing 81. The housing 81 supports, connects, and accommodates other components of the adapter 80 to be described later. The housing 81 may be discrete or integrated. In some examples, the housing 81 is formed by mating a first housing 811 with a second housing 812. Specifically, the first housing 811 and the second housing 812 may mate with each other roughly along a direction perpendicular to the extension planes of the first housing 811 and the second housing 812. The first housing 811 and the second housing 812 mate with each other to form the preceding accommodation space. A circuit board assembly 84 to be described later may be disposed in the accommodation space.

[0291] In the present application, one of the first interface 82 and the second interface 83 is used as an input terminal of the adapter 80, and the other one of the first interface 82 and the second interface 83 is used as an output terminal of the adapter 80. In different scenarios, whether one of the interfaces is used as the input terminal or the output terminal is variable. In some examples, to solve the interface adaptation problem between the charger 30a, the outdoor work vehicle 70, and the electrical device 10b in different scenarios, the first interface 82 and the second interface 83 may have different physical structures. In some examples, the difference in physical structure between the first interface 82 and the second interface 83 is reflected in the shapes. Specifically, the difference in interface shape may be reflected as the interface sections presenting different geometric figures, or as the interface sections presenting geometric figures of the same type but with different areas, or as inconsistent contours of the interface sections. In an example, one of the first interface 82 and the second interface 83 is a circular interface, and the other one of the first interface 82 and the second interface 83 is a square interface. In some other examples, the difference in physical structure between the first interface 82 and the second interface 83 may also be reflected in that the first interface 82 and the second interface 83 are made of different materials, use transmission media of different materials, have different pin arrangements, have different numbers of pins, or have male pins and female sockets, respectively.

[0292] In some other examples, the difference in physical structure between the first interface 82 and the second interface 83 is also reflected in electrical characteristics or interface protocols, including, but not limited to, the differences in voltage, current, and power between the first interface 82 and the second interface 83 and the differences in encoding and decoding methods and frame formats of communication signals transmitted through the first interface 82 and the second interface 83.

[0293] As shown in FIG. 56, in some examples, the adapter 80 or the housing 81 is not always located in the same straight line, that is, the plugging and unplugging direction allowed by the first interface 82 and the plugging and unplugging direction allowed by the second interface 83

are not collinear so that the adapter 80 is not easy to fall off after being connected and is easier for the user to operate during use. In some examples, the included angle between the plugging and unplugging direction of the first interface 82 and the plugging and unplugging direction of the second interface 83 is greater than or equal to 120° and less than or equal to 185° . Preferably, the included angle between the plugging and unplugging direction of the first interface 82 and the plugging and unplugging direction of the second interface 83 is greater than or equal to 140° and less than or equal to 165° . In an example, the included angle between the plugging and unplugging direction of the first interface 82 and the plugging and unplugging direction of the second interface 83 may be 135° , 150° , or 165° .

[0294] In some examples, based on the above, the housing 81 is formed by mating the first housing 811 with the second housing 812, one of the first interface 82 and the second interface 83 is supported by both the first housing 811 and the second housing 812, and the other one of the first interface 82 and the second interface 83 is supported solely by the first housing 811 or the second housing 812, thereby adapting to the design requirements of the different physical structures of the interfaces and the non-collinear plugging and unplugging directions of the interfaces as described above. As shown in FIGS. 56 to 58, halves of the outer circumference of the circular interface are covered by the first housing 811 and the second housing 812, respectively. After mating with each other, the first housing 811 and the second housing 812 can jointly support the circular interface. The entire outer circumference of the square interface is wrapped and supported by the second housing 812.

[0295] In some examples, the adapter 80 may further include a cable, the first interface 82 and/or the second interface 83 may not be supported by the housing 81 and may be disposed outside the housing 81, the first interface 82 and/or the second interface 83 may be connected to the housing 81 via the cable, and the cable may contain an electronic circuit for signal communication between the first interface 82 and/or the second interface 83 and the circuit board assembly 84 in the housing 81 and an electronic circuit for power transmission between the first interface 82 and the second interface 83 so that the user can use the adapter 80 to connect different devices more easily without causing problems such as difficulty in connection or easy falling off due to the orientation of the device interface. In some examples, the cable may be provided in the form of a spring wire to make related operations more convenient.

[0296] In a possible example, the adapter 80 may serve as an intermediary for power transmission between the charger 30a and the outdoor work vehicle 70, where the charger 30a is the power feeder and the outdoor work vehicle 70 is the power receiver. One of the first interface 82 and the second interface 83 is detachably connected to the charger 30a, and the other one of the first interface 82 and the second interface 83 is detachably connected to the outdoor work vehicle 70. The case where the first interface 82 is connected to the charger 30a and the second interface 83 is connected to the outdoor work vehicle 70 is used as an example for an exemplary description. However, it is to be understood that even for the same adapter 80, in cases of different chargers 30 and outdoor work vehicles 70, the corresponding connection relationship of the interfaces may be non-fixed. In one scenario, the first interface 82 is connected to the charger 30a and the second interface 83 is connected to the outdoor

work vehicle 70, while in another scenario, the first interface 82 is connected to the outdoor work vehicle 70 and the second interface 83 is connected to the charger 30a.

[0297] In this example, the output port 32a of the charger 30a may output the direct current, and the output power of the output port 32a is greater than or equal to 550 W and less than or equal to 5000 W. The first interface 82 of the adapter 80 is connected to the output port 32a of the charger 30a to acquire the electrical energy outputted by the charger 30a or further exchange communication signals with the charger 30a. The second interface 83 of the adapter 80 is connected to the outdoor work vehicle 70 to output electrical energy to the power supply device of the outdoor work vehicle 70. Specifically, the battery pack in the power supply device can be charged, or the second interface 83 further exchanges communication signals with the outdoor work vehicle 70. As shown in FIGS. 52, 53, and 56, the first interface 82 connected to the charger 30a and the second interface 83 connected to the outdoor work vehicle 70 have different physical structures, that is, the output port 32a of the charger 30a and an electrical interface 71 of the outdoor work vehicle 70 have different physical structures. For example, the first interface 82 is a circular interface, and the second interface 83 is a square interface. For other cases where the physical structures of the interfaces are different, reference may be made to the preceding relevant descriptions.

[0298] In some examples, the input port 31a of the charger 30a is connected to the alternating current such as the mains power, the charger 30a is provided with a voltage conversion module for performing alternating current-direct current conversion, and the voltage conversion module can convert the alternating current connected to the input port 31a into the direct current that satisfies the requirements and then output the direct current through the output port 32a. Specifically, the voltage conversion module of the charger 30a may include a rectifier circuit, a filter circuit, an isolator circuit, a transformer circuit, a control circuit, and the like.

[0299] In some examples, the direct current voltage outputted by the output port 32a of the charger 30a is greater than or equal to 10 V and less than or equal to 70 V. Preferably, the direct current voltage outputted by the output port 32a is greater than or equal to 21 V and less than or equal to 65 V. In an example, the direct current voltage outputted by the output port 32a may be 21 V, 48 V, or 60 V.

[0300] The present application correspondingly proposes a charging device 10, which is at least a combination of the charger 30a and the adapter 80. The charging device 10 serves as an intermediary for power transmission between the mains power and the outdoor work vehicle 70. An input interface of the charging device 10 is the input port 31a of the charger 30a and is connected to the alternating current such as the mains power, the output port 32a of the charging device 10 is the second interface 83 of the adapter 80 and is detachably connected to the electrical interface 71 of the outdoor work vehicle 70 to supply power to the power supply device in the outdoor work vehicle 70, and the output port 32a of the charger 30a and the first interface 82 of the adapter 80 are connected for power transmission inside the charging device 10. Moreover, the adapter 80 includes a communication module 842 in addition to the first interface 82 and the second interface 83. The communication module 842 communicates with the charger 30a and the outdoor work vehicle 70 through the first interface 82 and the second

interface **83**, respectively. The overall conversion efficiency of the adapter **80** is greater than or equal to 90%. The overall conversion efficiency of the adapter **80** described here is determined based on the electrical energy acquired by the adapter **80** from the charger **30a** through the first interface **82** and the electrical energy finally outputted by the adapter **80** to the outdoor work vehicle **70** through the second interface **83**. Preferably, the conversion efficiency of the adapter **80** is greater than or equal to 95%. In some examples, the rated operating current of the adapter **80** is greater than or equal to 8 A and less than or equal to 85 A. Preferably, the rated operating current of the adapter **80** is greater than or equal to 8 A and less than or equal to 25 A.

[0301] In some examples, to satisfy the scenario requirements of the power tool and the outdoor work vehicle, the output power of the output port **32a** of the charger **30a** is greater than or equal to 550 W and less than or equal to 5000 W. Preferably, the output power of the output port **32a** of the charger **30a** is greater than or equal to 1000 W and less than or equal to 5000 W. In an example, the output power of the output port **32a** of the charger **30a** may be 1600 W, 2500 W, or 4000 W.

[0302] In some examples, the communication module **842** of the adapter **80** can communicate with the charger **30a** and the outdoor work vehicle **70** at least via RS-485 communication and controller area network (CAN) communication. Specifically, the output port **32a** of the charger **30a** and/or the electrical interface **71** of the outdoor work vehicle **70** may comply with the RS-485 standard and can execute communication protocols such as Modbus that are compatible with the RS-485 standard, or the output port **32a** of the charger **30a** and/or the electrical interface **71** of the outdoor work vehicle **70** may execute the CAN protocol. In some examples, the output port **32a**, the electrical interface **71**, and the first interface **82** and the second interface **83** of the adapter **80** are all RS-485 interfaces.

[0303] In some examples, in the case where the charger **30a** and the outdoor work vehicle **70** perform communication methods of different standards/protocols, the control module **841** may achieve interface compatibility for the first interface **82** and the second interface **83** and perform conversions such as level values and data formats on the communication signals transmitted by a first communication subunit and a second communication subunit.

[0304] In another possible example, the adapter **80** may serve as an intermediary for power transmission between the charger **30a** and the electrical device **10b**, where the charger **30a** is the power feeder and the electrical device **10b** is the power receiver. One of the first interface **82** and the second interface **83** is detachably connected to the charger **30a**, and the other one of the first interface **82** and the second interface **83** is detachably connected to the electrical device **10b**. The case where the first interface **82** is connected to the charger **30a** and the second interface **83** is connected to the electrical device **10b** is used as an example for an exemplary description. However, it is to be understood that even for the same adapter **80**, in cases of different chargers **30** and electrical devices **10**, the corresponding connection relationship of the interfaces may be non-fixed. In one scenario, the first interface **82** is connected to the charger **30a** and the second interface **83** is connected to the electrical device **10b**, while in another scenario, the first interface **82** is connected to the outdoor work vehicle **10** and the second interface **83** is connected to the charger **30a**.

[0305] The present application correspondingly proposes a charging device **10**, which is at least a combination of the charger **30a** and the adapter **80**. The charging device **10** serves as an intermediary for power transmission between the mains power and the electrical device **10b**. An input interface of the charging device **10** is the input port **31a** of the charger **30a** and is connected to the alternating current such as the mains power, an output interface of the charging device **10** is the second interface **83** of the adapter **80** and is detachably connected to the electrical interface of the electrical device **10b** to charge battery packs **20d** mounted on the electrical device **10b**, and the output port **32a** of the charger **30a** and the first interface **82** of the adapter **80** are connected for power transmission inside the charging device **10**. The output port **32a** of the charger **30a** and the first interface **82** of the adapter **80** have the same physical structures, the first interface **82** and the second interface **83** of the adapter **80** have different physical structures, and the second interface **83** of the adapter **80** and the electrical interface of the electrical device **10b** have the same physical structures. For the description of the different physical structures of the first interface **82** and the second interface **83**, reference may be made to the above. In this example, one of the first interface **82** and the second interface **83** of the adapter **80** may be a circular interface, and the other one of the first interface **82** and the second interface **83** may be a square interface. For example, the first interface **82** is a circular interface, and the second interface **83** is a square interface.

[0306] In some examples, the electrical device **10b** may include the first electrical device **110** and the second electrical device **120**. The first electrical device **110** shown in FIG. 54 can charge at least two identical battery packs, and the second electrical device **120** shown in FIG. 55 can charge at least two different battery packs. For the description of the first electrical device **110** and the second electrical device **120**, reference may be made to the above. In some other examples, the electrical device **10b** may supply power to a power tool equipped with a motor. In some examples, the electrical device **10b** supplies power to the power tool equipped with the motor through the battery packs **20d** mounted on the electrical device **10b**. For example, fully charged battery packs may be connected to the power tool to supply power to the power tool. In some other examples, the electrical device **10b** supplies power to the power tool equipped with the motor by directly supplying power to the power tool with the charging device **10**, the electrical device **10b** has another electrical interface to which the power tool can be connected, and the electrical energy connected through the second interface **83** of the adapter **80** is outputted from the electrical device **10b** to the power tool through the electrical interface via a circuit that does not pass through the battery packs on the electrical device **10b**.

[0307] In another possible example, the adapter **80** may serve as an intermediary for power transmission between the outdoor work vehicle **70** and the electrical device **10b**, where one of the outdoor work vehicle **70** and the electrical device **10b** is the power feeder and the other one of the outdoor work vehicle **70** and the electrical device **10b** is the power receiver, that is, the outdoor work vehicle **70** may supply power to the electrical device **10b**, or the electrical device **10b** may supply power to the outdoor work vehicle **70**, that is, the identities of the outdoor work vehicle **70** and the electrical device **10b** as the power feeder and the power receiver may be interchanged. One of the first interface **82**

and the second interface **83** is detachably connected to the outdoor work vehicle **70**, and the other one of the first interface **82** and the second interface **83** is detachably connected to the electrical device **10b**. The case where the first interface **82** is connected to the electrical device **10b** and the second interface **83** is connected to the outdoor work vehicle **70** is used as an example for an exemplary description. However, it is to be understood that even for the same adapter **80**, in cases of different outdoor work vehicles **70** and electrical devices **10**, the connection relationship of the interfaces may be non-fixed. In one scenario, the first interface **82** is connected to the electrical device **10b** and the second interface **83** is connected to the outdoor work vehicle **70**, while in another scenario, the first interface **82** is connected to the outdoor work vehicle **70** and the second interface **83** is connected to the electrical device **10b**.

[0308] The present application correspondingly proposes the outdoor work vehicle **70** connected to the adapter **80**. The outdoor work vehicle **70** may include a power supply device, a traveling assembly, and a working attachment. The power supply device provides electrical energy for the operation of the traveling assembly and the working attachment and includes at least one battery pack, and the total energy of the power supply device is greater than or equal to 2 kW·h. The traveling assembly implements the traveling function and includes traveling wheels and a motor. The traveling wheels support the outdoor work vehicle **70** to travel on the ground, and the motor drives the traveling wheels. The working attachment implements a working function. The manned mower is used as an example, and the working attachment may be a cutting blade that performs the mowing operation. The adapter **80** connected to the outdoor work vehicle **70** may include the housing **81**, the first interface **82**, and the second interface **83**. The first interface **82** and the second interface **83** are disposed at two ends of the housing **81**, respectively. For details, reference may be made to the above.

[0309] After the outdoor work vehicle **70** connected to the adapter **80** is connected to the charger **30a**, the power supply device in the outdoor work vehicle **70** may be charged using the mains power. After the outdoor work vehicle **70** connected to the adapter **80** is connected to the electrical device **10b**, the battery packs **20d** mounted on the electrical device **10b** may be charged using the power supply device in the outdoor work vehicle **70**, or the power supply device in the outdoor work vehicle **70** may be charged using the battery packs **20d** mounted on the electrical device **10b**. The electrical interface **71** of the outdoor work vehicle **70** and the second interface **83** of the adapter **80** have the same physical structures, and the first interface **82** of the adapter **80** has the same physical structure as the electrical interface of the electrical device **10b** and/or the output port **32a** of the charger **30a**. In some examples, the physical structures of the first interface **82** and the second interface **83** of the adapter **80** may be different. For related descriptions, reference is made to the above. In an example, the first interface **82** is a circular interface, and the second interface **83** is a square interface.

[0310] In some examples, the adapter **80** mainly solves the adaptation problem of the interfaces in terms of physical structures and interface protocols. The voltage connected to the first interface **82** of the adapter **80** is basically the same as the voltage outputted by the second interface **83**.

[0311] In some examples, the adapter **80** has more complex and diverse functions, and in addition to the housing **81**, the first interface **82**, and the second interface **83**, the adapter **80** further includes the circuit board assembly **84** disposed in the accommodation space within the housing **81**. As shown in FIG. **59**, at least a control module **841** and a communication module **842** may be disposed on the circuit board assembly **84**. The communication module **842** may include a first communication submodule connected to the first interface **82** and a second communication submodule connected to the second interface **83**. When the interface of the adapter **80** is connected to the charger **30a**, the outdoor work vehicle **70**, or the electrical device **10b**, these devices can transmit communication signals through the interfaces in addition to transmitting electrical energy through the first interface **82** and the second interface **83**. The control module **841** may be connected to the communication module **842** to acquire the preceding communication signals to perform relevant control. In some examples, a power module **843** is further disposed on the circuit board assembly **84** and may supply power to the control module **841** and/or the communication module **842**. Specifically, the power module **843** converts the electrical energy acquired through the first interface **82** or the second interface **83** into a form that satisfies the power supply requirements of the modules, thereby supplying power to the control module **841** and/or the communication module **842**. In some examples, the power module **843** includes a buck submodule and an LDO submodule. Specifically, as shown in FIG. **59**, the power supply current flowing in from the first interface **82** or the second interface **83** passes through an anti-reverse diode and an anti-surge resistor for initial filtering, enters the buck submodule for a voltage reduction, is filtered after the first voltage reduction, enters the LDO submodule for a second voltage reduction, and is filtered after the second voltage reduction so that the power supply current can be used as an auxiliary power supply to supply electrical energy to the control module **841** and/or the communication module **842**. As for the specific implementation of the buck submodule and the subLDO module, many optional manners exist, and the details are not repeated here.

[0312] In some examples, the circuit board assembly **84** includes a first circuit board **8401** and a second circuit board **8402**, some of the control module **841**, the communication module **842**, and the power module may be disposed on the first circuit board **8401**, and other parts may be disposed on the second circuit board **8402**. In an example, the control module **841** may be disposed on the first circuit board **8401**, and the communication module **842** may be disposed on the second circuit board **8402**. In another example, the control module **841** and the communication module **842** may be disposed on the first circuit board **8401**, and the power module **843** may be disposed on the second circuit board **8402**. In some examples, in the adapter **80**, the plane where the first circuit board **8401** is located is parallel to the plane where the second circuit board **8402** is located, and the projections of the first circuit board **8401** and the second circuit board **8402** along the up and down direction at least partially overlap. The case where the first circuit board **8401** is above the second circuit board **8402** is used as an example. In an example, the projection of the first circuit board **8401** along the up and down direction may fall within the projection of the second circuit board **8402** along the up and down direction, thereby maintaining the compact structure

of the circuit board assembly **84** and effectively utilizing the accommodation space inside the adapter **80**.

[0313] In some examples, as shown in FIG. **58**, in addition to the housing **81**, the first interface **82**, the second interface **83**, and the circuit board assembly **84**, the adapter **80** further includes a sealing member **86**. Specifically, the adapter **80** may include a bracket and screws for supporting and fixing the circuit board assembly **84** and locking members for locking the harness for introducing current and/or communication signals onto the housing **81** and/or the bracket and may also include sealing rings each of which is between the first interface **82** or the second interface **83** and the housing **81** to implement waterproofing and dustproofing functions.

[0314] In a possible example, the adapter **80** also serves as an intermediary for electrical energy regulation between any two of the charger **30a**, the outdoor work vehicle **70**, and the electrical device **10b**, and the first interface **82** and the second interface **83** are detachably connected to the two, respectively. In some examples, the adapter **80** serves as an intermediary for electrical energy regulation between the charger **30a** and the outdoor work vehicle **70**. In some other examples, the adapter **80** serves as an intermediary for electrical energy regulation between the charger **30a** and the electrical device **10b**. In some other examples, the adapter **80** serves as an intermediary for electrical energy regulation between the outdoor work vehicle **70** and the electrical device **10b**. In the example where the charger **30a** is provided, the charger **30a** is the power feeder, and the outdoor work vehicle **70** or the electrical device **10b** is the power receiver. In the example where the charger **30a** is not provided, the identities of the outdoor work vehicle **70** and the electrical device **10b** as the power feeder and the power receiver may be interchanged. The case where the charger **30a** supplies power, the outdoor work vehicle **70** receives power, the first interface **82** is connected to the charger **30a**, and the second interface **83** is connected to the outdoor work vehicle **70** is used as an example for the description below.

[0315] In this example, the output port **32a** of the charger **30a** may output the direct current, and the output power of the output port **32a** is greater than or equal to 550 W and less than or equal to 5000 W. The first interface **82** of the adapter **80** is connected to the output port **32a** of the charger **30a** to acquire the current outputted by the charger **30a** and exchange communication signals with the charger **30a**. The second interface **83** of the adapter **80** is connected to the electrical interface **71** of the outdoor work vehicle **70** to output the current to the power supply device in the outdoor work vehicle **70** and exchange communication signals with the outdoor work vehicle **70**. Based on the above, in addition to the housing **81**, the first interface **82**, and the second interface **83**, the adapter **80** further includes the circuit board assembly **84**, and the control module **841**, the communication module **842**, and the power module **843** are disposed on the circuit board assembly **84**. The power module **843** may be used as an auxiliary power supply to convert electrical energy from the interface of the adapter **80** to supply power to the control module **841** and/or the communication module **842**, the first communication submodule connected to the first interface **82** and the second communication submodule connected to the second interface **83** are both connected to the control module **841**, and the control module **841** regulates the magnitude of the current transmitted between the first interface **82** and the second interface **83**.

[0316] In some examples, after the adapter **80** is connected to the charger **30a** and the outdoor work vehicle **70**, the control module **841** may call the first communication submodule and the second communication submodule to perform communication detection on the first interface **82** and the second interface **83**, respectively to determine whether the first interface **82**, the second interface **83**, the interface of the charger **30a** connected to the first interface **82**, and the electrical interface **71** on the outdoor work vehicle **70** connected to the second interface **83** have faults, data anomalies, or other issues. In the case where it is determined that the interface communication detection passes, the control module **841** allows current transmission between the first interface **82** and the second interface **83** or further acquires communication signals from the first interface **82** and/or the second interface **83** and based on the information carried by the communication signals, adjusts the magnitude of the current transmitted between the two interfaces.

What is claimed is:

1. An adapter, comprising:

a housing;

a first interface configured to be detachably connected to an output port of a charger; and

a second interface configured to be detachably connected to an outdoor work vehicle, wherein the outdoor work vehicle is configured to be powered by a power supply device to implement a function;

wherein the adapter is configured to acquire electrical energy outputted by the charger through the first interface and output the electrical energy through the second interface to charge the power supply device, output power of the output port is greater than or equal to 550 W and less than or equal to 5000 W, and the first interface and the second interface have different physical structures.

2. The adapter of claim 1, wherein the output port is configured to be detachably connected to an electrical device to charge a battery pack mounted to the electrical device, and a direct current voltage of the output port is greater than or equal to 21 V and less than or equal to 65 V.

3. The adapter of claim 1, wherein the charger comprises an alternating current-direct current conversion module, and the alternating current-direct current conversion module is configured to convert an alternating current connected to an input port of the charger into a direct current outputted by the output port.

4. The adapter of claim 1, wherein the housing comprises a first housing and a second housing, and the first housing and the second housing together form an accommodation space for accommodating a circuit board assembly.

5. The adapter of claim 4, wherein one of the first interface and the second interface is supported by both the first housing and the second housing, and another one of the first interface and the second interface is supported by the first housing or the second housing.

6. The adapter of claim 4, wherein an included angle between a plugging and unplugging direction of the first interface and a plugging and unplugging direction of the second interface is greater than or equal to 120° and less than or equal to 185°.

7. The adapter of claim 1, wherein the first interface and the second interface have different shapes.

8. The adapter of claim 7, wherein the first interface is a circular interface, and the second interface is a square interface.

9. The adapter of claim 1, wherein a voltage connected to the first interface is basically the same as a voltage outputted by the second interface.

10. The adapter of claim 1, wherein the outdoor work vehicle comprises a manned mower or an all-terrain vehicle.

11. A charging device, comprising:

a charger comprising an input port configured to be connected to mains power and an output port configured to output electrical energy; and

an adapter comprising a first interface configured to be detachably connected to the output port of the charger to receive the electrical energy outputted by the charger and a second interface configured to be detachably connected to an electrical device to supply the electrical energy to the electrical device;

wherein one of the first interface and the second interface is a circular interface, and another one of the first interface and the second interface is a square interface.

12. The charging device of claim 11, wherein output power of the output port of the charger is greater than or equal to 550 W and less than or equal to 5000 W.

13. The charging device of claim 11, wherein the electrical device comprises a first electrical device and a second electrical device, the first electrical device is configured to charge at least two identical battery packs, and the second electrical device is configured to charge at least two different battery packs.

14. The charging device of claim 11, wherein the electrical device is configured to supply power to a power tool equipped with a motor.

15. The charging device of claim 11, wherein the adapter comprises a power module, a control module, and a communication module, and the power module is configured to supply power to the control module and the communication module.

16. A combination, comprising:

an outdoor work vehicle comprising a working attachment configured to implement a working function, a power supply device comprising at least one battery pack, wherein a total energy of the power supply device is greater than or equal to 2 kW·h, and a traveling assembly comprising traveling wheels for supporting the outdoor work vehicle to travel on a ground and a motor for driving the traveling wheels; and

an adapter comprising a first interface configured to be detachably connected to an output port of a charger to charge the outdoor work vehicle or configured to be detachably connected to an electrical device to charge a battery pack mounted on the electrical device and a second interface configured to be detachably connected to the outdoor work vehicle to implement a charging or discharging function of the outdoor work vehicle.

17. The combination of claim 16, wherein the first interface and the second interface have different shapes.

18. The combination of claim 16, wherein the adapter further comprises a control module configured to adjust a magnitude of a current transmitted between the first interface and the second interface.

19. The combination of claim 18, wherein the adapter further comprises a communication module, the communication module comprises a first communication submodule and a second communication submodule, and the control module is configured to call the first communication submodule and the second communication submodule to perform communication detection and, when the communication detection passes, allow current transmission between the first interface and the second interface.

20. The combination of claim 19, wherein a communication manner allowed by the communication module comprises RS-485 communication and controller area network (CAN) communication.

* * * * *